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Refrigerator

Abstract

A refrigerator enables a user to easily introduce and remove goods into and from the refrigerator. Specifically, the refrigerator enables a drawer for receiving goods to be more conveniently used. The refrigerator includes a cabinet having a storage compartment with a food introduction port at the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, drawers disposed in the storage compartment and arranged vertically, a moving frame extending vertically to correspond to the height at which the drawers are disposed, the moving frame being configured to selectively push the drawers such that the drawers are moved toward the food introduction port, an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port, and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/316,761, filed on May 12, 2023, which is a continuation of U.S. application Ser. No. 17/585,023, filed on Jan. 26, 2022, now U.S. Pat. No. 11,725,874, which is a continuation of U.S. application Ser. No. 17/101,927, filed on Nov. 23, 2020, now U.S. Pat. No. 11,274,878, which is a continuation of U.S. application Ser. No. 16/825,009, filed on Mar. 20, 2020, now U.S. Pat. No. 10,876,789, which is a continuation of U.S. application Ser. No. 15/554,839, filed on Aug. 31, 2017, now U.S. Pat. No. 10,612,836, which is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/KR2016/012607, filed Nov. 3, 2016, which claims the benefit of Korean Application No. 10-2016-0001300, filed on Jan. 5, 2016 and Korean Application No. 10-2015-0154816, filed on Nov. 4, 2015. The disclosures of the prior applications are incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a refrigerator. Specifically, the present disclosure relates to a refrigerator that is capable of enabling a user to easily introduce or remove goods into or from the refrigerator. More specifically, the present disclosure relates to a refrigerator that is capable of enabling a drawer for receiving goods to be more conveniently used.

BACKGROUND

(2) In general, a refrigerator is an appliance that discharges cool air, generated using a refrigeration cycle that uses a compressor, a condenser, an expansion valve, and an evaporator, for lowering the temperature in the refrigerator to store foods in a frozen state or in a refrigerated state.

(3) A refrigerator generally includes a freezing compartment for storing foods or beverages in a frozen state and a refrigerating compartment for storing foods or beverages in a refrigerated state.

(4) Refrigerators may be classified into a top mount type refrigerator configured such that a freezing compartment is disposed on a refrigerating compartment, a bottom freezer type refrigerator configured such that a freezing compartment is disposed under a refrigerating compartment, and a side by side type refrigerator configured such that a freezing compartment and a refrigerating compartment are arranged side by side. Doors are provided at the freezing compartment and the refrigerating compartment. A user may access the freezing compartment or the refrigerating compartment by opening a corresponding one of the doors.

(5) In addition, there is a refrigerator configured such that a user may access the freezing compartment and the refrigerating compartment by opening a single door. In general, this type of refrigerator is a small-sized refrigerator configured such that the freezing compartment is provided in a predetermined space within the refrigerating compartment.

(6) Furthermore, there is a French type refrigerator, which is a modification of the top mount type refrigerator, configured such that the upper refrigerating compartment is opened and closed by left and right doors. Of course, the freezing compartment of the French type refrigerator may be opened and closed by left and right doors.

(7) In general, shelves, on which goods are placed, or receiving boxes, in which good are received, are disposed in the refrigerating compartment and the freezing compartment. The receiving boxes are generally provided to form independent storage spaces in the storage compartment. That is, the receiving boxes may be provided in order to store vegetables or fruits separately from other goods or to store meat or fish separately from other goods.

(8) In recent years, the capacity of refrigerators has been gradually increased. Accordingly, the forward and rearward width of the storage compartment is increased, with the result that it is not easy to withdraw goods that are stored deep inside the storage compartment. For this reason, most of the receiving boxes are configured to have a drawer form. That is, the user may pull the receiving boxes in order to take goods out from the receiving boxes. In particular, the drawer type receiving boxes are generally provided in the lower region of the refrigerator in order to improve user convenience.

(9) In addition, in recent years, a home bar, an ice maker, a shelf, and a door box have been increasingly frequently mounted at the rear of the door of the refrigerator in order to use the rear of the door as an additional storage space or an additional functional space. That is, the door has additional functions, such as the provision of additional storage space or the production and supply of ice or cold water, in addition to simply opening and closing the freezing compartment or the refrigerating compartment. For these reasons, the distance by which the rear of the door is inserted into the refrigerating compartment or the freezing compartment is further increased. As a result, the fronts of the shelves or the receiving boxes provided in the refrigerating compartment or the freezing compartment may interfere with the rear of the door.

(10) In order to reduce such interference, the fronts of the shelves or the receiving boxes may be positioned so as to be spaced apart rearward from the front of the main body of the refrigerator by a predetermined distance. That is, the fronts of the shelves or the receiving boxes may be positioned further inward in the freezing compartment or the refrigerating compartment. In a case in which the receiving boxes are configured to have a drawer shape, therefore, it may be difficult for the user to withdraw the receiving boxes while holding the fronts of the receiving boxes. In other words, the user must insert his/her hand more deeply into the storage compartment in order to withdraw the receiving boxes. Particularly, in a case in which the receiving boxes are provided in the lower part of the refrigerator, the user must withdraw the receiving boxes in a crouching style, which is very inconvenient.

(11) Supposing that the fronts (for example, handles) of the receiving boxes are positioned deeply in the storage compartment, rather than right in front of the user, when the user opens the door in order to withdraw the receiving boxes, such inconvenience may be easily understood.

(12) In order to solve the above problem, the applicant of the present application has proposed a storage structure configured to be interlocked with the door, which is disclosed in Korean Patent Application Publication No. 2010-0130357 (hereinafter, referred to as a "prior invention"). The storage structure according to the prior invention includes a link for mechanically interlocking the door and the storage structure. When the door is opened, therefore, the storage structure is withdrawn. That is, the storage structure is mechanically withdrawn to a position spaced apart forward from an initial position by a predetermined distance such that the user can withdraw a drawer provided in the storage structure more easily. When the opening angle of the door is

increased, therefore, the distance by which the drawer is withdrawn increases.

(13) However, the prior invention has a problem in that when the door is opened, the link is exposed outward, whereby the link blocks the movement path of the user. In addition, it is not possible to provide a refrigerator having an aesthetically pleasing appearance as the result of the provision of the link.

(14) In addition, in the drawer according to the prior invention, additional force is required in order to open the door. This is because the force necessary to pull the drawer as well as the force necessary to open the door are both required. A particularly high force may be required when the door is initially opened. This is because a force higher than a static frictional force of the drawer must be applied in order to withdraw the drawer. The static frictional force of the drawer is proportional to the load of the drawer. In a case in which a large amount of goods is stored in the drawer, therefore, it is difficult to open the door.

(15) In addition, the prior invention has a problem in that the storage structure, which substantially occupies the entire space of the storage compartment, moves forward and rearward, whereby the space for storing goods is somewhat reduced. That is, the space for storing goods may be much less than the entire volume of the storage compartment.

(16) Meanwhile, the user may not open the door slowly, but may open the door very quickly using a very high force. In this case, a very high force and impact may be applied to the link and the drawer. Of course, a very high force and impact may be applied to the elastic device. As a result, the door, the link, the connection between the link and the drawer, and the elastic device may be damaged.

(17) Meanwhile, the prior invention has a problem in that it is not possible to insert the storage structure to the initial position in a state in which the door is open. This is because the insertion of the storage structure is prevented by the link in a state in which the door is open. In a case in which a portion of the storage structure is used, therefore, the remaining portions of the storage structure, which are not used, remain withdrawn, which causes a loss of cool air.

(18) Generally, in conventional refrigerators, the drawer, particularly the drawer in the freezing compartment, is withdrawn and inserted along a rail. The rail is provided at the sidewall of the storage compartment, and the drawer is provided with a rail connection part. The rail connection part is formed in the shape of a roller. The drawer moves forward and rearward in a state in which the rail connection part is inserted in the rail.

(19) However, it is not easy to couple the above-mentioned type of drawer to the rail. Particularly, in a state in which heavy goods are received in the drawer, it is difficult for the user to fit the roller into the rail while holding the drawer.

(20) In addition, the user may completely separate the drawer from the refrigerator, rather than putting goods in the drawer or taking goods out from the drawer after withdrawing the drawer, as needed. In this case, it is very inconvenient for the user to separate the rail connection part from the rail and then to couple the rail connection part to the rail. Consequently, the user usually uses the drawer in a state in which the drawer is not separated from the refrigerator, unless there is some special reason otherwise.

SUMMARY

(21) The present invention has been made to fundamentally solve the above problems.

(22) It is an object of the present invention to provide a refrigerator configured such that when a user opens a door of the refrigerator, a drawer provided in a storage compartment is automatically withdrawn forward by a predetermined distance.

(23) It is another object of the present invention to provide a refrigerator configured such that a drawer is automatically moved from an initial position to a ready position by an electric driving unit, whereby no additional force beyond a user's force to open the door is necessary. That is, it is another object of the present invention to provide a refrigerator configured such that the force necessary to open a door and the force necessary to move a drawer from an initial position to a

ready position are individual or independent. Specifically, it is another object of the present invention to provide a refrigerator configured such that a door is opened by the manual application of a user's force to the door, and a drawer is moved from an initial position to a ready position by an electrical force regardless of whether a user's force is applied.

(24) It is another object of the present invention to provide a refrigerator configured such that a drawer electrically moves from an initial position to a ready position and such that the drawer non-electrically returns from the ready position to the initial position. In other words, it is another object of the present invention to provide a refrigerator configured such that a drawer moves from an initial position to a ready position using electrical energy and such that the drawer moves from the ready position to the initial position without using electrical energy.

(25) It is another object of the present invention to provide a refrigerator configured such that a drawer moves from an initial position to a ready position as the result of driving of a motor, and the drawer moves from the ready position to the initial position regardless of the driving of the motor.

(26) It is another object of the present invention to provide a refrigerator configured such that the driving force of a motor is selectively transferred to a drawer. In particular, it is another object of the present invention to provide a refrigerator configured such that the driving force of a motor is transferred to a drawer when the drawer is withdrawn, and the driving force of the motor is not transferred to the drawer when the drawer is inserted.

(27) It is another object of the present invention to provide a refrigerator configured such that a drawer automatically moves from an initial position to a ready position, and the drawer is manually moved from the ready position to the initial position.

(28) It is another object of the present invention to provide a refrigerator configured such that a speed at which a drawer moves from an initial position to a ready position is different from a speed at which the drawer moves from the ready position to the initial position. Specifically, it is another object of the present invention to provide a refrigerator configured such that a speed at which a drawer moves from a ready position to an initial position is higher than a speed at which the drawer moves from the initial position to the ready position.

(29) It is another object of the present invention to provide a refrigerator configured such that a drawer moves from a ready position to an initial position due to an elastic restoring force. In particular, it is another object of the present invention to provide a refrigerator configured such that an element impeding the insertion of a drawer is removed, whereby the drawer returns relatively rapidly using an elastic restoring force. Therefore, it is another object of the present invention to provide a refrigerator configured such that a drawer completely returns to an initial position while a door is being closed.

(30) It is another object of the present invention to provide a refrigerator configured such that a drawer is automatically withdrawn and automatically inserted. That is, it is another object of the present invention to provide a refrigerator configured such that an electric driving unit is driven to withdraw and insert a drawer. In particular, it is another object of the present invention to provide a refrigerator configured such that a speed at which a drawer is withdrawn and a speed at which the drawer is inserted are controlled to be different from each other, thereby minimizing the collision between the drawer and a door.

(31) It is another object of the present invention to provide a refrigerator configured such that a speed at which a door is opened and/or a closed is sensed in order to change the speed of an electric driving unit configured to move a drawer, particularly the speed of a motor.

(32) It is another object of the present invention to provide a refrigerator configured such that interference between a door and a drawer, configured to automatically move when the door is opened or closed, is considerably reduced using a sensor for very precisely sensing the opening angle of the door and/or the closing angle of the door. That is, it is another object of the present invention to provide a refrigerator configured such that an opening (or closing) angle of a door that is capable of minimizing interference between the door and a drawer is set, whereby it is possible

to very precisely sense whether the door is open (or closed) at the set angle.

(33) It is another object of the present invention to provide a refrigerator including a door opening sensor that is capable of flexibly corresponding to a door opening angle that varies depending upon the product models.

(34) It is another object of the present invention to provide a refrigerator configured such that it is sensed whether a door is open or closed at a consistent angle using a single sensor, which is easily applied to conventional refrigerators.

(35) It is another object of the present invention to provide a refrigerator configured such that an element for automatically withdrawing a drawer is not exposed in a storage compartment, whereby it is possible to protect an electric driving unit, to improve user convenience, and to provide the interior of the storage compartment with an aesthetically pleasing appearance.

(36) It is another object of the present invention to provide a refrigerator configured such that it is possible to simultaneously move a plurality of drawers from an initial position to a ready position using a single electric driving unit. To this end, it is another object of the present invention to provide a refrigerator including a moving frame that is capable of simultaneously transferring the driving force of a single electric driving unit to a plurality of drawers.

(37) It is another object of the present invention to provide a refrigerator including a moving frame that exhibits a high load distribution property, high durability, and high reliability in assembly.

(38) It is another object of the present invention to provide a refrigerator configured such that a drawer is automatically inserted and withdrawn with high reliability and durability. In particular, it is another object of the present invention to provide a refrigerator configured such that it is possible to minimize damage to an electric driving unit due to overload of the electric driving unit or repetitive use of the electric driving unit for a long period of time.

(39) It is another object of the present invention to provide a refrigerator configured such that a drawer configured to be automatically withdrawn, an electric driving unit configured to automatically withdraw the drawer, and relevant elements are easily assembled, and, in addition, are easily repaired as needed. In addition, it is another object of the present invention to provide a refrigerator configured such that it is possible to minimize the reduction in capacity of a storage compartment due to the above-mentioned elements.

(40) It is another object of the present invention to provide a refrigerator configured such that a plurality of drawers is simultaneously automatically withdrawn, and, in addition, the drawers are easily manufactured and maintained.

(41) It is another object of the present invention to provide a refrigerator configured such that a drawer is easily connected to or separated from a rail configured to support the drawer.

(42) It is another object of the present invention to provide a refrigerator configured such that only a basket for receiving goods is easily separated from and coupled to a drawer. That is, it is another object of the present invention to provide a refrigerator configured such that only a basket is easily separated from and coupled to a drawer in a state in which the connection between a rail and a rail connection part of the drawer is maintained.

(43) It is another object of the present invention to provide a refrigerator configured such that it is possible to maximally prevent a rail from being visibly exposed to a user.

(44) It is a further object of the present invention to provide a control method of a refrigerator that is capable of minimizing the load of a motor and flexibly corresponding to various environments in which a drawer is used.

Means for Solving the Problems

(45) To achieve these objects and other advantages and in accordance with the purpose of the invention, as embodied and broadly described herein, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, an electric driving unit for automatically moving the drawer to a ready position spaced apart forward from an initial position

by a predetermined distance when the door is opened, and a controller for controlling the driving of the electric driving unit.

(46) The drawer may be automatically and/or electrically withdrawn.

(47) The drawer may be manually moved from the ready position to the initial position when the door is closed.

(48) The drawer may be non-electrically moved from the ready position to the initial position when the door is closed.

(49) The drawer may be elastically moved from the ready position to the initial position when the door is closed.

(50) The drawer may be inserted and withdrawn regardless of whether a user's force is applied to the door when the door is opened and closed.

(51) The drawer may be automatically inserted when the door is closed. The electric driving unit may be driven to automatically insert the drawer. That is, the electric driving unit may be controlled to be driven in order to insert and withdraw the drawer. The motor may be driven in a clockwise direction in order to withdraw the drawer, and the motor may be driven in a counterclockwise direction in order to insert the drawer.

(52) The speed at which the motor is driven may be controlled by the controller.

(53) The speed at which the motor is driven to insert the drawer and the speed at which the motor is driven to withdraw the drawer may be controlled to be different from each other. Of course, the speed at which the motor is driven to insert the drawer and the speed at which the motor is driven to withdraw the drawer may be controlled to be the same. Impact applied to the drawer and the door when the door is opened may be greater than impact applied to the drawer and the door when the door is closed. In order to prevent or minimize the impact applied to the drawer and the door, therefore, the speed at which the motor is driven to insert the drawer and the speed at which the motor is driven to withdraw the drawer may be controlled such that the speed at which the motor is driven to insert the drawer is higher than the speed at which the motor is driven to withdraw the drawer.

(54) The speed at which the motor is driven to move the drawer may be variably set depending on the speed at which the door is moved. When the speed at which the door is closed is increased, the speed at which the motor is driven to insert the drawer may be controlled to be further increased. When the speed at which the door is opened is increased, the speed at which the motor is driven to withdraw the drawer may be controlled to be further increased.

(55) The drawer may be withdrawn from the initial position to the ready position regardless of the increase in opening angle of the door. That is, the drawer may not move even when the opening angle of the door is increased, and the drawer may be withdrawn at a specific opening angle of the door or within a specific opening angular range of the door. On the other hand, the drawer may not move even when the closing angle of the door is decreased, and the drawer may be inserted at the specific opening angle of the door or within the specific opening angular range of the door. In other words, the drawer may be inserted and withdrawn without being mechanically interlocked with the door.

(56) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a sensor for sensing whether the door is open, an electric driving unit for moving the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open, and a rail configured to allow the drawer to move forward and rearward relative to the storage compartment.

(57) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a sensor for sensing whether the door

is open, an electric driving unit for moving the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open, a rail configured to allow the drawer to move forward and rearward relative to the storage compartment, and an elastic device configured to be elastically deformed when the drawer moves from the initial position to the ready position and configured to provide an elastic restoring force to the drawer when the drawer moves from the ready position to the initial position.

(58) The drawer may be moved from the ready position to the initial position by the elastic restoring force of the elastic device. The drawer may be moved only by the elastic restoring force regardless of whether a user's force is applied to the door.

(59) The elastic device may be constantly connected with the drawer within a section between the initial position and the ready position of the drawer.

(60) The elastic device may be selectively connected with the drawer. When the connection between the elastic device and the drawer is released, the drawer may be moved regardless of elastic deformation and elastic restoration of the elastic device.

(61) The elastic device may be disconnected from the drawer when the drawer is withdrawn further forward from the ready position. As the result of the release of the connection between the elastic device and the drawer, the drawer may be further withdrawn manually.

(62) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a sensor for sensing whether the door is open, a motor assembly, a moving frame configured to be moved forward and rearward by driving of the motor assembly, the moving frame pushing the drawer when being moved forward, and a controller for controlling the driving of the motor assembly.

(63) The controller may control the motor assembly to move the moving frame forward when it is sensed that the door is open. The controller may control the movement of the moving frame such that the drawer is moved to a ready position spaced apart forward from an initial position by a predetermined distance.

(64) The moving frame may return after moving the drawer to the ready position. That is, the moving frame may return rearward after pushing the drawer such that the drawer is moved to the ready position. Here, the position to which the moving frame returns may also be referred to as the initial position. At this time, the controller may control the motor assembly to be driven in the reverse direction. When the moving frame returns to the initial position, therefore, the pushing force applied to the drawer may be removed or released.

(65) This means that the force resisting the return of the drawer is substantially removed or released. Consequently, the drawer may easily return to the initial position when the door is closed.

(66) Of course, an elastic device may be provided such that the drawer is returned to the initial position by an elastic restoring force of the elastic device. In this case, the force resisting the return of the drawer is removed, and the drawer is returned to the initial position by the elastic restoring force of the elastic device, whereby the drawer may return at a very high speed. As a result, it is possible to prevent or minimize the collision between the drawer and the door.

(67) In addition, in a case in which the elastic device is provided, it is possible to return the drawer to the initial position even in a state in which the door is open. In a case in which a plurality of drawers is provided, the drawers may be inserted to the initial position after use, thereby minimizing the loss of cool air.

(68) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a sensor for sensing whether the door is open, a drawer provided in the storage compartment, the drawer being configured to move to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open, an elastic device configured to be elastically deformed when the drawer moves from the initial position to the

ready position and configured to provide an elastic restoring force to the drawer such that the drawer moves from the ready position to the initial position, and an electric driving unit for moving the drawer from the initial position to the ready position and elastically deforming the elastic device.

(69) A force to withdraw the drawer forward and a force to elastically deform the elastic device may be generated by driving of the electric driving unit. That is, the force to withdraw the drawer forward and the force to elastically deform the elastic device may have no relation to the magnitude of the user's force applied to open the door.

(70) In addition, a force to insert the drawer rearward may also have no relation to the user's force to open the door. That is, the drawer may be inserted rearward by the elastic restoring force of the elastic device.

(71) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment so as to be movable forward and rearward, and a rail for supporting the drawer such that the drawer moves forward and rearward relative to the storage compartment, wherein the drawer includes a basket for receiving goods and a drawer frame provided with a basket location part, on which the basket is located, and a rail coupling part located on the rail so as to be coupled to the rail.

(72) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment so as to be movable forward and rearward, the drawer including a basket for receiving goods and a drawer frame provided with a rail coupling part, and a rail coupled to the rail coupling part for supporting the drawer such that the drawer moves forward and rearward relative to the storage compartment, wherein the rail coupling part is formed in a channel shape such that the rail coupling part is located on the rail downward from above so as to surround the rail, and the rail is provided at the rear thereof with a catching part, into which a rear end of the rail coupling part is inserted, and the rail is provided at the front thereof with an elastic protrusion, which is inserted into a mounting hole provided at the front end of the rail coupling part.

(73) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a sensor for sensing whether the door is open, a drawer provided in the storage compartment so as to be movable forward and rearward, the drawer including a basket for receiving goods and a drawer frame provided with a rail coupling part and a catching member protruding toward a sidewall of the storage compartment, a rail coupled to the rail coupling part for supporting the drawer such that the drawer moves forward and rearward relative to the storage compartment, the rail being supported by the sidewall of the storage compartment via a front rail bracket and a rear rail bracket, and an electric driving unit including a transfer member protruding from the sidewall of the storage compartment toward the drawer frame, the electric driving unit moving the catching member based on the movement of the transfer member to move the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open, wherein the transfer member is configured to move between the front rail bracket and the rear rail bracket within a section between the initial position and the ready position of the drawer in order to avoid the interference between the transfer member and the front and rear rail brackets.

(74) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, the drawer including a catching member, a sensor for sensing whether the door is open, a support assembly configured to be coupled to the drawer for supporting the drawer so as to be movable forward and rearward relative

to the storage compartment, the support assembly being separably coupled to a left sidewall or a right sidewall of the storage compartment, wherein the support assembly includes a support cover configured to be coupled to the sidewall of the storage compartment, a motor assembly mounted to the inside surface of the support cover facing the sidewall, a rail mounted to the outside surface of the support cover for supporting the drawer so as to be movable forward and rearward, and a moving frame mounted to the support cover so as to move forward and rearward in a space between the sidewall and the support cover by driving of the motor assembly, the moving frame including a transfer member configured to push the catching member at the rear of the catching member through the support cover.

(75) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, the drawer including a catching member, a sensor for sensing whether the door is open, a support assembly configured to be coupled to the drawer for supporting the drawer so as to be movable forward and rearward relative to the storage compartment, the support assembly being separably coupled to a left sidewall or a right sidewall of the storage compartment, wherein the support assembly includes a support cover configured to be coupled to the sidewall of the storage compartment, the support cover being provided with a slit extending forward and rearward, a motor assembly mounted to the inside surface of the support cover facing the sidewall, a rail mounted to the outside surface of the support cover for supporting the drawer so as to be movable forward and rearward, and a moving frame mounted to the support cover so as to move forward and rearward in a space between the sidewall and the support cover by driving of the motor assembly, the moving frame including a transfer member configured to push the catching member at the rear of the catching member through the slit.

(76) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a magnet provided at the door, the magnet being configured to turn about a rotary shaft of the door with a predetermined turning radius as the door is opened, and a reed switch provided above or under the magnet such that the reed switch is spaced apart from the magnet, the reed switch being fixed to the cabinet regardless of the hinged rotation of the door, the reed switch having a critical point of effective magnetic intensity for contact point switching when an opening angle of the door reaches a predetermined opening angle.

(77) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a sensor including a magnet provided at the door, the magnet being configured to turn about a rotary shaft of the door with a predetermined turning radius as the door is opened and a reed switch fixed to the cabinet, a contact point of the reed switch being switched at a critical point of effective magnetic intensity due to the magnet, the sensor being configured to sense that the door is open when an opening angle of the door reaches a predetermined opening angle, and an electric driving unit for moving the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open.

(78) In another aspect of the present invention, a control method of a refrigerator, including a motor, a drawer provided in a storage compartment defined in a cabinet so as to movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, includes determining a condition for automatically withdrawing the drawer in a ready state (a determination step), upon determining at the determination step that the condition is satisfied, driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position (a withdrawal step), and driving the motor in a reverse direction to return the transfer member rearward (a returning step).

(79) The control method may further include continuously driving the motor in the one direction to stop the withdrawal of the drawer after the withdrawal step (a stopping step). The returning step may be performed after the stopping step.

(80) The refrigerator may further include a door for opening and closing the storage compartment and a sensor for sensing that the door is open when an opening angle of the door is a predetermined opening angle. The condition for automatically withdrawing the drawer may include generating a door opening signal through the sensor.

(81) The refrigerator may further include a door switch for sensing whether the door is in tight contact with the cabinet to sense whether the door is open or closed. The door switch may be provided separately from the sensor. In terms of function, the door switch may be provided to control lighting in the storage compartment, and the sensor may be provided to control the motor, which is related to the movement of the drawer.

(82) The sensor may not only sense whether the door is open but may also sense whether the door is closed. Sensing whether the door is open may be sensing whether the door has been opened to a predetermined angle. Sensing whether the door is closed may be sensing whether the door has been closed to a predetermined angle. The opening angle of the door at which it is sensed that the door is open and the closing angle of the door at which it is sensed that the door is closed may be the same. For example, the opening angle of the door at which it is sensed that the door is open and the closing angle of the door at which it is sensed that the door is closed may be 90 degrees.

(83) The control method may further include an urgent returning step to prevent or minimize the collision between the drawer and the door when the door is closed. The urgent returning step may be performed to protect the drawer when the drawer is withdrawn or a motor assembly is driven to withdraw the drawer.

(84) Specifically, the control method may further include stopping the withdrawal step or the stopping step and driving the motor in the reverse direction to return the transfer member rearward when it is sensed through the sensor that the door is closed during the withdrawal step or the stopping step (an urgent returning step).

(85) When the urgent returning step starts to be performed in a state in which the motor is driven in the one direction, the driving of the motor may be stopped for a predetermined time, and the motor may then be driven in the reverse direction.

(86) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a plurality of drawers disposed in the storage compartment, the drawers being arranged vertically, a moving frame extending vertically so as to correspond to the height at which the drawers are disposed, the moving frame being configured to selectively push the drawers such that the drawers are moved toward the food introduction port, an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port, and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

(87) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer disposed in the storage compartment, a moving frame configured to selectively push the drawer such that the drawer is moved toward the food introduction port, an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port, and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open, wherein the drawer remains separated from the moving frame at a position at which the front part of the drawer is withdrawn after escaping from the food introduction port.

(88) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected

to the cabinet for opening and closing the storage compartment, a drawer disposed in the storage compartment, a moving frame configured to selectively push the drawer such that the drawer is moved toward the food introduction port, an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port, a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open, and an elastic device coupled to one side of the drawer and the inside wall of the storage compartment for selectively generating an elastic restoring force, wherein the drawer is returned by the elastic restoring force.

(89) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer disposed in the storage compartment, a moving frame coupled to the drawer for moving the drawer forward and rearward, an electric driving unit coupled to the moving frame, and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open or closed, wherein the controller controls the electric driving unit to be driven at a higher speed when the door is closed than when the door is opened.

(90) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer disposed in the storage compartment, a fixed rail coupled to the sidewall of the storage compartment for supporting the load of the drawer, a moving rail movably coupled to the fixed rail, the moving rail being coupled to the side surface of the drawer, a moving frame for selectively pushing the drawer to move the drawer toward the food introduction port, an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port, and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

(91) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a support cover mounted to the inside wall of the storage compartment so as to define the sidewall of the storage compartment, the support cover being provided with a through part, a plurality of rails mounted to the outside surface of the support cover, the rails being arranged vertically, a plurality of drawers disposed in the storage compartment such that the drawers are inserted or withdrawn through the food introduction port along the rails, the drawers being arranged vertically, a moving frame disposed inside the support cover, the moving frame extending vertically so as to correspond to a height at which the drawers are disposed, the moving frame being configured to selectively push the drawers through a transfer member extending through the through part such that the drawers are moved toward the food introduction port, an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port inside the support cover, and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

(92) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a sensor configured to sense whether the door is open, and an electric driving unit configured to drive the drawer such that the drawer is withdrawn forward when it is sensed that the door is open.

(93) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a plurality of drawers vertically arranged in the storage compartment, a sensor configured to sense whether the door is open, a moving frame configured to push the respective drawers while being moved forward, and an electric driving unit configured to move the moving frame forward such that the drawers are withdrawn forward when it is sensed that the door is open.

(94) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a fixed rail coupled to a sidewall of the storage compartment for supporting the load of the drawer, a moving rail movably coupled to the fixed rail, the moving rail being coupled to a side surface of the drawer, a moving frame configured to push the drawer while being moved forward, and an electric driving unit configured to move the moving frame forward such that the drawer is withdrawn forward when it is sensed that the door is open.

(95) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a support cover having an outside surface, which defines an inner surface of the storage compartment, and an inside surface, which faces a sidewall of the storage compartment, the support cover being provided with a through part, a rail mounted to the outside surface of the support cover, a drawer configured to be inserted and withdrawn through the food introduction port along the rail, a moving frame provided between the inside surface of the support cover and the sidewall of the storage compartment, the moving frame being configured to push the drawer through a transfer member extending through the through part such that the drawer is moved forward, and an electric driving unit provided between the inside surface of the support cover and the sidewall of the storage compartment for moving the moving frame forward.

(96) In another aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a sensor configured to sense that the door is open when the door is turned and opened by a predetermined angle, an elastic device configured to be elastically deformed when the drawer is withdrawn and to be elastically restored when the drawer is inserted, and an electric driving unit for driving the drawer in one direction such that the drawer is withdrawn forward to generate a force for moving the drawer forward when it is sensed that the door is open.

(97) In a further aspect of the present invention, a refrigerator includes a cabinet having a storage compartment with a food introduction port formed in the front thereof, a door hingedly connected to the cabinet for opening and closing the storage compartment, a drawer provided in the storage compartment, a rail configured to allow the drawer to move forward and rearward relative to the storage compartment, a sensor configured to sense that the door is open when the door is turned and opened by a predetermined angle, a moving frame configured to selectively push the drawer such that the drawer is moved toward the food introduction port, and an electric driving unit configured to drive the drawer such that the drawer is withdrawn forward to a ready position when it is sensed that the door is open.

(98) The features of the above embodiments may be integrated into other embodiments unless the features are inconsistent or exclusive.

(99) According to an embodiment of the present invention, it is possible to provide a refrigerator configured such that when a user opens a door of the refrigerator, a drawer provided in a storage compartment is automatically withdrawn forward by a predetermined distance. That is, it is possible to provide a refrigerator configured such that a drawer automatically moves from an initial position to a ready position. The ready position is a position to which the drawer is withdrawn forward from the initial position by a predetermined distance. That is, since the drawer in the storage compartment can be withdrawn to a position closer to the user, it is possible for the user to very conveniently use the drawer. In other words, since the drawer automatically moves from the initial position to the ready position, which is closer to the user, it is possible for the user to grasp the drawer in order to withdraw the drawer, thereby improving user convenience.

(100) According to another embodiment of the present invention, it is possible to provide a

refrigerator configured such that a drawer is automatically moved from an initial position to a ready position by an electric driving unit, whereby no additional force other than a user's force is necessary in order to open the door. That is, it is possible to provide a refrigerator configured such that a force necessary to open a door and a force necessary to move a drawer from an initial position to a ready position are individual or independent. Specifically, it is possible to provide a refrigerator configured such that a door is opened by the manual application of a user's force to the door, and a drawer is moved from an initial position to a ready position by an electrical force regardless of a user's force. Consequently, it is possible for the user to conveniently use the drawer without using any additional force.

(101) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer electrically moves from an initial position to a ready position, and the drawer non-electrically returns from the ready position to the initial position. Specifically, it is possible to provide a refrigerator configured such that a drawer moves from an initial position to a ready position using electrical energy, and the drawer moves from the ready position to the initial position without using electrical energy. Consequently, it is possible to reduce electrical energy consumption.

(102) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer moves from an initial position to a ready position as the result of driving of a motor, and the drawer moves from the ready position to the initial position regardless of the driving of the motor. Consequently, it is possible to reduce electrical energy consumption.

(103) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that the driving force of a motor is selectively transferred to a drawer. In particular, it is possible to provide a refrigerator configured such that the driving force of a motor is transferred to a drawer when the drawer is withdrawn, and the driving force of the motor is not transferred to the drawer when the drawer is inserted. Consequently, it is possible to reduce electrical energy consumption.

(104) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer automatically moves from an initial position to a ready position, and the drawer is manually moved from the ready position to the initial position. Consequently, it is possible to reduce electrical energy consumption.

(105) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that the speed at which a drawer moves from an initial position to a ready position is different from the speed at which the drawer moves from the ready position to the initial position. Specifically, it is possible to provide a refrigerator configured such that the speed at which a drawer moves from a ready position to an initial position is higher than the speed at which the drawer moves from the initial position to the ready position. Consequently, it is possible to minimize the incidence of collision between the drawer and the door when the drawer returns while the door is being closed.

(106) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer moves from a ready position to an initial position due to an elastic restoring force. In particular, it is possible to provide a refrigerator configured such that an element impeding the insertion of a drawer is removed, whereby the drawer is returned relatively rapidly by an elastic restoring force. Therefore, it is possible to provide a refrigerator configured such that a drawer completely returns to an initial position while a door is being closed. In addition, it is possible to minimize the incidence of collision between the drawer and the door when the drawer returns while the door is being closed.

(107) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer is automatically withdrawn and automatically inserted. That is, it is possible to provide a refrigerator configured such that an electric driving unit is driven

to withdraw and insert a drawer. In particular, it is possible to provide a refrigerator configured such that the speed at which a drawer is withdrawn and the speed at which the drawer is inserted are controlled to be different from each other, thereby minimizing the incidence of collision between the drawer and a door.

(108) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that the speed at which a door is opened and/or closed is sensed in order to change the speed of an electric driving unit configured to move a drawer, particularly the speed of a motor. Consequently, it is possible to prevent the collision between the drawer and the door as the result of the drawer being withdrawn too fast when the door is opened or to minimize the amount of time the user waits for the withdrawal of the drawer to the ready position as the result of the drawer being withdrawn too slow when the door is opened. In addition, it is possible to prevent the collision between the drawer and the door as the result of the drawer being inserted too slowly when the door is closed or to minimize the application of impact to the drawer as the result of the drawer being inserted too fast when the door is closed.

(109) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that interference between a door and a drawer, configured to automatically move when the door is opened or closed, is considerably reduced using a sensor for very precisely sensing the opening angle of the door and/or the closing angle of the door. That is, it is possible to provide a refrigerator configured such that the opening (or closing) angle of a door at which interference between the door and a drawer is minimized is set, whereby it is possible to very precisely sense whether the door is open (or closed) at the set angle.

(110) According to another embodiment of the present invention, it is possible to provide a refrigerator including a door opening sensor that is capable of flexibly corresponding to a door opening angle that varies depending upon the product models. Consequently, it is possible to reduce manufacturing cost.

(111) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that it is sensed using a single sensor whether a door is open or closed at a consistent angle, which is easily applied to conventional refrigerators. Consequently, it is possible to reduce manufacturing cost and to construct simple control logic.

(112) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that an element for automatically withdrawing a drawer is not exposed in a storage compartment, whereby it is possible to protect an electric driving unit, to improve user convenience, and to provide the interior of the storage compartment with an aesthetically pleasing appearance.

(113) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that it is possible to simultaneously move a plurality of drawers from an initial position to a ready position using a single electric driving unit. To this end, it is possible to provide a refrigerator including a moving frame that is capable of simultaneously transferring the driving force of a single electric driving unit to a plurality of drawers. The moving frame is not provided to support the load of the drawer. That is, the moving frame is provided simply to simultaneously withdraw the drawers. Consequently, it is possible to minimize the load applied to the electric driving unit.

(114) According to another embodiment of the present invention, it is possible to provide a refrigerator including a moving frame that exhibits even load distribution, high durability, and high reliability in assembly. Consequently, it is possible to uniformly withdraw a plurality of drawers without deviation.

(115) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer is automatically inserted and withdrawn with high reliability and durability. In particular, it is possible to provide a refrigerator configured such that it is possible to minimize damage to an electric driving unit attributable to overload of the electric

driving unit or repetitive use of the electric driving unit for a long period of time.

(116) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer configured to be automatically withdrawn, an electric driving unit configured to automatically withdraw the drawer, and relevant elements are easily assembled, and, moreover, are easily repaired as needed. In addition, it is possible to provide a refrigerator configured such that it is possible to minimize the reduction in capacity of a storage compartment due to the above-mentioned elements.

(117) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a plurality of drawers is simultaneously automatically withdrawn, and, in addition, the drawers are easily manufactured and maintained.

(118) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that a drawer is easily connected to or separated from a rail configured to support the drawer.

(119) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that only a basket for receiving goods is easily separated from and coupled to a drawer. That is, it is possible to provide a refrigerator configured such that only a basket is easily separated from and coupled to a drawer in a state in which the connection between a rail and a rail connection part of the drawer is maintained. Consequently, it is possible to improve user convenience.

(120) According to another embodiment of the present invention, it is possible to provide a refrigerator configured such that it is possible to maximally prevent a rail from being visibly exposed to a user. Consequently, it is possible to provide a refrigerator configured such that it is possible to maximally prevent the constraint of the drawer due to foreign matter introduced into the rail and to provide an aesthetically pleasing appearance.

(121) According to a further embodiment of the present invention, it is possible to provide a control method of a refrigerator that is capable of minimizing the load of a motor and flexibly corresponding to various environments in which a drawer is used. In particular, it is possible to provide a control method of a refrigerator that is capable of minimizing the collision between a drawer and a door when the door is closed very fast after being opened. In addition, it is possible to minimize the overload that may be applied to the motor due to the collision between the drawer and the door, thereby improving durability.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front view showing a refrigerator according to an embodiment of the present invention;

(2) FIG. 2 is a view showing a lower storage compartment of the refrigerator shown in FIG. 1;

(3) FIG. 3 is a schematic conceptual view defining the position of a drawer relative to a storage compartment;

(4) FIG. 4 is an exploded view showing a support assembly according to an embodiment of the present invention;

(5) FIG. 5 is a view showing a state in which a rail is mounted to a support cover in the support assembly shown in FIG. 4;

(6) FIG. 6 is a view showing an initial position of a motor assembly and a moving frame in the support assembly shown in FIG. 4;

(7) FIG. 7 is a view showing a ready position of the motor assembly and the moving frame in the support assembly shown in FIG. 4;

(8) FIG. 8 is a view of the moving frame shown in FIG. 4;

- (9) FIG. 9 is an enlarged view of part “A” shown in FIG. 7;
- (10) FIG. 10 is an enlarged view showing a connection between a catching member of the drawer and a transmission member of the moving frame;
- (11) FIG. 11 is an enlarged sectional view showing a connection between the drawer and the support assembly;
- (12) FIG. 12 is an exploded view showing a support assembly according to another embodiment of the present invention;
- (13) FIG. 13 is a view showing a state in which a rail and an elastic device are mounted to a support cover in the support assembly shown in FIG. 12;
- (14) FIG. 14 is an enlarged sectional view showing a connection between the drawer and the support assembly;
- (15) FIG. 15 is a front view showing an example of the elastic device;
- (16) FIG. 16 is a side view showing a connection between the lower part of the drawer and the support assembly at an initial position of the drawer;
- (17) FIG. 17 is a side view showing the connection between the lower part of the drawer and the support assembly at a ready position of the drawer;
- (18) FIG. 18 is a view showing a support assembly or a sidewall and a plurality of drawers according to another embodiment of the present invention;
- (19) FIG. 19 is an exploded view of the support assembly shown in FIG. 18;
- (20) FIG. 20 is a view showing a connection between the moving frame and the drawer shown in FIG. 18;
- (21) FIG. 21 is an enlarged view showing a connection between a hanging member and the drawer shown in FIG. 20;
- (22) FIG. 22 is an enlarged view showing the hanging member and the drawer shown in FIG. 23 in a disconnected state;
- (23) FIG. 23 is a view showing a drawer that is applicable to an embodiment of the present invention;
- (24) FIG. 24 is a view showing a rail that is applicable to an embodiment of the present invention;
- (25) FIG. 25 is a view showing a state in which the drawer shown in FIG. 23 and the rail shown in FIG. 24 are coupled to each other;
- (26) FIG. 26 is an enlarged view of part “B” shown in FIG. 25 after the drawer and the rail are coupled to each other;
- (27) FIG. 27 is a view showing an embodiment of a sensor shown in FIG. 2 and a state in which the sensor is mounted;
- (28) FIG. 28 is a view showing another embodiment of the sensor shown in FIG. 2 and a state in which the sensor is mounted;
- (29) FIG. 29 is a block diagram showing a control construction that is applicable to an embodiment of the present invention; and
- (30) FIGS. 30 to 36 are flowcharts showing steps of a control method that is applicable to an embodiment of the present invention.

DETAILED DESCRIPTION

(31) Hereinafter, embodiments according to the present invention will be described in detail with reference to the accompanying drawings.

(32) FIG. 1 is a front view showing a refrigerator 1 according to an embodiment of the present invention. Specifically, an example of a four-door refrigerator including an upper refrigerating compartment 11 and lower freezing compartments 12 and 13 is shown in FIG. 1. For the convenience of description, left and right doors for the upper refrigerating compartment 11 and a left door for the lower left freezing compartment 12 are omitted. That is, only a right door 20 for the lower right freezing compartment 13 is shown in FIG. 1. Of course, this embodiment may be applied to a side-by-side type refrigerator in addition to the refrigerator with the above-stated

construction. That is, this embodiment may be applied to any refrigerator that includes doors for opening and closing storage compartments and drawers configured to move forward and rearward in the storage compartments.

(33) The refrigerator includes a cabinet **10**, in which the storage compartments **11**, **12**, and **13** are defined, and doors **20** hingedly connected to the cabinet **10** for opening and closing the storage compartments **11**, **12**, and **13**. The door **20** is turned with respect to the cabinet to open the storage compartments. Consequently, the door **20** may be a turnable door. If a plurality of storage compartments is provided, as previously described, a plurality of doors may be provided accordingly.

(34) The refrigerating compartment **11** may be partitioned from the freezing compartments **12** and **13** by a horizontal partition wall **14**. In addition, the left freezing compartment **12** and the right freezing compartment **13** may be partitioned from each other by an additional sidewall or partition wall **16**. For the sake of convenience, the partition wall **16** may be referred to as a vertical partition wall. The refrigerating compartment **11**, the left freezing compartment **12**, and the right freezing compartment **13** may be opened and closed by individual doors.

(35) Drawers **30**, each of which includes a basket **31** for receiving goods, may be provided in the storage compartments **11**, **12**, and **13**, particularly in the freezing compartments **12** and **13**. Each drawer **30** may include a drawer frame **32**. The basket **31** may be coupled to the drawer frame **32**. In some cases, the drawer may define a front opening at its front surface through which its contents can be accessed by the user. Additionally, or alternatively, the drawer may define an upper opening at its upper surface through which its contents can be accessed by the user.

(36) A plurality of drawers **30** may be provided such that the drawers **30** are arranged vertically. FIG. **1** shows an example in which three drawers **30a**, **30b**, and **30c** are provided in each of the right and left freezing compartments such that the drawers **30a**, **30b**, and **30c** are arranged vertically. Of course, this embodiment may be applied to an example in which drawers are provided in the refrigerating compartment in addition to the freezing compartments. Furthermore, one, two, or three doors may be connected to one door so as to operate automatically, or four or more doors may be connected to one door so as to operate automatically.

(37) In this embodiment, it is possible to provide a refrigerator configured such that the drawers **30** can move automatically when the door **20** is opened and/or closed for user convenience. For example, it is possible to provide a refrigerator configured such that the drawers **30**, provided in the right freezing compartment **13**, can move automatically when the right freezing compartment **13** is opened or closed by the door **20**. Automatic movement of the drawers may be applied to any one of the freezing compartments **12** and **13** or to both the freezing compartments **12** and **13**. In addition, the automatic movement of the drawers may be applied to the refrigerating compartment, in addition to the freezing compartments.

(38) FIG. **2** is a view showing the lower freezing compartments **12** and **13** of the refrigerator shown in FIG. **1**. The left freezing compartment door, which is not seen in FIG. **2**, is in a closed state, and the right freezing compartment door **20** is in an open state.

(39) As shown in FIG. **2**, the drawers **30** provided in the left freezing compartment are inserted further inward than the drawers provided in the right freezing compartment. In FIG. **2**, the position of the drawers **30** provided in the left freezing compartment may be referred to as an initial position, and the position of the drawers **30** provided in the right freezing compartment may be referred to as a ready position.

(40) In other words, the drawers may be positioned at the initial position in a state in which the door **20** is closed, and the drawers may be positioned at the ready position in a state in which the door **20** is open. Of course, in the state in which the door **20** is open by a predetermined angle or more, the drawers **30** may be moved from the initial position to the ready position.

(41) The initial position is a position at which interference between the door **20** and the drawers **30** is eliminated and the drawers are inserted into the storage compartment such that the door **20** can

be completely closed, and the ready position is a position at which the drawers **30** are withdrawn such that a user can easily hold the drawers **30**. Consequently, the ready position may be a position spaced apart forward from the initial position. In addition, the ready position may be a position at which the drawers are automatically withdrawn when the door is opened.

(42) Hereinafter, the positions at which a drawer **30** is inserted and withdrawn and the distances by which the drawer **30** is inserted and withdrawn will be described in detail with reference to FIG. 3.

(43) FIG. 3 shows three positions of the drawer **30** in a state in which the door **20** is open. For the convenience of description, the positions of the drawer **30** may be set on the basis of the front of the drawer **30** or a handle **35** of the drawer **30**.

(44) Position P1 may be an initial position. The initial position is a position at which the drawer **30** is inserted such that a door basket **25** does not interfere with the drawer **30** in a state in which the door **20** is closed.

(45) Position P2 may be a position spaced apart forward from position P1. Position P2 may be a position at which the drawer **30** is withdrawn by a predetermined distance such that the user can easily withdraw the drawer **30**. Position P2 may be a ready position. This is because position P2 is a position at which the drawer is ready such that the user can easily withdraw the drawer. In some cases, Position P2 may correspond to a withdrawal distance of between approximately 100 mm to 120 mm from Position P1.

(46) Position P3 may be a position spaced apart forward from position P2. The P3 position may be a position at which the drawer is maximally withdrawn. That is, position P3 may be the maximum withdrawal position, at which the drawer **30** is maximally withdrawn in a state in which the drawer **30** is not separated from the storage compartment. The reason for this is that when the drawer **30** is completely separated from the refrigerator, the drawer is not normally positioned.

(47) Consequently, the movement of the drawer **30** from position P1 to position P3 may be referred to as the withdrawal of the drawer **30**, and the movement of the drawer **30** from position P3 to position P1 may be referred to as the insertion of the drawer **30**.

(48) As shown in FIG. 3, receiving boxes or baskets **25**, which define additional door storage regions **21**, may be provided at the rear of the door **20**. In order to eliminate interference between the door **20**, particularly the basket **25** provided in the door **20**, and the drawer **30**, the drawer **30** is positioned at the initial position (position P1) in a state in which the door **20** is closed. When the user open the door **20** in order to withdraw goods, the drawer **30** moves forward from the initial position to the ready position (position P2) such that the user can more easily withdraw the drawer **30**. As a result, the front of the drawer **30** or the handle **35** becomes closer to the user such that the user can more easily withdraw the drawer **30**.

(49) That is, the initial position may be a position at which the drawer has been maximally moved inward into the storage compartment, and the ready position may be a position spaced apart forward from the initial position by a predetermined distance. At the ready position, it is not necessary for the user to reach deep into the storage compartment in order to grasp the handle **35**, whereby it is very convenient for the user to manipulate the drawer **30**.

(50) In order to introduce goods into the drawer **30** or to remove goods from the drawer **30**, the user may withdraw the drawer **30** up to the maximum withdrawal position (position P3).

(51) As shown in FIG. 3, the drawer **30** may not escape from an opening **17** defined in the storage compartment even at the ready position. The opening may be a food introduction port. That is, when the door **20** is opened, the drawer **30**, particularly the front of the drawer **30** or the handle **35**, may be positioned further rearward than the opening **17**. That is, the drawer **30** may remain positioned in the storage compartment. This is because the reason that the user opens the door **20** is not necessarily to use or withdraw the drawer **30**. For example, the user may open the door **20** in order to use the door storage region **21**. In addition, in a case in which a plurality of drawers **30** is provided, only a specific one of the drawers may be withdrawn. If the remaining drawers, which the user does not wish to withdraw, are withdrawn from the storage compartment, cool air may be

lost.

(52) As will be described hereinafter, embodiments of the present invention may provide a refrigerator configured such that the drawer can move automatically based on the extent to which the user opens the door, particularly at a specific door opening angle. Consequently, it is possible to prevent unnecessary movement of the drawer, thereby reducing a loss of cool air and a loss of energy. In addition, it is possible to provide a refrigerator configured such that interference between the drawer and the door is minimized.

(53) In addition, when the door is opened, cool air is inevitably discharged from the storage compartment. When the drawer is withdrawn, cool air is discharged from the basket of the drawer. That is, as the withdrawal distance of the drawer increases, the loss of cool air from the basket is accelerated. Particularly, when the drawer is withdrawn further forward than the food introduction port, the loss of cool air may be accelerated still further. In order to minimize the loss of cool air in the basket at the ready position, therefore, the drawer **30** may be prevented from escaping from the interior of the storage compartment at the ready position.

(54) For example, the ready position may be a position spaced apart forward from the initial position by about 120 mm. Of course, the distance between the ready position and the initial position may be set differently based on the shape of the refrigerator, the position of the drawer, the distance by which the door storage region **21** is inserted into the storage compartment, the capacity of the refrigerator, etc. However, the ready position may be a position at which the front of the drawer **30** or the handle **35** does not escape from the opening **17** of the storage compartment. That is, the front of the drawer **30** or the handle **35** may be positioned further inward than the opening **17** without escaping from the opening **17**.

(55) In this embodiment, the refrigerator may be configured such that when the door **20**, which opens and closes the storage compartment, is opened, the drawer, which is provided in the storage compartment, can move automatically (can be withdrawn automatically) from the initial position to the ready position. That is, the refrigerator may be configured such that when the door is opened, the front of the drawer may move automatically toward the food introduction port **17**.

Consequently, the refrigerator according to this embodiment may further include an electric driving unit for moving the drawer. In addition, the refrigerator according to this embodiment may further include a sensor for sensing a condition under which the electric driving unit is to be operated.

(56) That is, in this embodiment, the refrigerator may be configured such that the drawer is withdrawn from position P1 to position P2 using electrical energy. In addition, in this embodiment, the refrigerator may be configured such that the drawer is automatically withdrawn regardless of force necessary for the user to open the door. In addition, in this embodiment, the refrigerator may be configured such that the drawer is automatically withdrawn using a driving force generated from a motor.

(57) Hereinafter, a support assembly for automatically withdrawing the drawer while movably supporting the drawer will be described in detail with reference to FIG. **4**.

(58) FIG. **4** shows a support assembly **100** that can be coupled to a freezing compartment sidewall of the refrigerator shown in FIG. **1**. Specifically, support assemblies **100** are positioned at the left and right sides of the partition wall **16**, which is one of the freezing compartment sidewalls. In this case, freezing compartments are provided at the left and right sides of the freezing compartment partition wall **16**. In a case in which one freezing compartment is provided, the partition wall **16** may be a left heat insulation wall (a left sidewall) or a right heat insulation wall (a right sidewall) of the freezing compartment. In a case in which freezing compartments are provided at the left and right sides of the partition wall **16**, the support assembly **100** may be provided in only one of the freezing compartments, not both the freezing compartments. In any case, the support assembly **100** may be mounted to the freezing compartment sidewall.

(59) In the case in which a single storage compartment is provided, opposite sidewalls of the storage compartment may be heat insulation walls. A heat insulation wall may be a wall, the inner

space of which is filled with an insulating material. In the case in which left and right storage compartments are provided, the left and right storage compartments may be partitioned by a partition wall. In this case, the left sidewall of the left storage compartment may be a heat insulation wall, and the right sidewall of the left storage compartment may be formed by the partition wall. In the case in which the right sidewall of the right storage compartment is a heat insulation wall, the left sidewall of the right storage compartment may be formed by the partition wall. The partition wall may be a non-heat insulation wall.

(60) In a case in which the left and right freezing compartments are separated from each other, as shown in FIG. 1, the support assembly **100** may be mounted to the left or right sidewall. The left sidewall or the right sidewall may be a heat insulation wall. However, it may be disadvantageous to mount the support assembly **100** to the heat insulation wall from the aspect of heat insulation. In addition, in a case in which the conventional thickness of the heat insulation wall is maintained, the inner space of the storage compartment in which the support assembly **100** is mounted may be reduced. For this reason, the support assembly **100** may be mounted to the freezing compartment partition wall **16**, at which heat insulation is not critical, selected from among the freezing compartment sidewalls.

(61) Of course, the support assembly **100** may not be mounted to the sidewall of the freezing compartment but may be mounted to the sidewall of the refrigerating compartment. In this case, the drawer provided in the refrigerating compartment may be automatically withdrawn. In addition, the refrigerating compartment may be partitioned into left and right parts, in the same manner as in the freezing compartment. Even in this case, a partition wall may be provided to partition the refrigerating compartment into left and right parts. A support assembly may be mounted to one of the sidewalls that define the refrigerating compartment. However, the support assembly may be mounted to the partition wall in order to prevent the reduction of heat insulation efficiency and to minimize the reduction in space of the storage compartment.

(62) The partition wall **16** may be a partition wall for partitioning the left and right freezing compartments from each other. The partition wall **16** may be symmetrical. That is, support assemblies having the same shape may be mounted to the left and right sides of the partition wall **16** in the same fashion. Consequently, drawers may be provided in the left and right freezing compartments such that the drawers can be automatically inserted and withdrawn.

(63) Hereinafter, an embodiment in which support assemblies **100** are mounted to the left and right sides of the sidewall defining the storage compartment, particularly the partition wall **16** for partitioning the left and right freezing compartments from each other, will be described in detail.

(64) The support assembly **100** supports the drawer **30** such that the drawer **30** can move forward and rearward. In addition, the drawer **30** may be automatically withdrawn through the support assembly **100**.

(65) The support assembly **100** may include a support cover **110**. The support cover **110** may be mounted to one sidewall of the freezing compartment, particularly the partition wall **16**.

Specifically, the support cover **110** may be mounted to the left sidewall or the right sidewall of the freezing compartment, rather than the upper sidewall, the lower sidewall, and the rear sidewall of the freezing compartment. As will be described hereinafter, various components, including the electric driving unit, may be mounted to the support cover **110**. The support assembly **100** may be mounted to one side wall of the freezing compartment, particularly the partition wall **16**, as a single assembly, or may be separated from the partition wall **16** through the support cover **110**. That is, the support assembly **100** may be integrally coupled to the sidewall of the storage compartment or may be separated from the sidewall of the storage compartment. Consequently, it is possible to very simply manufacture the support assembly **100** and to easily maintain the support assembly **100**. This is because, as will be described hereinafter, the drawer **30** can be separated from a rail **120**, and then the support assembly **100**, including the support cover **110**, can be separated from the partition wall **16**. On the other hand, the support assembly may be manufactured, the support

assembly may be mounted to the partition wall **16**, and the drawer **30** may be coupled to the rail **120**.

(66) The support cover **110** may include an outside surface **111** and an inside surface **112**. The inside surface **112** of the support cover **110** may be coupled to the sidewall so as to face the sidewall. The outside surface **111** may be exposed in the storage compartment. Consequently, the outside surface **111** may define the inner surface of the storage compartment. The sidewall may be a partition wall.

(67) The rail **120** may be mounted to the support cover **110**. Specifically, the rail **120** may be mounted to the outside surface **111** of the support cover **110**. The rail **120** may be provided at each of the left and right sides of the storage compartment. Consequently, one of the rails may be mounted to the outside surface of the support cover **110**, and the other rail may be mounted to the sidewall of the storage compartment. The rail **120** may be provided such that the drawer can move forward and rearward in the storage compartment. That is, the drawer **30** may be supported such that the drawer **30** can move forward and rearward in the storage compartment along the rail **120**. The drawer **30** may slide forward and rearward along the rail **120**. Consequently, the rail **120** may support the load of the drawer **30**, and the load of the drawer **30** may be transferred to one sidewall of the freezing compartment or the partition wall **16** via the rail **120**. Embodiments of the rail **120** and the structure in which the rail **120** and the drawer **30** are coupled to each other will be described later.

(68) In a case in which a plurality of drawers **30** is provided, a plurality of rails **120** may be provided. Consequently, a plurality of rails **120** may be mounted to a single support cover **110**. Specifically, a plurality of drawers **30** may be provided such that the drawers **30** are arranged vertically, and therefore a plurality of rails **120** may be provided such that the rails **120** are arranged vertically.

(69) Meanwhile, as the result of the inside surface **112** of the support cover **110** being mounted to the sidewall of the storage compartment or the partition wall **16** so as to face the sidewall of the storage compartment or the partition wall **16**, a predetermined space **130** is defined between the support cover **110** and the sidewall of the storage compartment or the partition wall **16**. The predetermined space **130** may be an unexposed space in the storage compartment. Consequently, elements provided in the predetermined space **130** may not be exposed in the interior of the storage compartment. For this reason, the predetermined space **130** may be referred to as an isolation space. In addition, the predetermined space **130** may be an electric driving unit mounting space, in which the electric driving unit will be mounted, as will be described hereinafter.

(70) The predetermined space, isolation space, or electric driving unit mounting space **130** is provided between an inside surface **112** of the support cover **110** and the partition wall **16**. As a result, elements mounted to the inside surface **112** of the support cover **110** are not exposed in the storage compartment. Consequently, the side of the support cover **110** facing the partition wall **16** may be the inside of the support cover **110**. On the other hand, elements mounted to an outside surface **111** of the support cover **110** may be exposed in the storage compartment. For example, the rail **120** may be mounted to the outside surface **111** of the support cover **110**, and therefore the rail **120** may be exposed in the storage compartment. Consequently, the side of the support cover **110** that faces the storage compartment may be the outside of the support cover **110**.

(71) An electric driving unit **150** may be provided on the inside surface **112** of the support cover **110**. That is, the electric driving unit **150** may be provided in the predetermined space **130**. Consequently, the electric driving unit **150** may not be exposed in the storage compartment. This is because the support cover **110** covers the electric driving unit **150**. That is, the electric driving unit **150** is provided inside the support cover **110**.

(72) Specifically, the electric driving unit **150** may not be mounted to the partition wall **16** but may be mounted to the inside surface of the support cover **110**. When the support cover **110** is separated from the partition wall **16**, therefore, the electric driving unit **150** may be separated from the

partition wall **16**.

(73) The electric driving unit **150** is operated to move the drawer **30** from the initial position to the ready position. That is, the electric driving unit **150** may be operated to move the drawer toward the food introduction port **17**. To this end, the electric driving unit **150** may include a motor assembly **160** for generating force necessary to move the drawer **30** and a moving frame **170** for applying force to the drawer **30** so as to move the drawer **30**. Specifically, the moving frame **170** may be configured to selectively push the drawers.

(74) The moving frame **170** may be configured such that the moving frame **170** is moved forward and rearward by the operation of the motor assembly **160**. Here, a direction in which the moving frame **170** is moved forward and rearward may be the same as the direction in which the drawer **30** is moved forward and rearward. That is, the moving frame **170** may be configured such that the moving frame **170** is moved by the motor assembly **160** in a direction identical to the direction in which drawer **30** is moved. In some cases, the speed at which the moving frame **170** is moved forward may be lower than the speed at which the moving frame **170** is moved rearward.

(75) Specifically, the moving frame **170** may be movably mounted to the support cover **110**. For example, the moving frame **170** may be mounted to the inside surface of the support cover **110** such that the moving frame **170** can be moved forward and rearward. The moving frame **170** may be supported so as to be slidable with respect to the support cover **110**. Since the moving frame **170** is substantially positioned in the predetermined space **130**, the structure of the moving frame **170** and the movement of the moving frame **170** cannot be seen in the storage compartment. However, the movement of the moving frame **170** must be transferred to the drawer **30**, which is provided in the storage compartment. For this reason, an element for transferring force, for example, a transfer member, which will be described hereinafter, may be exposed to the outside surface of the support cover **110**. That is, the transfer member may extend from the inside of the support cover **110** to the outside of the support cover **110**.

(76) Consequently, the transfer member may extend from the inside surface **112** to the outside surface **11** of the support cover **110**.

(77) In other words, the moving frame **170** may be an element for transferring force generated by the motor assembly **160**, which is positioned in the predetermined space **130**, to the drawer **30**, which is positioned outside the predetermined space **130**. The details of the moving frame **170** will be described later.

(78) As shown in FIG. 4, a through part **16a** may be formed through the partition wall **16**. The through part **16a** may be formed such that the motor assembly **160** is disposed through the through part **16a**. The motor assembly **160** may have a predetermined horizontal width, as shown in FIG. 4. In a case in which the motor assembly **160** is positioned in the predetermined space **130** with the result that the motor assembly **160** is isolated from the storage compartment, therefore, the horizontal width of the partition wall **16** or the horizontal width of the support cover **110** may be excessively increased, which may reduce the inner space of the storage compartment. Consequently, it is possible to prevent the inner space of the storage compartment from being reduced due to the motor assembly **160** by the provision of the through part **16a**.

(79) Specifically, in a case in which the support assemblies **110** are positioned on the left and right sides of the partition wall **16**, a portion of the left motor assembly (the motor assembly for moving the drawer in the left freezing compartment) may be positioned in the right space **130** through the through part **16a**. Similarly, a portion of the right motor assembly (the motor assembly for moving the drawer in the right freezing compartment) may be positioned in the left space **130** through the through part **16a**. The two motor assemblies **160** may be vertically arranged side by side. That is, the two motor assemblies **160** may be vertically arranged side by side such that the horizontal widths of the motor assemblies **160** partially overlap each other. As a result, owing to the support assemblies, it is possible to minimize the effect in thickness of the motor assemblies **160** as compared with a case in which the two motor assemblies **160** are horizontally arranged side by side

at the same height.

(80) The through part **16a** may be formed so as to correspond to the external shape of the motor assembly **160**. Consequently, the motor assembly **160** may be fixed and supported in the through part **16a**. In a case in which the motor assemblies are provided at the left and right sides of the partition wall **16**, the through part **16a** may be formed so as to have a vertically extending length. One of the motor assemblies is disposed through the upper side of the through part **16a**, and the other motor assembly is disposed through the lower side of the through part **16a**. That is, the two motor assemblies **160** may be vertically arranged through the through part **16a**. When the support cover **110** is coupled to the partition wall **16**, the through part **16a** may be covered by the support cover **110**.

(81) The through part **16a** may be formed in a case in which the motor assemblies are provided on the left and right sides of the partition wall **16**. In a case in which the motor assembly is provided at only one sidewall of the storage compartment, however, a recessed part may be formed in place of the through part. Consequently, a portion of the horizontal width of the motor assembly may be inserted into the recessed part, whereby it is possible to minimize the reduction of the inner space of the storage compartment due to the motor assembly.

(82) Meanwhile, the motor assembly **160** includes a motor **162** configured to be operated by electrical energy. To this end, an electric cable for supplying electrical energy must be connected to the motor assembly **160**. The electric cable may be connected from a power supply device of the refrigerator to the motor assembly **160**.

(83) An upper opening **16b** for electric cable connection may be formed in the partition wall **16**. The electric cable connected to the power supply device may extend to the upper opening **16b** of the vertical partition wall **16** through the horizontal partition wall **14** shown in FIG. 1. An electric cable through part **16c** may be formed through the partition wall **16**. Consequently, the electric cable may further extend from the upper opening **16b** to the electric cable through part **16c**. An electric cable **16d** may extend from the electric cable through part **16c** to the right side (one side) and may then be terminated using an electric cable coupling part **16e**. The electric cable coupling part **16e** may be an electric cable coupling part configured to be connected to the right motor assembly **160**. In the same manner, an electric cable and an electric cable coupling part may also be provided at the left side (the other side) of the electric cable through part **16c**.

(84) The electric cable may extend through the horizontal partition wall **14** and the vertical partition wall **16** before the support assembly **100** is mounted to the partition wall **16**, and may then extend through the electric cable through part **16c**. The electric cable coupling part **16e** may be formed at the end of the electric cable.

(85) The electric cable coupling part **16e** is positioned in the predetermined space **130**, which was previously described. Before the support assembly **100** is mounted to the partition wall **16**, therefore, the motor assembly **160** is connected to the partition wall **16** through the electric cable coupling part **16e**. Subsequently, the support assembly **100** may be fixed to the partition wall **16** through a fastening part **118** formed at the support cover **110** and a fastening part **16f** formed at the partition wall **16**. The fastening parts **118** and **16f** may be formed in the shape of a boss for screw coupling. On the other hand, screws may be removed in order to separate the support assembly **100** from the partition wall **16**. Subsequently, the motor assembly **160** is separated from the electric cable coupling part **16e**, whereby the support assembly **100** is completely separated from the partition wall **16**. That is, the structural and electrical connection between the support assembly **100** and the partition wall **16** may be released.

(86) Consequently, it is possible to very easily perform coupling, separation, and connection between the support assembly **100** and the partition wall **16** through the structure of the partition wall **16** or the sidewall of the storage compartment, the structure of the support assembly **100**, and the electric cable connection structure via the sidewall or the partition wall.

(87) As previously described, the motor assembly **160** includes the motor **162**. In general, motors

are formed to have a cylindrical shape. A direction in which a rotary shaft of the motor extends may be perpendicular to the sidewall of the storage compartment or the partition wall **16**. As a result, the horizontal width of the motor assembly **160** may be increased due to the size of the motor (the height of the cylindrical motor).

(88) As shown in FIG. **4**, the support cover **110** may be provided with a motor avoidance recess **119**. For example, a circular motor avoidance recess **119** may be formed in the support cover **110** such that the circular motor avoidance recess **119** corresponds in shape to the motor. The motor avoidance recess **119** may receive at least a portion of the motor. Consequently, it is possible to enlarge the part of the motor assembly **160** corresponding to the motor without increasing the horizontal width of the motor assembly **160**. In order to eliminate the interference between the motor avoidance recess **119** and the enlarged part of the motor assembly **160**, the motor avoidance recess **119** may be formed in the support cover **110**.

(89) For the right support cover, the motor avoidance recess **119** protrudes rightward from the right support cover. The protruding motor avoidance recess **119** may interfere with other elements mounted to the support cover. In order to solve this problem, the motor avoidance recess **119** may be formed between the rails **120**.

(90) On the assumption that three rails **120** are mounted to the support cover **110**, the motor avoidance recess **119** may be formed in the right support cover between the middle rail and the lower rail. On the other hand, the motor avoidance recess **119** may be formed in the left support cover between the upper rail and the middle rail.

(91) The motor assembly **160** may be more securely coupled to the support cover **110** by the provision of the motor avoidance recess **119**. In addition, the motor assembly **160** may be formed between the rails in order to minimize the reduction of the inner space of the storage compartment due to the extension of the predetermined space **130**.

(92) In a case in which the horizontal width of the motor assembly **160** is further increased in consideration of the size of the motor, for example in a case in which the horizontal width of a housing **161** of the motor assembly **160** is further increased, it is necessary to reduce the inner space of the storage compartment in order to avoid the interference between the motor assembly **160** and the rail.

(93) Hereinafter, a structure that supports the drawer and applies force to the drawer will be described in detail with reference to FIG. **5**. FIG. **5** is an enlarged view showing the upper part of the support assembly shown in FIG. **4**. As shown in FIG. **5**, a rail **120** corresponding to one drawer **30** and a structure for transferring force to the drawer **30** are provided at the support cover **110**.

(94) Referring to FIG. **5**, the drawer **30**, while not shown in FIG. **5**, is supported such that the drawer **30** can move forward and rearward along the rail **120**. In general, the user may pull or push the drawer in order to withdraw or insert the drawer **30**. The rail **120** is generally provided in order for the user to easily withdraw or insert the drawer **30** using the minimum force. To this end, the rail **120** may be mounted to the outside surface **111** of the support cover **110**, i.e. the surface of the support cover **110** facing the interior of the storage compartment.

(95) As previously described, the electric driving unit **150**, particularly the motor assembly **160**, may be mounted to the inside surface **112** of the support cover **110**. Here, the inside surface **112** of the support cover **110** may be the surface of the support cover **110** facing the sidewall or the partition wall **16**. Consequently, it is necessary to provide a structure for transferring force or displacement generated at the inside surface **112** of the support cover to the outside surface **111** of the support cover.

(96) To this end, a through part **113** may be formed in the support cover **110**. That is, the through part **113** may be formed through the support cover **110**. It is possible to transfer the movement of the moving frame **170** to the drawer **30** through the through part **113**.

(97) Specifically, the moving frame **170** may include a transfer member **171** for transferring force to the drawer **30**. The transfer member **171** may be a portion of the moving frame **170**.

Alternatively, the transfer member **171** may be connected to the moving frame **170**. In addition, the transfer member **171** may be selectively connected to the moving frame **170**. In any case, the movement of the moving frame **170** may be transferred to the drawer **30** via the transfer member **171**.

(98) The transfer member **171** may extend through the through part **113**. That is, the moving frame **170** may be moved on the inside surface **112** of the support cover, whereas the transfer member **171** may be moved on the outside surface **111** of the support cover through the through part **113**. Consequently, the transfer member **171** is moved forward and rearward through the through part **113**.

(99) Since the transfer member **171** is moved forward and rearward, the through part **113** may be formed so as to define a movement path of the transfer member **171**. For this reason, the through part **113** may be referred to as a slit that is formed so as to extend forward and rearward. The transfer member **171** transfers the force generated by the electric driving unit, particularly the movement of the moving frame **170**, to the drawer **30**. That is, the transfer member **171** may push the drawer **30** such that the drawer **30** can move along the rail. In other words, the drawer **30** may move automatically even when the user does not manipulate the drawer.

(100) As shown in FIG. 5, the load of the drawer may be transferred to the support cover **110** via the rail **120**. The load transferred to the support cover **110** may be transferred to the sidewall or the partition wall to which the support cover **110** is mounted. Consequently, the load of the drawer may not substantially affect the electric driving unit **150**. In other words, load applied to the electric driving unit **150** in order to withdraw the drawer may have little to no relationship to the load of the drawer.

(101) In addition, the load of the drawer may not be transferred to the electric driving unit **150**, particularly the moving frame **170**. Specifically, since the direction in which the moving frame **170** is moved is substantially perpendicular to the direction in which the load of the drawer is applied, it is possible to minimize the effect in movement of the moving frame **170** attributable to the increase in load of the drawer. In other words, the moving frame **170** is decoupled from the drawer in the vertical direction such that the weight of the drawer may not be vertically transferred to the moving frame **170**. Even when the load of the drawer is increased, therefore, the moving frame **170** may be smoothly moved forward and rearward.

(102) Hereinafter, a mechanism between the electric driving unit **150** and the drawer **30** will be described in detail with reference to FIGS. 6 and 7. FIG. 6 is a view showing the support assembly **100** at the initial position of the drawer when viewed from inside the support cover **110**, and FIG. 7 is a view showing the support assembly **100** at the ready position of the drawer when viewed from inside the support cover **110**. Of course, the support cover **110** may be fixed to the sidewall of the storage compartment or the partition wall regardless of the movement of the drawer.

(103) The motor assembly **160** may be provided in the predetermined space or electric driving unit mounting space **130** between the inside surface **112** of the support cover **110** and the partition wall **16**. Specifically, the motor assembly **160** may be mounted to the inside surface **112** of the support cover **110**. Consequently, the motor assembly **160** may be fixed to the support assembly **100** regardless of the movement of the drawer.

(104) The motor assembly **160** may include a housing **161**, in which power generating and transfer elements, such as a motor **162** and a gear **162a**, may be received. The housing **161** is fixed to the support cover **110** such that the motor assembly **160** is stably supported by the support cover **110**. As previously described, the part of the housing **161** corresponding to the motor **162** may protrude further outward than the remaining parts of the housing **161** due to the shape of the motor **162**, and may be located in the motor avoidance recess **119**.

(105) A plurality of gears **162a** may be provided in order to reduce the rotational speed and to transfer torque.

(106) The electric driving unit **150** may include a connection member **163**. The motor assembly

160 may include the connection member **163**. The connection member **163** may be provided between the motor assembly **160**, particularly the housing **161** of the motor assembly **160**, and the moving frame **170**. That is, the connection member **163** may be provided in order to interconnect the motor assembly **160** and the moving frame **170**.

(107) The connection member **163** may be configured such that the distance by which the connection member **163** is withdrawn from the motor assembly **160**, particularly the housing **161**, is changeable. That is, the distance by which the connection member **163** is withdrawn may be changed. When the distance by which the connection member **163** is withdrawn from the housing **161**, which is fixed, is increased, the distance between the housing **161** and the moving frame **170** increases. On the other hand, when the distance by which the connection member **163** is withdrawn from the housing **161**, which is fixed, is decreased, the distance between the housing **161** and the moving frame **170** decreases. Consequently, the motor assembly **160** may drive the connection member **163** such that the distance by which the connection member **163** is withdrawn is changed, and may move the moving frame **170** as the result thereof.

(108) One side of the connection member **163** may be positioned so as to move relative to the motor assembly **160**, and the other side of the connection member **163** may be positioned so as to move together with the moving frame **170**. That is, the other side of the connection member **163** may be coupled to the moving frame **170**. A connection member coupling part **174** may be formed at the moving frame **170**. The connection member **163** is coupled to the moving frame **170** via the connection member coupling part **174**. Consequently, the movement of the connection member **163** may result in the movement of the moving frame **170**.

(109) Specifically, the connection member **163** may be formed in the shape of a rack, and the motor assembly **160** may be formed in the shape of a pinion. That is, one of the gears **162a** may be a pinion gear, which may be connected to the connection member **163**. For example, the clockwise rotation of the motor **162** may be converted into the forward movement of the connection member **163** through the gears **162a**, and the counterclockwise rotation of the motor **162** may be converted into the rearward movement of the connection member **163** through the gears **162a**. Of course, the directions in which the motor is rotated and the directions in which the connection member is moved may be reversed based on the configuration of the gears.

(110) Consequently, the distance by which the connection member is withdrawn may be increased or decreased according to the forward and reverse driving of the motor assembly **160**. The driving of the motor assembly **160** may push or pull the connection member **163** and thus may push or pull the moving frame.

(111) The moving frame **170** may be configured to transfer driving force generated by the motor assembly **160** to the drawer **30**. Basically, therefore, the moving frame **170** is moved by the driving of the motor assembly **160**. Specifically, the moving frame **170** may be movably provided on the inside surface **112** of the support cover **110**.

(112) As previously described, a plurality of drawers may be provided in the storage compartment. When the door is opened, all of the drawers may move from the initial position to the ready position. The movement of the drawers may be performed simultaneously. Consequently, the moving frame **170** may be configured to transfer a driving force to all of the drawers.

(113) In order to transfer a driving force to the drawers, which are arranged vertically, the moving frame **170** may extend in a vertical direction. For example, the moving frame **170** may extend upward and downward. That is, the moving frame **170** may extend vertically so as to correspond to the height of the drawers, which are arranged vertically. In addition, the moving frame **170** may be provided with a plurality of transfer members **171**. In the same manner, the transfer members may be disposed at a single moving frame **170** such that the transfer members are arranged vertically. One transfer member **171** may be provided so as to correspond to one drawer **30**. As a result, all of the drawers, which are arranged vertically, may be moved by a single moving frame **170**. That is, the moving frame may move forward to push the drawers.

(114) In FIGS. 6 and 7, an example in which three transfer members **171** are formed at a single moving frame **170** is shown. This means that a single moving frame **170** is moved in order to move three transfer members **171** which are arranged vertically. That is, three transfer members **171** may be simultaneously moved from the initial position to the ready position by moving a single moving frame **170**. As a result, it is possible to simultaneously move a plurality of drawers through a single motor assembly **160**, a single connection member **163**, and a single moving frame **170**. That is, it is possible to easily move a plurality of drawers even though only one electric driving unit **150** that is operably connected to a single door **20** is provided. Consequently, it is possible to realize simple and easy control logic. In addition, the motor assembly **160**, the connection member **163**, and the moving frame **170** may not be provided for each drawer. As a result, it is possible to minimize the reduction in capacity of the storage compartment. Of course, it is possible to minimize the increase in manufacturing cost and to realize very easy installation and maintenance.

(115) The moving frame **170** may be supported on the inside surface **112** of the support cover **110** such that the moving frame **170** can move forward and rearward. More specifically, the moving frame **170** may be supported such that the moving frame **170** can slide forward and rearward.

(116) As previously described, the moving frame **170** may be configured to move a plurality of drawers **30**. To this end, the moving frame **170** may be formed in the shape of a plate that extends vertically. That is, the moving frame **170** may be formed in the shape of a plate that extends in a vertical direction. In addition, the deviation in movement between the upper and lower parts of the moving frame **170** may be minimized when the moving frame **170** is moved.

(117) FIG. 8 is a perspective view of the moving frame **170**. The moving frame **170** is configured to have a structure for simultaneously withdrawing three drawers.

(118) Transfer members **171** may be provided at the upper end, the lower end, and the middle of the moving frame **170**. To this end, the moving frame **170** may extend vertically so as to correspond to the height of the drawers **30**.

(119) The moving frame **170** may be formed in the shape of a plate that extends in a vertical direction. The moving frame **170** may have a relatively small thickness. In order to increase the rigidity of the moving frame **170**, therefore, a plurality of ribs **170a** may be formed at the moving frame **170**. The ribs may include horizontal ribs and vertical ribs. In addition, the ribs may be formed in the shape of a lattice.

(120) Specifically, the moving frame **170** may be formed in the shape of a plate that has a predetermined width in a direction in which the moving frame **170** is moved, i.e. in a forward and rearward direction. Of course, the moving frame **170** may be formed in the shape of a rectangle that has a height greater than a forward and rearward width. As previously described, the vertical height of the moving frame **170** may be formed so as to correspond to the height at which the drawers are arranged. In addition, the moving frame **170** may be formed in the shape of a thin plate having a relatively small thickness. Consequently, it is possible to minimize the reduction of the inner space of the storage compartment due to the thickness of the moving frame **170** and to move the drawers **30** while exhibiting sufficient rigidity. This is because force is applied to the moving frame **170** in the forward and rearward direction, not in the thickness direction, when the moving frame **170** pushes the drawer **30**.

(121) In addition, the moving frame **170** may be provided with a sliding support part **172**. A pair of sliding support parts **172** may be formed at the upper end of the moving frame **170**, and a pair of sliding support parts **172** may be formed at the lower end of the moving frame **170**. In addition, a pair of sliding support parts **172** may be formed at the middle of the moving frame **170**.

Consequently, the moving frame **170** may move in a state in which the moving frame **170** is supported by at least four upper, lower, left, and right support points. The moving frame **170** may have two upper support points, two lower support points, and two middle support points. As a result, it is possible to prevent the moving frame from being twisted when the moving frame **170** is moved forward and rearward.

(122) The forward and rearward width of the upper end and the lower end of the moving frame **170** may be increased in order to form the support parts **172** at the upper end and the lower end of the moving frame **170**. The transfer members **171** may be formed at the extension parts of the moving frame **170**.

(123) On the other hand, the middle of the moving frame **170**, at which the transfer member is formed, may not extend horizontally. As a result, the transfer member **171** formed at the middle of the moving frame **170** may become separated from the moving frame **170** when the transfer member **171** is used for a long period of time. That is, a connection between the transfer member **171** and the moving frame **170** may be broken or damaged. This is because the transfer member **171** may protrude from the moving frame **170** and may be bent and thus broken or damaged when the transfer member **171** is used for a long period of time.

(124) In order to solve this problem, a reinforcement rib or reinforcement protrusion **171a** may be formed between the moving frame **170** and the transfer member provided at the middle of the moving frame **170**. The reinforcement rib may be formed parallel to a direction in which force is applied to the reinforcement rib. A plurality of reinforcement ribs may be formed, or the reinforcement protrusion **171a** may extend from the transfer member.

(125) Meanwhile, when the moving frame **170** is used for a long period of time, the middle part of the moving frame **170** may extend toward the drawer or in the opposite direction. That is, the middle part of the moving frame **170** may become convex. In this case, the transfer member may be constrained in the slit **113**, or may escape from the slit **113**. Particularly, in a case in which the transfer member escapes from the slit **113**, it is not possible for the transfer member to transfer force to the drawer **30**.

(126) For this reason, the middle part of the moving frame **170** as well as the upper and lower ends of the moving frame **170** may be slidably supported.

(127) In order to more smoothly move the moving frame **170**, guide bars **114** may be formed at the support cover **110**. The guide bars **114** may be formed so as to correspond to the upper and lower ends of the moving frame **170**. To this end, the guide bars may include an upper guide bar and a lower guide bar. More specifically, the guide bars **114** may be formed so as to correspond to the upper end, the middle, and the lower end of the moving frame **170**. In the same manner, the sliding support parts **172** may be formed at the upper end, the middle, and the lower end of the moving frame **170**.

(128) Each of the sliding support parts **172** may be formed so as to surround a corresponding one of the guide bars **114**. Consequently, the sliding support parts **172** may slide forward and rearward in a state in which the sliding support parts **172** surround the respective guide bars **114**.

(129) FIG. **9** is a partially enlarged view showing the sliding support part **172** and the guide bar **114**. Specifically, FIG. **9** is an enlarged view of part "A" shown in FIG. **7**. FIG. **9** shows that the sliding support part **172** surrounds the guide bar **114**.

(130) As shown in FIG. **9**, a liner **173** may be interposed between the guide bar **114** and the sliding support part **172**. The liner may be made of a polyoxymethylene (POM) material. That is, the liner may be made of engineered plastic such as polyacetal or polyoxymethylene. The POM material exhibits high mechanical strength, high wear resistance, low frictional resistance, and high lubricity. For this reason, it is possible for the guide bar **114** to support the moving frame **170** such that the moving frame **170** can move smoothly even when the guide bar **114** is used for a long period of time. Of course, the guide bar **114** may be coated with a lubricant such as grease.

(131) As shown in FIG. **9**, the sliding support part **172** slides forward and rearward along the guide bar **114**. At this time, the sliding support part **172** may not move smoothly due to the loading and twisting of the moving frame **170**.

(132) Friction may be concentrated on the upper inside surface and the lower inside surface of the liner **173** due to the loading of the moving frame **170**. In addition, friction may be concentrated on the left inside surface and the right inside surface of the liner **173** due to the twisting of the moving

frame **170**, which may be caused by the force applied to the transfer member **171** protruding from the moving frame **170**.

(133) For this reason, friction avoidance recesses **173a**, **173b**, **173c**, and **173d** may be formed in the upper inside surface, the lower inside surface, the left inside surface, and the right inside surface of the liner **173**, respectively. It is possible to minimize the frictional force between the liner and the guide bar by the provision of the friction avoidance recesses **173a**, **173b**, **173c**, and **173d**, whereby the moving frame may move smoothly while being securely supported.

(134) In particular, the friction avoidance recesses **173a**, **173b**, **173c**, and **173d** may be filled with grease, by which the frictional force may be further minimized. In addition, since sufficient grease is supplied to a friction part, the moving frame may move smoothly even when the moving frame is used for a long period of time.

(135) The moving frame **170** may simultaneously withdraw a plurality of drawers. In other words, the moving frame **170** may simultaneously push a plurality of drawers without temporal or positional deviation. If temporal or positional deviation occurs, the moving frame **170** may be twisted. As a result, the moving frame **170** may not move smoothly, and excessive stress may be concentrated on a specific part of the moving frame.

(136) Consequently, it may be very important to mount the moving frame **170** in position. To this end, the guide bar **114** may be mounted in position first. To this end, guide bar fixing parts **114a** may be provided.

(137) As shown in FIGS. **6** and **7**, the guide bar fixing parts **114a** may be formed at two upper points and two lower points of the support frame. The two guide bars may be mounted in position without upward and downward deviation or forward and rearward deviation, owing to the provision of the guide bar fixing parts **114a**. The moving frame may also be mounted in position through the guide bars.

(138) In order for the transfer members **171**, provided at the moving frame, to simultaneously transfer force to the drawers, the drawers must be mounted in position without deviation, which will be described hereinafter in detail when the detailed structure of the drawers is described.

(139) Meanwhile, in FIG. **6**, the motor assemblies **160** are mounted lower than the upper and lower centers of the support cover **110** such that the motor assemblies **160** are mounted vertically through the partition wall **16**, as described with reference to FIG. **4**. That is, the motor assemblies **160** may be mounted to the opposite support cover **110** at higher positions than the upper and lower centers of the support cover **110**.

(140) The connection member **163** may push or pull the upper part or the lower part of the moving frame **170** at positions other than the upper and lower center parts of the moving frame **170**, due to the position of the motor assembly **160**. Basically, therefore, the connection member **163** applies force to the moving frame such that the moving frame is twisted. In order to minimize the application of force to the moving frame **170** at eccentric positions, rather than the upper and lower center parts of the moving frame **170**, the connection member **163** includes an extension part **164**. The extension part **164** may extend upward or downward from the end of the connection member **163** (i.e. the end of the connection member **163** that is connected to the moving frame).

(141) The extension part **164** may be formed so as to extend through the upper and lower center parts of the moving frame **170**. That is, the extension part **164** shown in FIG. **6** may extend further upward from the upper and lower center parts of the moving frame **170**, and the opposite extension part **164** may extend further downward from the upper and lower center parts of the moving frame **170**. As a result, it is possible to minimize twisting of the moving frame **170** even when the upper and lower centers of the connection member **163** are not aligned with the upper and lower centers of the moving frame **170**. The connection member **163** may be coupled to the moving frame as the result of the coupling between the extension part **164** and the connection member coupling part **174**.

(142) A plurality of connection member coupling parts **174** may be provided in order to uniformly

transfer force and displacement applied through the connection member **163** to the upper and lower parts of the moving frame **170**. In addition, it is possible to uniformly transfer a driving force generated by the electric driving unit to the moving frame **170** through the extension part **164**.

(143) As previously described, the through part **113** is formed in the support cover **110**. The through part **113** may be referred to as a slit-shaped through part or a slit. The number of through parts **113** may be the same as the number of drawers **30**. The through parts **113** are formed through the support cover **110**. As shown in FIGS. **6** and **7**, the through parts **113** are formed in the support cover **110** so as to extend horizontally. The transfer member **171** moves leftward and rightward along the slit **113**. In the refrigerator, the transfer member **171** moves forward and rearward along the slit **113**. Since the transfer member is formed through the support cover **110**, the transfer member may be connected to the drawer **30** provided on the outside surface **111** of the support cover **110**. That is, the transfer member **171** may be coupled to the drawer **30**, or may contact the drawer **30**. The transfer member **171** may be connected to the drawer **30** in order to directly apply force to the drawer **30**.

(144) As shown in FIG. **6**, the distance between the motor assembly **160** and the moving frame **170** at the initial position of the drawer and the moving frame is relatively small. In this state, the moving frame **170** is biased to the left side. In other words, the moving frame is more deeply positioned in the storage compartment of the refrigerator.

(145) When the motor assembly **160** is driven, the distance between the motor assembly **160** and the moving frame **170** is increased. That is, the connection member **163** pushes the moving frame **170** such that the moving frame **170** moves forward. At this time, the transfer member **171** pushes the drawer **30** as the result of the movement of the moving frame **170**, whereby the drawer **30** is moved to the ready position. In other words, the support assembly **100** is moved from the initial position shown in FIG. **6** to the ready position shown in FIG. **7**. That is, the connection member **163** and the moving frame **170** shown in FIG. **6** are positioned at the initial position, and the connection member **163** and the moving frame **170** shown in FIG. **7** are positioned at the ready position.

(146) On the other hand, the relative position between the support cover **110** and the housing **161** of the motor assembly is not changed. Consequently, the moving frame **170** may be provided between the initial position and the ready position so as to be moved forward and rearward by the motor assembly **160**. In addition, the moving frame **170** may be connected to the drawer in order to apply force to the drawer in a direction in which the drawer is withdrawn from the initial position to the ready position.

(147) As shown in FIG. **10**, the connection between the drawer **30** and the moving frame, particularly the transfer member **171**, may be referred to as catching. In addition, the connection release may be referred to as catching release.

(148) Specifically, the drawer **30** is provided with a catching member **33**. The catching member **33** may protrude toward the support cover **110**. The transfer member **171** may protrude toward the drawer **30** through the support cover **110**.

(149) The transfer member **171** is positioned at the rear of the catching member **33**. As a result, a pushing force may be transferred to the transfer member **171**, but a pulling force may not be transferred to the transfer member **171**. That is, the transfer member **171** pushes the catching member **33** forward while moving forward. When the transfer member **171** is moved rearward, however, the connection between the catching member **33** and the transfer member **171** is released. Consequently, the transfer member **171** may selectively push the catching member **33**. More specifically, the transfer member **171** pushes the catching member **33** forward when the transfer member **171** moves forward, but does not push the catching member when the transfer member **171** moves rearward.

(150) The catching member **33** may be formed at a rail coupling part **37** of the drawer **30**. That is, the catching member **33** may be formed at the rail coupling part **37**, at which the drawer **30** is

coupled to the rail **120**. Consequently, the position at which the drawer is pushed is substantially identical to the position at which the drawer is coupled to the rail **120**.

(151) Hereinafter, an automatic movement mechanism of the drawer **30** will be described in detail with reference to FIG. **11**. FIG. **11** is an enlarged sectional view showing a connection between the drawer **30** and the support assembly **100**.

(152) The drawer **30** may include a basket **31** for receiving goods and a drawer frame **32** provided outside the basket **31**. The basket **31** may be supported by the rail **120** via the drawer frame **32** such that the basket **31** can move along the rail **120**. The basket **31** and the drawer frame **32** may be moved as one body. The drawer frame **32** may be provided at the lower side of the basket **31**.

(153) The rail **120** may include a fixed rail **122** and a moving rail **121**. The rail **120** may be coupled to the sidewall of the storage compartment or the partition wall **16** via rail brackets **123** and **124**. In addition, the rail **120** may be coupled to the support cover **110**.

(154) The fixed rail **122** may be configured to support the load of the drawer **30**. The moving rail **121** may be configured to move forward and rearward relative to the fixed rail **122**. For example, the moving rail **121** may be configured to slide relative to the fixed rail **122**.

(155) The moving rail **121** is coupled to the rail coupling part **37** of the drawer **30**. Consequently, the moving rail **121** and the drawer **30** may move forward and rearward as one body.

(156) The rail coupling part **37** is formed at the side of the drawer. In addition, the rail coupling part **37** may be provided to locate the moving rail **121** in the upper part thereof. That is, the rail coupling part **37** may be provided such that the upper part of the moving rail **121** is located in the rail coupling part **37** while the upper part of the moving rail **121** is surrounded by the rail coupling part **37**.

(157) The structure of the drawer **30** and the coupling structure between the drawer **30** and the rail **120** will be described hereinafter in detail.

(158) The drawer frame **32** is positioned at one side (the left side) of the support cover **110**, and the moving frame **170** is positioned at the other side (the right side) of the support cover **110**. The transfer member **171** of the moving frame **170** may extend to the vicinity of the drawer frame **32** through the slit **113** formed in the support cover **110**.

(159) The moving frame **170** may be selectively connected to the drawer frame **32**. That is, the moving frame **170** may be configured to selectively push the drawer through the drawer frame **32**. As the result of the connection between the moving frame **170** and the drawer frame **32**, the movement of the moving frame **170** may be converted into the movement of the drawer frame **32**, i.e. the drawer **30**. On the other hand, as the result of the disconnection between the moving frame **170** and the drawer frame **32**, conversion of the movement of the moving frame **170** into the movement of the drawer **30** may be prevented.

(160) Specifically, the drawer **30** may be provided with a catching member **33**. The catching member **33** may be referred to as a first catching member **33** such that the catching member **33** is distinguished from another catching member, which will be described hereinafter. The first catching frame **33** may be formed at the drawer frame **32**, and may extend toward the moving frame **170**. Consequently, the first catching frame **33** may be formed at the side surface of the drawer **30**. More specifically, the first catching frame **33** may be formed at the lower side surface of the drawer **30**.

(161) As previously described, the moving frame **170** may move forward from the initial position to the ready position. The movement of the moving frame **170** is converted into the movement of the drawer **30** from the initial position to the ready position. The movement of the moving frame **170** from the initial position to the ready position may be performed by the application of force to the moving frame **170** from the rear of the moving frame **170**. Consequently, the moving frame **170** may push the drawer **30** such that the drawer **30** is moved from the initial position to the ready position.

(162) To this end, as described with reference to FIG. **10**, the first catching member **33** may be

positioned at the front of the transfer member **171**. In addition, the moving frame **170** contacts the drawer **30** in a state in which the moving frame **170** and the drawer **30** are in the initial position. As the transfer member **171** moves from the initial position to the ready position, therefore, the transfer member **171** may continuously push the first catching member **33**. As a result, the drawer **30** may also be moved from the initial position to the ready position.

(163) On the other hand, in a state in which the drawer **30** is at the ready position, the transfer member **171** may return rearward, which may be referred to as the return of the transfer member to the initial position. That is, at this time, the connection or catching between the transfer member **171** and the first catching member **33** is released. Consequently, the drawer **30** remains at the ready position, and the transfer member **171**, particularly the moving frame **170**, may return rearward.

(164) In addition, as shown in FIG. **11**, the sectional area of the transfer member **171** may be greater than the sectional area of the catching member **33**. That is, the relatively large transfer member **171** may transfer force to the relatively small catching member **33**. Consequently, force is stably transferred from the transfer member **171** to the catching member **33**. In addition, the transfer member **171** may further extend upward and downward from the catching member **33**. Consequently, the transfer member **171** may stably transfer force to the entire contact surface of the catching member **33**, even when the catching member **33** droops.

(165) As previously described, at the initial position, the drawer **30** may remain in the storage compartment. When the user wishes to use the drawer **30**, the user opens the door **20** and pulls the drawer **30** such that at least a portion of the drawer is withdrawn from the storage compartment. The above-mentioned maximum withdrawal position may be a position at which the drawer **30** is maximally withdrawn forward in a state in which the drawer **30** is supported by the rail **120**. The maximum withdrawal position may be preset through the rail **120**. That is, the distance between the ready position and the maximum withdrawal position may be preset.

(166) Basically, the drawer **30** may be supported by the rail **120** such that the drawer **30** can move between the initial position and the maximum withdrawal position. As previously described, the drawer **30** may automatically move from the initial position to the ready position by driving of the electric driving unit **150**.

(167) The drawer may be manually withdrawn from the ready position to the maximum withdrawal position (a position spaced apart forward from the ready position by a predetermined distance). That is, the connection between the moving frame **170** and the drawer **30** is released between the ready position and the maximum withdrawal position such that the drawer **30** can be manually withdrawn.

(168) When the door **20** is opened, the drawer **30** may automatically move to the ready position such that the user can easily withdraw the drawer **30**. In order to use the drawer **30**, the user may further manually withdraw the drawer **30** away from the ready position. After the use of the drawer **30**, the user may manually insert the drawer **30** into the storage compartment. For example, the user may manually push the drawer **30** to the ready position or to the vicinity of the ready position. Of course, the user may manually push the drawer **30** to the initial position.

(169) That is, automatic withdrawal of the drawer **30**, which is achieved simultaneously when the door is opened, may be performed from the initial position to the ready position, and the drawer **30** may be manually withdrawn from the ready position to the maximum withdrawal position.

(170) Meanwhile, when the drawer **30** is withdrawn to the ready position, the motor assembly **160** may be operated to move the connection member **163** to the initial position. Consequently, the insertion of the drawer **30** may be performed manually. Manual insertion may be performed as follows. The user may insert the drawer **30** while directly holding the drawer **30**. Alternatively, when the door **20** is closed, the door **20** may push the drawer **30** such that the drawer **30** is inserted. The reason for this is that the user's force is used to insert the drawer.

(171) For example, the user may directly push the drawer **30** from the maximum withdrawal position to the initial position such that the drawer **30** is inserted. Alternatively, the user may

directly push the drawer **30** from the maximum withdrawal position to the ready position such that the drawer **30** is inserted, and may then close the door **20** such that the drawer **30** is pushed to the initial position. When the door **20** is closed, the door basket provided at the rear of the door **20** pushes the drawer **30**. When the user manually closes the door **20**, therefore, the user must use force greater than the force required to insert the drawer in order to close the door **20**.

(172) In the above, the mechanism between the drawer **30** and the support assembly **100** has been described in terms of automatic withdrawal of the drawer **30**. That is, an embodiment in which the drawer **30** is automatically withdrawn using electrical energy has been described. In the above embodiment, it is not necessary for the user to apply force to the drawer **30** in order to withdraw the drawer **30**.

(173) Meanwhile, effort on the part of the user may be minimized when the drawer **30** is inserted as well as when the drawer **30** is withdrawn. That is, it is not necessary for the user to apply force to the drawer **30** in order to insert the drawer **30**.

(174) In this embodiment, it is possible to provide a refrigerator configured such that the drawer **30** can be inserted automatically for user convenience. In particular, it is possible to provide a refrigerator configured such that the drawer **30** can be automatically pushed from the ready position or the vicinity of the ready position to the initial position. That is, for the same reason that no force from the user is needed to automatically withdraw the drawer **30**, no force from the user may be needed to automatically insert the drawer **30**. In addition, the door basket provided at the rear of the door **20** may prevent impact from being applied to the drawer, and it may not be necessary for the user to apply force to the drawer in order to insert the drawer, except for the force necessary to close the door **20**.

(175) To this end, an automatic drawer insertion device may be provided. Accordingly, the drawer may be inserted back into the initial position without the user having to provide the required closing force. For example, as described in this embodiment, an elastic device **180** may be provided. The elastic device **180** may be mounted to the sidewall of the storage compartment. In the same manner as in the previous embodiment, the elastic device **180** may be mounted to the support cover **110**.

(176) Hereinafter, an embodiment including an elastic device **180** will be described in detail with reference to FIGS. **12** to **14**. In FIGS. **12** to **14**, an elastic device **180** is added, unlike FIGS. **4**, **5**, and **11**. Consequently, a detailed description of elements corresponding to the elements described with reference to FIGS. **4**, **5**, and **11** will be omitted.

(177) The elastic device **180** may be configured to provide an elastic restoring force to the drawer **30**. In particular, the elastic device **180** may be configured such that the elastic device **180** is elastically deformed when the drawer **30** is withdrawn and such that the elastic device **180** provides an elastic restoring force to the drawer **30** when the drawer **30** is inserted. That is, the withdrawal of the drawer may be automatically performed by the electric driving unit, and the insertion of the drawer may be automatically performed by the elastic device **180**. In other words, the withdrawal of the drawer may be automatically performed by the driving force of the motor, and the insertion of the drawer may be automatically performed by the elastic restoring force of the elastic device **180**.

(178) In addition to the rail **120**, the elastic device **180** may be mounted to the support cover **110**. Specifically, a set including a pair of rails **120**, a slit **113**, and an elastic device **180** may be provided at a single drawer **30**. In a case in which a plurality of drawers **30** is provided, a plurality of sets may also be provided. In addition, in the same manner as the rail **120**, the elastic device **180** may be mounted to the outside surface **111** of the support cover **110**.

(179) The rail **120** may be mounted to the lower side of the slit **113**, formed in the support cover **110**, and the elastic device **180** may be mounted to the upper side of the slit **113**.

(180) The elastic device **180** may be configured to provide an elastic restoring force to the drawer **30** when the drawer **30** returns from the ready position or the vicinity of the ready position to the

initial position. The drawer **30** may be automatically returned to the initial position by the elastic restoring force.

(181) To this end, the elastic device **180** may be selectively connected to the drawer **30**. That is, the elastic device **180** may be configured to selectively hold the drawer.

(182) Specifically, the elastic device **180** may include a hanging member **181**. The hanging member **181** may be selectively connected to the drawer **30**. More specifically, the drawer may be provided with a second catching member **34**. That is, as shown in FIG. **14**, a second catching member **34**, which is connected to the hanging member **181** of the elastic device **180**, may be provided in addition to the first catching member **33**, which is connected to the transfer member **171** of the moving frame.

(183) The second catching member **34** may be provided at the upper side of the first catching member **33**. Specifically, the second catching member **34** may protrude from the drawer frame **32** toward the support cover **110**.

(184) Hereinafter, an embodiment of the elastic device **180** will be described in detail with reference to FIG. **15**.

(185) The elastic device **180** includes a spring **187** as an example of an elastic member. In addition, a hanging member **181** is provided in order to elastically deform the spring **187**. The hanging member is coupled to the catching member of the drawer **30** such that the hanging member is displaced in response to the movement of the drawer **30**. As the result of this displacement, the elastic device **180** may be elastically deformed or may generate an elastic restoring force.

(186) A connection member **189** may be interposed between the hanging member **181** and the spring **187**. One end of the spring **187** is fixed, and the other end of the spring **187** is coupled to the connection member **189**. The connection member **189** is coupled to the hanging member **181**. Consequently, the hanging member **181** and the connection member **189** are moved as one body, whereby the spring **187** may be elastically deformed or elastically restored.

(187) The elastic device **180** includes a housing **182**. The housing **182** receives the spring **187** and the hanging member **181**. Of course, the housing **182** may receive the connection member **189**. The housing **182** may be provided with a plurality of coupling parts **188**. The elastic device **180** may be fixedly coupled to the sidewall of the storage compartment, the partition wall, or the support cover **110** via the coupling parts **188**.

(188) The hanging member **181** is basically configured to move while being displaced linearly. A slot **183** is formed in order to guide the movement of the hanging member **181**. The hanging member **181** is provided with a guide protrusion **181a**. The guide protrusion **181a** moves along the slot **183**.

(189) As the hanging member **181** moves forward, the connection between the hanging member **181** and the drawer is released at a specific position. To this end, an inclined slot **185** may be formed at the front of the slot **183**. During the forward movement of the hanging member **181**, the guide protrusion **181a** may move upward along the inclined slot **185**. The upward movement of the guide protrusion **181a** is achieved by the rotation of the hanging member **181**. To this end, the hanging member **181** may be provided with a rotating protrusion **181b**, which forms the rotational center of the hanging member **181**.

(190) The rotating protrusion **181b** moves forward and rearward along the slot **183** in the same manner as the guide protrusion **181a**. The rotating protrusion **181b** may be positioned at the rear of the guide protrusion **181a**. When the guide protrusion **181a** moves upward along the inclined slot **185**, therefore, the hanging member **181** is rotated about the rotating protrusion **181b**.

(191) When the rotating protrusion **181b** is caught by the inclined slot **185**, the connection between the drawer **30** and the elastic device **180** is released. When the drawer **30** is inserted, however, the drawer may be reconnected with the elastic device **180**. At this time, the drawer **30** may be automatically inserted by the elastic restoring force.

(192) Here, it should be noted that the insertion speed of the drawer may be configured to be higher

than the withdrawal speed of the drawer. For example, the speed at which the clastic device **180** inserts the drawer back into the storage compartment, or in some cases the speed at which the drawer is inserted back into the storage compartment by being driven by a motor, may be higher than the withdrawal speed at which the drawer is withdrawn from the storage compartment by the driving force of the motor. In some cases, the insertion speed may be approximately 1.5 to 2.0 times higher than the withdrawal speed. This difference in speed can help prevent, for example, the door from crashing into the drawer when the drawer is being inserted back into the storage compartment.

(193) The connection member **189** may also be provided with a guide protrusion **189a**, which is configured to move along the slot **183**.

(194) The housing **182** may be provided with a guide slot **182a** for guiding the movement of the second catching member **34**, which is selectively connected to the hanging member **181**. An enlarged opening **182b** may be formed in the front of the guide slot **182a**. The catching member **34** moves forward along the guide slot **182a** and escapes from the enlarged opening **182b** at a specific position. That is, after the connection between the drawer and the elastic device is released, the drawer may be withdrawn further forward. In other words, the drawer may be further withdrawn.

(195) Meanwhile, when the withdrawn drawer is inserted, the catching member **34** must enter the guide slot **182a**. In order to more easily perform such entry, the enlarged opening **182b** is provided.

(196) The housing **182** may be provided with a positioning part **183a** for placing the hanging member **181** in position. The positioning part **183a** may be configured to position the hanging member **181** in position after the hanging member **181** returns rearward.

(197) When the hanging member **181** is not positioned in position at the rear, as previously described, the drawer may not be completely inserted to the initial position. Consequently, the hanging member **181** may return rearward due to the elastic restoring force of the spring, and then the hanging member **181** may be prevented from moving forward by a repulsive force.

(198) To this end, the positioning part **183a** may be formed in the shape of a rib.

(199) The positioning part **183a** may be formed at the rear of the slot **183**, or may be inclined rearward in a direction in which the height of the slot **183** is reduced. Of course, the height of the slit at the positioning part **183a** may be smaller than the height of the slit at the other parts.

Consequently, the guide protrusion **181a** of the hanging member **181** may return to the proper position while pushing the positioning part **183a** upward due to the elastic restoring force.

Subsequently, the positioning part **183a** pushes the guide protrusion **181a** due to the elastic restoring force thereof. Consequently, the guide protrusion **181a** may be maintained in position.

(200) A communication part **183b** may be formed in the end of the positioning part **183a** and the slot **183**. The guide protrusion **181a** may be inserted into the slot **183** through the communication part **183b**.

(201) The left side diameter and the right side diameter of the guide protrusion **181a** are greater than the height of the slot **183**. However, the diameter of the part corresponding to the slot **183** is equal to or less than the height of the slot **183**. Consequently, the guide protrusion **181a** moves forward and rearward along the slot **183** but does not escape from the slot **183**. As a result, it is not easy to insert the guide protrusion **181a** into the slot **183**.

(202) The communication part **183b** may be a structure for inserting the guide protrusion **181a** into the slot **183**.

(203) However, the guide protrusion **181a** may escape from the slot **183** through the communication part **183b**. That is, if the upper part of the positioning part **183a** is plastically deformed, the guide protrusion **181a** may be constrained by the positioning part **183a**. As a result, the drawer **30** may not be automatically withdrawn.

(204) In order to minimize the breakage or bending of the positioning part **183a**, therefore, the positioning part **183a** may be formed in the shape of a cantilever. In addition, the thickness of the fixed end of the positioning part **183a** may be greater than the thickness of the free end of the

positioning part **183a**. For example, the thickness of the positioning part **183a** may be gradually decreased from the fixed end to the free end thereof.

(205) As a result, the positioning part **183a** may be reinforced while the positioning part **183a** is easily elastically deformed. In addition, the positioning part **183a** may be reinforced by increasing the horizontal width of the fixed end of the positioning part **183a**.

(206) Consequently, it is possible to prevent the breakage of the positioning part **183a** even when the positioning part **183a** is used for a long period of time.

(207) Meanwhile, the hanging member **181** may be an element that requires high strength and low friction. In the same manner as the liner **173**, therefore, the hanging member **181** may be made of a POM material.

(208) Hereinafter, a mechanism for automatically inserting the drawer using the elastic device **180** will be described in detail with reference to FIGS. **16** and **17**. FIG. **16** is a side view showing the drawer **30** and the support assembly **100** at the initial position of the drawer **30**, and FIG. **17** is a side view showing the drawer **30** and the support assembly **100** at the ready position of the drawer **30**. For the convenience of description, the upper part of the drawer **30** is omitted.

(209) As shown in FIGS. **16** and **17**, the hanging member **181** of the elastic device **180** may be selectively connected to the second catching member **34** provided at the drawer **30**. At the initial position of the drawer **30**, the connection between the elastic device **180** and the drawer **30** may be released. As the drawer **30** moves to the ready position, the elastic device **180** and the drawer may be connected with each other.

(210) Specifically, the drawer **30** is automatically withdrawn forward by the operation of the electric driving unit **150**. That is, the drawer **30** is withdrawn from the initial position to the ready position. When the drawer **30** is withdrawn, the second catching member **34** provided at the drawer **30** is connected to the hanging member **181** to move the hanging member **181** forward. The hanging member **181** moves forward together with the second catching member **34**. As a result, the spring **187** may be elastically deformed. For example, the spring **187** may be extended. The drawer **30** is automatically inserted due to the elastic restoring force generated at this time.

(211) More specifically, the drawer **30** may be automatically withdrawn while overcoming the clastic force of the elastic device **180** by the operation of the electric driving unit **150**. When the force applied to the drawer **30** by the electric driving unit **150** is removed, the elastic restoring force generated by the elastic device **180** may be applied to the drawer **30**. As a result, the hanging member **181** pulls the second catching member **34** of the drawer **30** due to the clastic restoring force. Consequently, the drawer **30** may be automatically inserted even if the user does not apply additional force in order to insert the drawer **30**.

(212) The transfer member **171** may be provided so as to push only the first catching member, whereas the hanging member **181** may be provided so as to be pulled by the second catching member **34** and to pull the second catching member **34**. That is, the hanging member **181** is pulled by the second catching member **34** when the drawer **30** is withdrawn, whereas the hanging member **181** pulls the second catching member **34** when the drawer **30** is inserted. In other words, the hanging member **181** and the second catching member **34** may be connected with each other when the drawer **30** is inserted or withdrawn. When the drawer **30** is inserted, the drawer **30** may not be manually inserted but may be automatically inserted, whereby the user may very conveniently manipulate the drawer. Of course, automatic insertion of the drawer **30** may be performed from the ready position or the vicinity of the ready position to the initial position, rather than from the maximum withdrawal position to the initial position.

(213) When the drawer **30** moves from the initial position to the ready position, the clastic device **180** is elastically deformed. When the drawer **30** moves from the ready position to the initial position, the elastic device **180** provides an elastic restoring force to the drawer **30**. The elastic device **180** may be continuously elastically deformed and elastically restored from the initial position to the ready position. In this case, the elastic restoring force may be provided to the drawer

30 until the drawer **30** completely returns to the initial position.

(214) When the drawer **30** starts to move from the initial position, however, a relatively large load may be applied to the electric driving unit **150** due to a static frictional force of the drawer **30**. The drawer **30** must move while overcoming the static frictional force of the drawer **30** and the elastic force of the elastic device **180**. As a result, larger load may be applied to the electric driving unit **150**. For this reason, it is necessary to release the connection between the drawer **30** and the elastic device **180** at the initial position of the drawer **30**. In addition, at the initial position, the drawer **30** may move forward by a predetermined distance such that the drawer **30** is connected with the elastic device **180**.

(215) FIG. **16** shows the initial position of the drawer **30**. In the state shown, the connection between the drawer **30** and the elastic device **180** may be released. Specifically, the connection between the second catching member **34** of the drawer **30** and the hanging member **181** of the elastic device **180** may be released. At this time, the connection between the first catching member **33** of the drawer **30** and the transfer member **171** of the moving frame **170** may be maintained. This is because it is necessary for the forward movement of the transfer member **171** to be transferred to the first catching member **33** of the drawer **30** without delay.

(216) When the door is opened, the electric driving unit **150** is operated, with the result that the transfer member **171** pushes the first catching member **33** forward. Consequently, the drawer **30** is withdrawn forward, and the second catching member **34** also moves forward. The second catching member **34** is connected to the hanging member **181** of the elastic device **180** while moving forward. Consequently, the hanging member **181** moves forward together with the second catching member **34**. The spring **187** may be elastically deformed by the forward movement of the second catching member **34**.

(217) The position at which the second catching member **34** and the hanging member **181** are connected with each other may be preset to be between the initial position and the ready position of the second catching member **34**. Here, the position at which the second catching member **34** and the hanging member **181** are connected with each other may be referred to as an elasticity start position. For example, on the assumption that the distance between the initial position and the ready position is 120 mm, the elasticity start position may be set to be 30 mm forward from the initial position. Until the drawer moves from the initial position to the elasticity start position, therefore, the resistance attributable to the spring **187** is not transferred to the electric driving unit **150**. Subsequently, as the drawer moves further forward from the elasticity start position, resistance attributable to the spring **187** is further increased and transferred to the electric driving unit **150**.

(218) The elasticity start position may be set by changing the shape of the slots **183** and **184** formed in the housing **182** of the elastic device **180** and the connection relationship between the slots and the hanging member **181**.

(219) FIG. **17** is a side view showing the ready position of the drawer.

(220) When the door is opened, the electric driving unit **150** moves the transfer member **171** forward. As the result of the forward movement of the transfer member **171**, the first catching member **33** of the drawer **30** also moves forward. The electric driving unit **150** moves the transfer member **171** forward until the drawer **30** reaches the ready position.

(221) As shown, at the ready position of the drawer **30**, the elastic device applies force to the drawer **30** in a direction in which the drawer **30** is inserted. In this embodiment, therefore, control may be performed so as to maintain the operation of the electric driving unit **150** (for example, so as to drive the electric driving unit **150** in a clockwise direction) at the ready position. That is, the operation of the electric driving unit **150** may be maintained such that the electric driving unit **150** pushes the drawer **30** in a state in which the door **20** is open. In other words, the driving of the motor assembly **160** may be maintained while the door is kept open such that the moving frame **170** is maintained at the ready position. Of course, as will be described hereinafter, when it is sensed that the door **20** has been closed, the motor assembly **160** may be reversely driven (for

example, may be driven in a counterclockwise direction) such that the moving frame returns to the initial position.

(222) On the other hand, if the operation of the electric driving unit **150** is maintained at the ready position of the drawer **30**, the electric driving unit **150** may be overloaded. This is because the connection member **163** does not move forward any longer, whereby the motor idles.

Consequently, the electric driving unit **150** may move the drawer **30** to the ready position, may continue to operate for a predetermined time, and may be reversely driven to return the moving frame **170** to the initial position. That is, the transfer member **171** may be returned rearward.

(223) The predetermined time may be determined in consideration of the time taken by the user to select a specific drawer and withdraw the selected drawer. For example, the electric driving unit **150** may move the moving frame **170** to the ready position, may stay at the ready position for about 10 seconds, and may then be reversely driven.

(224) As previously described, automatic withdrawal of the drawer **30** by the electric driving unit **150** may be performed from the initial position to the ready position. Consequently, the withdrawal of the drawer from the ready position to the maximum withdrawal position may be manually performed. That is, the user may directly pull the drawer **30** in order to withdraw the drawer **30**.

(225) For example, in the case in which a plurality of drawers is provided vertically, the drawers may be automatically withdrawn to the ready position. In the state in which the drawers are at the ready position, the user may further withdraw one of the drawers before a predetermined time. After the predetermined time, the remaining drawers, which have not been withdrawn further, may be automatically inserted by the elastic device. In the case in which the further withdrawn drawer is a lower drawer, a space for access to the interior of the drawer may be increased due to the insertion of an upper drawer. Consequently, it is possible for the user to more easily access the storage space in the drawer. When the drawer **30** is withdrawn from the ready position to the maximum withdrawal position, the elastic device may be elastically deformed in a direction in which the withdrawal of the drawer is impeded. When the drawer **30** is manually withdrawn, therefore, the connection between the elastic device **180** and the drawer may be released.

(226) To this end, an inclined slot **185** may be formed in the slot formed in the housing **182** of the elastic device. Specifically, the inclined slot **185** may be formed at the front of one of the two slots **183** and **184**, which are vertically arranged side by side, e.g. the slot **184**. For the sake of convenience, the inclined slot **185** may be referred to as a first inclined slot **185** such that the inclined slot **185** is distinguished from another inclined slot, which will be described hereinafter. The first inclined slot **185** is positioned at the front of the slot **184**. When the user withdraws the drawer **30** slightly forward from the ready position, the hanging member **181** may be constrained in the first inclined slot **185**. At this time, the hanging member **181** is rotated, whereby the coupling between the hanging member **181** and the second catching member **34** is released. The position at which the coupling between the hanging member **181** and the second catching member **34** is released may be referred to as an elasticity end position. In this embodiment, therefore, the elasticity start position may be in front of the initial position, and the elasticity end position may be in front of the ready position.

(227) When the coupling between the hanging member **181** and the second catching member **34** is released, the user may easily manually withdraw the drawer to the maximum withdrawal position without impedance from the elastic device **180**.

(228) In this embodiment, the moving frame **170** may remain at the ready position in a state in which the door is open, as previously described. In a state in which the door is open, therefore, the user may withdraw the drawer, and may then insert the drawer **30** to the ready position. That is, the drawer may be reconnected to the elastic device.

(229) At this time, the user may not insert the drawer **30** to the ready position. In this case, the elastic restoring force of the elastic device **180** is not transferred to the drawer **30**, since the connection between the drawer **30** and the elastic device **180** is released.

(230) In this embodiment, however, the distance between the elasticity end position and the ready position is relatively small, as previously described. When the door is closed, therefore, the door may push the drawer **30** rearward. That is, the drawer **30** may be pushed to restore the connection between the second catching member **34** and the hanging member **181**. Since the moving frame **170** returns to the initial position when the door is closed, the force acting to withdraw the drawer **30** is removed. As a result, the hanging member **181** pulls the second catching member **34** due to the elastic restoring force of the elastic device **180** such that the drawer **30** automatically returns to the initial position.

(231) Meanwhile, the electric driving unit may be controlled such that the transfer member **171** stays at the ready position for about 10 seconds, as previously described. The user may further withdraw a specific drawer in order to take goods out of the drawer and may then manually insert the specific drawer to the ready position. On the assumption that the time taken at this time is about 12 seconds, the specific drawer may be inserted to the ready position, and at the same time the specific drawer may be automatically returned to the initial position by the elastic restoring force.

(232) As previously described, the clastic start position may be set to be identical to the initial position. In this case, however, impact may be applied to the drawer **30** when the drawer **30** returns to the initial position. Furthermore, in this case, the elastic restoring force of the spring may be reduced over time since the clastic variation of the spring is relatively increased (i.e. the clastic section of the spring is increased).

(233) Consequently, the elastic start position may be set to be spaced apart forward from the initial position such that the initial return speed of the drawer is relatively high while the final return speed of the drawer is relatively low.

(234) The initial return speed is related to the door closing speed. For example, if the door is very rapidly closed and the initial return speed of the drawer is lower than the door closing speed, the door may apply impact to the drawer. On the other hand, if the final return speed of the drawer is higher than the door closing speed, the drawer may apply great impact to the rail **120**. For this reason, it may be necessary for the drawer to rapidly return at the early stage and to softly and slowly return at the late stage. That is, the drawer may be returned by inertia at the late stage.

(235) Meanwhile, when the drawer **30** is withdrawn, the second catching member **34** is connected to the hanging member **181** at the elasticity start position. On the other hand, when the drawer **30** is inserted, the connection between the second catching member **34** and the hanging member **181** is released at the elasticity start position. These operations may be performed in the same manner as the operations at the elasticity end position. In the same manner, a second inclined slot **186** may be formed at the rear end of the upper slot **183** such that the hanging member **181** can be rotated at the elasticity start position. The hanging member **181** is rotated in the first inclined slot in the counterclockwise direction such that the connection between the hanging member **181** and the first inclined slot is released, whereas the hanging member **181** is rotated in the second inclined slot **186** in the clockwise direction such that the connection between the hanging member **181** and the second inclined slot is released.

(236) According to the above embodiment, the withdrawal of the drawer from the initial position to the ready position may be automatic withdrawal of the drawer performed by the driving of the electric driving unit, and the insertion of the drawer from the ready position to the initial position may be automatic insertion of the drawer performed by the elastic device. The automatic withdrawal of the drawer may be performed using the driving force of the motor based on electrical energy, and the automatic insertion of the drawer may be performed using the elastic restoring force of the spring.

(237) According to the above embodiment, the withdrawal of the drawer from the ready position to the maximum withdrawal position may be manual withdrawal of the drawer performed by the user, and the insertion of the drawer from the maximum withdrawal position to the ready position may be manual insertion of the drawer performed by the user.

(238) According to the above embodiment, the spring of the elastic device may start to be elastically deformed at the elasticity start position, which is spaced apart forward from the initial position of the drawer by a predetermined distance, and the elastic deformation may be continuously performed from the elasticity start position to the ready position of the drawer.

(239) According to the above embodiment, the spring may be continuously elastically deformed until the drawer reaches the elasticity end position, which is spaced apart forward from the ready position of the drawer by a predetermined distance. The withdrawal of the drawer from the ready position to the elasticity end position may be performed manually. The drawer may be manually withdrawn from the elasticity end position to the maximum withdrawal position of the drawer. At this time, the connection between the spring and the drawer may be released. When the user manually inserts the drawer, therefore, the drawer may be automatically inserted to the initial position by the elastic restoring force even if the drawer is inserted to the elasticity end position. That is, the drawer may be automatically inserted to the initial position by the elastic restoring force even if the drawer is not manually inserted to the ready position.

(240) Hereinafter, another embodiment of the present invention will be described in detail with reference to FIGS. **18** and **19**.

(241) In this embodiment, a drawer **30** in a single storage compartment may be automatically withdrawn. For example, in a case in which storage compartments are formed on opposite sides of a partition wall **16**, a drawer **30** in one of the storage compartments may be automatically withdrawn. In addition, in this embodiment, an electric driving unit **150** is mounted to the left sidewall or the right sidewall of the storage compartment.

(242) Basically, this embodiment is identical to the previous embodiment with respect to the mechanism for automatically withdrawing the drawer **30**. Of course, this embodiment is also identical to the previous embodiment with respect to the mechanism for automatically inserting the drawer **30**. Consequently, this embodiment may be identical or similar to the previous embodiment with respect to the basic functions and connection relationships thereof, even though this embodiment may be different in structure or shape from the previous embodiment.

(243) At least one drawer **30** may be movably supported by the sidewall of the storage compartment or the partition wall via a rail. In this embodiment, a support assembly **100** or a support cover **110** may be defined by the sidewall of the storage compartment or the partition wall. Alternatively, the support assembly **100** or the support cover **110** may be mounted to the sidewall of the storage compartment or the partition wall.

(244) An electric driving unit **150** is mounted to the support cover **110**, and a moving frame **170** is connected to the electric driving unit **150**. Consequently, the moving frame **170** may be moved forward and rearward by the driving of the electric driving unit **150**.

(245) In this embodiment, the support cover **110** may be provided with a through part or slit **113**, a recess **16g**, a motor avoidance recess **119**, and a plurality of fastening parts **118**, in the same manner as in the previous embodiment. In addition, an elastic device **180** may be provided in order to automatically insert the drawer **30** using an elastic restoring force.

(246) In this embodiment, a moving frame cover **190** may be further provided. The recess **16g** may be formed in order to mount the motor assembly **160** and, in addition, to mount and move the moving frame **170**. The recess **16g** may be further extended to define a predetermined space **130**.

(247) Consequently, the moving frame cover **190** may be configured to cover a space in which the moving frame moves. That is, the moving frame cover **190** may be coupled to the support cover **110** to protect the moving frame.

(248) In addition, the moving frame **170** may be provided with rollers **176**. The rollers **176** may be formed at corners of the moving frame **170**. The rollers **176** may slidably support the moving frame **170** relative to the moving frame cover **190**. As a result, the moving frame **170** may be stably supported such that the moving frame **170** can move forward and rearward.

(249) Hereinafter, this embodiment will be described in more detail with reference to FIGS. **20** to

22.

(250) In this embodiment, automatic withdrawal and automatic insertion of the drawer may be performed using a single transfer member. The basic mechanism and construction of this embodiment may be similar or identical to those of the previous embodiment. Therefore, features of this embodiment that differ from the previous embodiment will be described hereinafter in detail.

(251) In this embodiment, a motor assembly **160** and a moving frame **170** may be provided, in the same manner as in the previous embodiment. The motor assembly **160** may be mounted to one sidewall of the storage compartment. The motor assembly **160** is connected to the moving frame **170**. The motor assembly **160** is operated to move the moving frame **170** forward and rearward in the same manner as in the previous embodiment.

(252) In addition, in this embodiment, an elastic device **180** may be provided, in the same manner as in the previous embodiment. The elastic device **180** may be mounted to one sidewall of the storage compartment. The elastic device **180** of this embodiment may be identical to the clastic device of the previous embodiment. In this embodiment, however, the clastic device **180** may be configured such that a hanging member **181** does not protrude downward or upward but protrudes laterally. In other words, the hanging member **181** may protrude toward the drawer **30**. This means that the hanging member **181** of the clastic device **180** extends to the drawer through the slit **113**.

(253) The drawer **30** may be provided with a catching member **36**. The catching member **36** is pushed to move the drawer **30** forward. The catching member **36** may be selectively connected to the hanging member **181** of the elastic device **180**. As the result of the forward movement of the drawer **30**, therefore, the catching member **36** moves the hanging member **181** forward. That is, the catching member **36** of this embodiment may correspond to the second catching member **34** of the previous embodiment in that the catching member **36** is selectively connected to the hanging member **181**.

(254) However, the catching member **36** is configured to be moved forward by a transfer member **171** provided at the moving frame **170**. That is, the catching member **36** of this embodiment may correspond to the first catching member **33** of the previous embodiment. In this embodiment, therefore, the catching member **36** is connected to the elastic member **180** and to the moving frame **170**.

(255) Specifically, as shown in FIG. **21**, the transfer member **171**, which is provided at the moving frame **170**, may be configured to push the hanging member **181** of the elastic device **180**. That is, the transfer member **171** pushes the hanging member, which is connected to the catching member **36**, with the result that the transfer member **171** pushes the catching member **36**. In other words, the transfer member **171** may be configured to push the hanging member **181** at one side of the rear of the hanging member **181**, which protrudes toward the drawer.

(256) For example, in FIG. **21**, a state in which the drawer **30** is inserted to the ready position is shown. That is, at the initial position, the transfer member **171** may simultaneously push the hanging member **181** of the elastic device **180** and the catching member **36** of the drawer to automatically withdraw the drawer to the ready position.

(257) For example, in FIG. **22**, a state in which the drawer **30** is manually withdrawn forward from the ready position is shown. When the user manually withdraws the drawer **30**, the catching member **36** of the drawer **30** moves the hanging member **181** forward. As a result, the connection between the hanging member **181** and the transfer member **171** is released. In addition, as the drawer is withdrawn forward from the ready position by a predetermined distance, the connection between the catching member **36** and the hanging member **181** is released. That is, as the hanging member **181** enters the inclined slot **185**, the connection between the catching member **36** and the hanging member **181** is released. As a result, the connection between the elastic device **180** and the drawer **30** is released. Consequently, it is possible for the user to easily manually withdraw the drawer **30**. Of course, even in this embodiment, the ready position may be set to correspond to the

position at which the connection between the clastic device **180** and the drawer **30** is released.

(258) In this embodiment, the moving frame **170** may be provided with a connection member coupling part **174**, which is coupled to a connection member. In addition, the moving frame **170** may be provided with rollers **176** for allowing the moving frame **170** to stably move forward and rearward.

(259) Two rollers **176** may be formed at the upper end of the moving frame, and two rollers **176** may be formed at the lower end of the moving frame. Consequently, the moving frame may stably move in a state in which the moving frame is supported by four support points. Of course, the rollers **176** may be provided on one sidewall of the storage compartment such that the rollers **176** can be rolled.

(260) The moving frame **170** may be provided between the elastic device **180** and the drawer such that the moving frame **170** can move forward and rearward. In particular, the transfer member **171** of the moving frame **170** may be provided between the catching member **36** of the drawer and the clastic device **180** such that the transfer member **171** can move forward and rearward.

(261) The moving frame **170** may be formed in the shape of a plate. In order to reduce the weight of the moving frame **170**, a plurality of slits **175** may be formed in the moving frame **170**. In a case in which the moving frame **170** is configured to move three vertically arranged drawers, three transfer members **171** may be provided. At this time, two of the transfer members **171** may be provided at the upper end and the lower end of the moving frame. The middle transfer member **171** may be formed through one of the slits **175**. Specifically, the middle transfer member **171** may be formed through the middle slit **177**. That is, the portion of the moving frame **170** in which the slits are not formed excluding the slit **177** may be the middle transfer member. The middle transfer member **171** may be the vertically middle portion of the moving frame **170** for interconnecting the upper and lower parts of the moving frame **170**.

(262) Meanwhile, even in this embodiment, it is necessary to minimize the reduction of the inner space of the storage compartment due to the electric driving unit **150**. To this end, the electric driving unit **150** may be positioned at the sidewall of the storage compartment or the partition wall. In addition, the electric driving unit **150** may be positioned between the rails. This is because the rails may protrude toward the storage compartment and the drawer may be coupled to the protruding rails. A predetermined space may be defined between the upper rail and the lower rail due to the protruding structure of the rails, and the electric driving unit **150** may be mounted in the space.

(263) Hereinafter, an embodiment of a drawer configured to be easily separated from the refrigerator such that the drawer can be conveniently used will be described in detail with reference to FIGS. **23** to **26**. The drawer that can be automatically withdrawn and automatically inserted as described above may be embodied based on the drawer according to this embodiment. That is, a drawer that can be used independently of the previous embodiments and can be applied to the previous embodiments will be described in detail. The drawer may equally apply to a general drawer that is manually withdrawn and manually inserted.

(264) The drawer **30** may include a basket **31** and a drawer frame **32**. The basket **31** may be configured to receive goods. The drawer frame **32** may be configured to support the basket **31**.

(265) Specifically, the drawer frame **32** may include a basket location part **38** and rail coupling parts **37** coupled to rails **120**.

(266) The basket **31** is located in the basket location part **38** such that the basket **31** is coupled to the drawer frame **32**. In particular, the basket **31** may be moved vertically downward in the basket location part **38** such that the basket **31** is coupled to the drawer frame **32**. On the other hand, the basket **31** may be moved vertically upward in the basket location part **38** such that the basket **31** is separated from the drawer frame **32**. Consequently, the basket **31** may be very easily coupled to and separated from the drawer frame **32**.

(267) The drawer frame **32** may be provided at the middle part thereof with an opening **38a**. The

basket location part **38** may be formed around the opening **38a**.

(268) The basket **31** may include an upper basket **31a** and a lower basket **31b**. The lower basket **31b** is inserted through the opening **38a**, and the upper basket **31a** is located in the basket location part **38**. The upper basket **31a** and the lower basket **31b** may be integrally formed.

(269) The rail coupling parts **37** may be provided at the left and right sides of the drawer frame **32** such that the rail coupling parts **37** extend forward and rearward. The rail coupling parts **37** and the rails may not be visible.

(270) To this end, the horizontal width of the upper basket may be greater than the horizontal width of the lower basket such that the upper basket can cover the rail coupling parts **37** when viewed from above. In addition, the drawer frame **32** may include a drawer decoration part **39**. The drawer decoration part **39** may be provided at the front of the drawer frame **32**. In particular, the drawer decoration part **39** may be provided at the lower part of the front of the basket **31** such that the drawer decoration part **39** extends leftward and rightward. That is, the drawer decoration part **39** may be provided at the front of the drawer **30** in order to cover the rail coupling parts **37**.

(271) Meanwhile, the catching member **33** may be formed at the drawer frame **32**. The drawer may be commonly used in the left freezing compartment and the right freezing compartment. Catching members **33** may be formed at the left and right sides of the drawer frame **32**. In a case in which the drawer **30** is disposed in the right freezing compartment, as previously described, only the left catching member **33** may be used.

(272) The drawer frame **32** may be provided with a catching member mounting part **34a**, to which a catching member **34**, configured to be coupled to the elastic device, is mounted. In the same manner, catching member mounting parts **34a** may be formed at the left and right sides of the drawer.

(273) The catching member **34**, which is configured to be coupled to the elastic device, is provided to automatically insert the drawer. If it is not necessary to automatically insert the drawer, therefore, the catching member **34** may be omitted. Consequently, the catching member **34** may be separably provided at the drawer frame **32** even though the catching member **33** is integrally formed at the drawer frame **32**.

(274) Hereinafter, a rail **120** will be described in detail with reference to FIG. **24**.

(275) The rail **120** is configured to support the drawer **30** such that the drawer **30** can move forward and rearward. Consequently, the rail **120** may include a moving rail **121**, configured to move forward and rearward together with the drawer **30**. In addition, the rail coupling part **37** may be coupled to the moving rail **121**.

(276) Specifically, the rail coupling part **37** may be formed in the sectional shape of a channel such that the rail coupling part **37** can be located while surrounding the moving rail **121**. That is, the rail coupling part **37** may be coupled to the moving rail **121** in a state in which the rail coupling part **37** is located on the moving rail **121**.

(277) A catching part **125** may be formed at the rear end of the moving rail **121**. The catching part **125** may be configured such that the rear end of the rail coupling part **37** is inserted into the catching part **125**. When the rail coupling part **37** is inserted into the catching part **125**, therefore, the rearward movement and the upward movement of the rail coupling part **37** may be limited at the rear end of the rail coupling part **37**.

(278) An elastic protrusion **128** may be formed at the front end of the moving rail **121**. The rail coupling part **37** may be provided at the front end thereof with a mounting hole **37a**, into which the elastic protrusion **128** is inserted.

(279) The moving rail **121** may be provided with an elastic protrusion bracket **126**, which is coupled to the moving rail **121** in order to form the elastic protrusion **128**. A mounting part **126a** may be provided at one side of the elastic protrusion bracket **126**, and an elastic protrusion support part **126b** may be provided at the other side of the elastic protrusion bracket **126**. Consequently, the elastic protrusion bracket **126** may be coupled to the moving rail **121** through the mounting part

126a.

(280) A cutout part **127** may be formed between the elastic protrusion support part **126b** and the mounting part **126a**. The elastic protrusion **128** may be bent from the elastic protrusion support part **126b**. Consequently, the elastic protrusion **128** may be elastically deformed through the cutout part **127** with respect to the elastic protrusion support part **126b**.

(281) Specifically, the elastic protrusion support part **126b** may be formed horizontally, and the elastic protrusion **128** may be bent vertically downward from the elastic protrusion support part **126b**. Consequently, the elastic protrusion **128** may be elastically deformed in a direction in which the angle between the elastic protrusion **128** and the elastic protrusion support part **126b** is decreased. That is, the elastic protrusion **128** may be elastically deformed toward the left and right middle parts of the drawer.

(282) Meanwhile, the rail **120** may include a fixed rail **122**. The fixed rail **122** is disposed under the moving rail **121** to support the moving rail **121** such that the moving rail **121** can slide.

(283) The rail **120** may be fixed to the sidewall of the storage compartment or the support cover **110** via rail brackets **123** and **124**. The rail brackets may include a front rail bracket **123** and a rear rail bracket **124**. That is, at least two support points may be formed at the front and rear of a single rail. The rail **120** is provided at each side of the drawer **30**.

(284) The front rail bracket **123** and the rear rail bracket **124** are spaced apart from each other by a predetermined distance in a forward and rearward direction. The transfer member **171** may be movably provided between the front rail bracket **123** and the rear rail bracket **124**. That is, a section of the transfer member **171** between the initial position and the ready position is positioned between the front rail bracket **123** and the rear rail bracket **124**. Consequently, interference between the transfer member **171** and the brackets **123** and **124** is prevented. This means that the slit **113** in the support cover **110** is formed between the front rail bracket and the rear rail bracket.

(285) Hereinafter, the coupling structure between the drawer **30** and the rail **120** will be described in more detail with reference to FIGS. **25** and **29**. FIG. **26** is an enlarged view of part “B” shown in FIG. **25** after the drawer **30** is coupled to the rail **120**.

(286) In order to couple the drawer frame **32** to the rail **120**, as shown in FIG. **25**, the user may move the drawer frame **32** rearward in a state in which the front of the drawer is higher than the rear of the drawer. That is, the user may move the drawer rearward in a state in which the drawer is inclined rearward. At this time, the rear end of the rail coupling part **37** is inserted and caught into the catching part **125**, which is provided at the rail.

(287) As shown in FIGS. **11** and **14**, the rail coupling part **37** may be mounted on the rail **120**, particularly the moving rail **121**, while surrounding the moving rail **121**. When the rail coupling part **37** may be mounted on the moving rail **121**, therefore, the leftward and rightward movement of the drawer is limited.

(288) Subsequently, in a state shown in FIG. **25**, the user may move the front of the drawer frame **32** downward. At this time, the elastic protrusion **128** is elastically deformed toward the left and right middle parts of the drawer. When the rail coupling part **37** is completely mounted on the moving rail **121**, the elastic protrusion **128** is elastically restored and is then inserted into the mounting hole **37a**. The mounting hole **37a** may be formed in a side flange **37b** configured to cover the outside surface of the moving rail. Consequently, the elastic protrusion **128** may be fixed in the mounting hole **37a**. As a result, the drawer frame **32** may be fixed to the moving rail **121** by the catching part **125** and the elastic protrusion **128**.

(289) Meanwhile, the drawer **30** may be separated from the rail **120** in the reverse order.

(290) The user may lift the front of the drawer **30** upward while pushing the elastic protrusion **128** on each side of the drawer **30**. At this time, the elastic protrusion **128** may escape from the mounting hole **37a**, with the result that the rail coupling part **37** provided at the front of the drawer **30** may be separated from the rail **120**. Subsequently, the user may lift the drawer **30** upward while pulling the drawer **30** forward. At this time, the rail coupling part **37** provided at the rear of the

drawer **30** may be separated from the catching part **125** of the rail **120**. Consequently, it is possible for the user to easily couple the drawer **30** to the rail **120** and, in addition, to easily separate the drawer **30** from the rail **120**.

(291) Meanwhile, in FIG. **25**, the drawer **30** is positioned at the initial position. As shown, the catching member **33** and the catching member mounting part **34a** are positioned between the front rail bracket **123** and the rear rail bracket **124** at the initial position of the drawer **30**. In particular, the catching member **33** and the catching member mounting part **34a** are positioned so as to be closer to the rear rail bracket **124**.

(292) When the drawer **30** moves to the ready position, the catching member **33** and the catching member mounting part **34a** move forward, with the result that the catching member **33** and the catching member mounting part **34a** may be closer to the front rail bracket **123**. Consequently, the catching member **33** and the catching member mounting part **34a** may always be positioned between the front rail bracket **123** and the rear rail bracket **124** within a section defined between the ready position and the initial position of the drawer **30**.

(293) Hereinafter, a sensor **40** for sensing a condition in which the electric driving unit **150** is operated will be described briefly. FIG. **2** shows an example of the sensor **40**.

(294) Specifically, the sensor **40** is configured to sense whether the door **20** is open. When the sensor **40** senses that the door **20** is open, the electric driving unit **150** is operated to move the drawer **30** from the initial position to the ready position. The electric driving unit **150** generates force for moving the drawer forward. That is, when the sensor **40** senses that the door **20** is open, the electric driving unit **150** may drive the drawer in one direction to withdraw the drawer forward.

(295) The door **20** may be a swing type door configured to rotate about a vertical shaft. That is, on the assumption that the opening angle of the door **20** when the door **20** completely closes the food introduction port **17** is 0 degrees, the door **20** may be rotated such that the opening angle of the door **20** exceeds 90 degrees.

(296) However, in a case in which the door **20** is even slightly separated from the food introduction port **17**, the door **20** may be considered to be open. For example, when the tight contact between the door **20** and the food introduction port **17** is released, the door **20** may be considered to be open. More specifically, as shown in FIG. **2**, when the tight contact between a gasket **22** provided at the rear of the door **20** and the cabinet **10** is released, the door **20** may be considered to be open. In this state, cool air may be lost. When the tight contact between the gasket **22** and the cabinet **10** is maintained, the door **20** may be considered to be closed.

(297) In order to sense whether the door **20** is in tight contact with the cabinet **10**, a door switch, which will be described hereinafter, may be provided. The door switch may be operably connected to a lighting device provided in the storage compartment. That is, upon determining that the tight contact between the door **20** and the cabinet **10** is released through the door switch, the lighting device may be controlled to be turned on. When it is determined through the door switch that the door **20** has come into tight contact with the cabinet **10** after the lighting device is turned on, the lighting device may be controlled to be turned off.

(298) In general, the door switch may be configured to have a structure that can be mechanically switched based on the distance between the door and the cabinet. The distance is very small. The reason for this is that the distance necessary to distinguish between the maintenance and release of tight contact between the cabinet and the door is very small. Consequently, the door switch may sense whether the door is open or closed substantially irrespective of the opening angle of the door.

(299) Meanwhile, the door switch may be configured so sense whether the door is open or closed based on the determination as to whether cool air leaks outside. On the other hand, the sensor **40** may be configured to sense whether the door is open or closed based on interference between the drawer and the door during the movement of the drawer and the door.

(300) The door switch and the lighting device will be described hereinafter in detail.

(301) In this embodiment, however, the opening of the door is based on the withdrawal of the

drawer, as previously described. That is, when the user opens the door **20** in order to withdraw and use the drawer, the drawer may move to the ready position. Consequently, the opening angle of the door at which it is sensed that the door is open is an important factor to consider. That is, the opening angle of the door required in order to automatically withdraw the drawer or the opening angle of the door at which automatic withdrawal of the drawer starts may be an important factor to consider.

(302) For example, when the user wishes to withdraw only goods received in the door storage region **21** provided at the rear of the door **20**, the door may be opened by 40 to 50 degrees. In this case, the drawer may remain at the initial position. That is, the drawer may remain in a state of not being withdrawn. The reason for this is that it is not necessary to withdraw the drawer when the drawer is not used since cool air may leak from the drawer. In addition, it is not necessary to withdraw the drawer when the drawer is not used since the drawer may collide with the rear of the door **20**.

(303) For example, when the door is opened by 40 or 50 degrees, the door switch may determine that the door has been opened. That is, in the case in which a door switch for determining only whether the door is open or closed is provided, a condition for automatic withdrawal of the drawer may be satisfied after the door switch senses that the door is open. Consequently, it is possible to realize more effective and stable control logic based on the relationship between the door switch and the sensor **40**, a description of which will follow. When the user wishes to withdraw and use the drawer, the user may know from experience that the door must be opened by 90 degrees or more. This is because when the opening angle of the door is less than 90 degrees, the drawer is caught by the door or the basket **25** provided at the rear of the door before the drawer is completely withdrawn. In order to completely withdraw the drawer without interference with the door, the door should typically be opened by 100 degrees or more.

(304) Consequently, the opening angle of the door at which the drawer is automatically moved may be 80 degrees or more, preferably about 90 degrees. In some cases, the opening angle of the door at which the drawer is automatically moved may be equal to or greater than 90 degrees. It may be sensed that the door is open at the above-mentioned opening angle of the door in order to drive the electric driving unit. This is because it takes a predetermined time for the drawer to move to the ready position. That is, the opening angle of the door at which it is sensed that the door is open may be less than the opening angle of the door at which the interference between the drawer and the door is completely eliminated. Of course, the opening angle of the door at which it is sensed that the door is open and the drawer starts to move automatically may be set such that the door does not interfere with the drawer at the ready position.

(305) Meanwhile, the opening angle of the door at which the drawer is automatically moved may be an angle at which the drawer does not interfere with the basket **25** provided at the rear surface of the door in a state in which the drawer is withdrawn to the ready position. As shown in FIG. 2, the basket **25** may protrude perpendicularly from the rear of the door. In a state in which the door is open by 90 degrees, therefore, the basket **25** may escape from the opening **17**. This means that the drawer **30** does not interfere with the basket **25** until the drawer is withdrawn to the opening **17**. Of course, in a state in which the drawer is further withdrawn in a state in which the door is open by 90 degrees, interference between the drawer **30** and the basket **25** may occur. In order to completely withdraw the drawer, therefore, it is necessary to increase the opening angle of the door, as previously described.

(306) For this reason, the sensor **40** may be a sensor that is capable of precisely sensing a predetermined opening angle of the door that is required in order to withdraw the drawer. To this end, the sensor **40** may include a magnet **42** and a reed switch **41**. Of course, the sensor may include only the reed switch **41**, or the reed switch **41** may sense the predetermined opening angle of the door using the magnet **42**. Other types of sensors or switches that are able to detect changes in the magnetic field may be used.

(307) As the distance between the sensor, such as the reed switch **41**, and the magnet **42** varies, the sensed magnetic field varies. For example, the intensity of the magnetic force applied to the reed switch **41** varies according to the opening angle of the door. The distance between the reed switch **41** and the magnet **42**, i.e. the distance at which the contact of the reed switch **41** is changed, may be precisely predicted by changing the magnetic force of the magnet **42**.

(308) Hereinafter, a sensor **40** that can be applied to the embodiments of the present invention will be described in detail with reference to FIGS. **27** and **28**. FIGS. **27** and **28** are enlarged sectional views showing the lower part of the door.

(309) In FIG. **27**, which shows an embodiment of the sensor **40**, the positional relationship between the reed switch **41** and the magnet **42** is shown. Specifically, in FIG. **27**, the door **20** is opened by 90 degrees. That is, the door **20** is rotated open by 90 degrees with respect to a hinge cover **45** positioned in the vicinity of a leg **2** fixed to the ground. FIG. **27** is a view showing the lower part of the door when viewed upward from the ground.

(310) The reed switch **41** may be provided at the hinge cover **45**, and the magnet **42** may be provided at the lower part of the door **20**. Specifically, the magnet **42** may be provided at a cap decoration part **24**. Of course, the reed switch **41** may be provided at the door **20**, and the magnet **42** may be provided at the hinge cover. However, a reed switch has a contact point and is configured to transmit a door opening signal or a door closing signal to the outside. Consequently, the reed switch may be positioned at a fixed member, such as the hinge cover **45**.

(311) Since the door **20** is rotated about a hinge shaft **23**, i.e. a rotational shaft of the door, the vertical distance between the reed switch **41** and the magnet **42** is uniform regardless of the opening angle of the door **20**. As the opening angle of the door **20** is changed, however, the horizontal distance between the reed switch **41** and the magnet **42** is changed. That is, the magnet **42** is rotated about the hinge shaft **23** at a consistent radius, and the horizontal distance between the reed switch **41** and the magnet **42** changes depending upon the rotational angle of the magnet.

(312) In a state in which the door **20** is closed, the magnet **42** is positioned in the vicinity of the reed switch **41**. In a state in which the door **20** is closed, therefore, the magnetic force of the magnet **42** may affect the reed switch **41**. As the opening angle of the door is increased, the magnet **42** approaches the reed switch **41** and then moves away from the reed switch **41**. That is, as the opening angle of the door is further increased, the magnetic force of the magnet **42** does not affect the reed switch **41**. For example, when the opening angle of the door reaches 90 degrees, therefore, the contact point of the reed switch **41** may be changed. As the result of the change of the contact point of the reed switch **41**, a door opening on signal, which is a drawer withdrawal condition, may be generated. That is, when the door is opened by a predetermined angle, it may be sensed that the door has been opened. In other words, in a section between a state in which the door **20** is closed and a state in which the opening angle of the door **20** is less than a predetermined angle (for example, 90 degrees), the magnetic force of the magnet **42** affects the reed switch **41**, with the result that the contact point of the reed switch **41** is maintained. When the opening angle of the door **20** reaches the predetermined angle, the magnetic force of the magnet **42** does not affect the reed switch **41**, with the result that the contact point of the reed switch **41** is changed.

(313) In other words, when the opening angle of the door **20** reaches the predetermined angle, the reed switch has a critical point of effective magnetic intensity for contact point switching. That is, when the reed switch reaches the critical point, the contact point of the reed switch **41** is changed, which means that it is sensed that the door has been opened.

(314) Consequently, it is possible to generate a door opening on signal, which is a drawer withdrawal condition, at a desired door opening angle (a predetermined door opening angle) by the provision of a sensor that is capable of sensing whether the door is open using the reed switch **41** and the magnet **42**.

(315) As previously described, when the door is opened and the opening angle of the door **20** reaches the predetermined degrees, the magnetic force of the magnet **42** does not affect the reed

switch **41**. That is, the reed switch escapes from the critical point. This means that when the door starts to be closed, in a state in which the door is open at an angle greater than the predetermined angle, and then the opening angle of the door reaches the predetermined angle, the magnetic force of the magnet **42** affects the reed switch **41**. Consequently, the door opening angle, at which it is sensed that the door is open, and the door closing angle, at which it is sensed that the door is closed, may be set to be substantially equal. In addition, it is possible for a single sensor **40** to sense whether the door is open or closed based on substantially the same angle.

(316) In other words, it is possible for the sensor **40** to sense whether the door is open or whether the door is closed based on the same angle. For example, after it is sensed that the door is open at a particular angle, the subsequent sensing at the same angle may be sensing that the door is closed. In addition, after it is sensed that the door is closed at a particular angle, the subsequent sensing at the same angle may be sensing that the door is open. That is, it is possible to sense whether the door is open or closed using a single sensor **40**. Alternatively, a sensor for sensing whether the door is open and a sensor for sensing whether the door is closed may be separately provided in place of a single sensor. In this case, the sensors may be of the same type. This is because it is necessary to eliminate the interference between the door and the drawer when the door is closed as well as when the door is opened, as previously described. That is, as will be described hereinafter, it is necessary to automatically insert the drawer before the door interferes with the drawer even when the drawer is automatically inserted.

(317) In a case in which a single magnet **42** is used, as shown in FIG. **27**, however, there may be tolerance in the door opening angle preset for individual products. For example, in a case in which a single magnet **42** and a single reed switch **41** are mounted in a state in which a door opening angle is set to 90 degrees, the door opening angle for individual products may have tolerance. That is, it may be sensed that the door is open when the door opening angle is 85 degrees for some products, it may be sensed that the door is open when the door opening angle is 90 degrees for some products, and it may be sensed that the door is open when the door opening angle is 95 degrees for some products. As a result, the door opening angle for individual products may vary. This variation in the door opening angle may result from variation in magnetic force of the magnet **42**, variation in mounting of the magnet **42** and the reed switch **41**, etc.

(318) In addition, in a case in which a single magnet **42** is used, it is not easy to change the door opening angle. This is because the door opening angle may be 90 degrees for some models while the door opening angle may be 85 degrees for some models.

(319) Consequently, it is necessary to provide a sensor **40** that is capable of flexibly changing the door opening angle while reducing the variation in a predetermined door opening angle. In order to solve a problem caused by a sensor **40** including a single magnet **42**, the present invention provides a sensor using a plurality of magnets.

(320) Hereinafter, another embodiment of the sensor **40** will be described in detail with reference to FIG. **28**. Basically, this embodiment is very similar to the previous embodiment. In this embodiment, however, a plurality of magnets **42** may be provided

(321) Even in this embodiment, the reed switch **41** may be provided at the hinge cover **45**, and the magnets **42** may be provided at the cap decoration part **24** of the door **20**. The cap decoration part **24** may be provided to define the lower surface of the door **20**. Consequently, the magnets **42** are positioned in the door **20**.

(322) The reed switch **41** may be fixedly provided at the cabinet **10**. For example, the reed switch **41** may be provided at the hinge cover **45**, which protrudes forward from the cabinet **10**. The door **20**, particularly the cap decoration part **24**, is rotated open and closed about the hinge shaft **23** while having a predetermined vertical gap at the upper part of the hinge cover **45**.

(323) As an example, the magnets **42** may include a horizontal magnet **42a** and a vertical magnet **42b**. The horizontal magnet **42a** may be a magnet that is provided parallel to the front surface of the door **20** or the front surface of the cabinet **10**, and the vertical magnet **42b** may be a magnet that

is substantially perpendicular to the horizontal magnet **42a**. Of course, the vertical magnet **42b** may be positioned at an obtuse angle relative to the horizontal magnet **42a** in order to form an arc shape together with the horizontal magnet **42a**.

(324) In a state in which the door **20** is closed, the horizontal magnet **42a** may be positioned so as to be parallel to one surface of the reed switch **41**. The reed switch **41** may be formed in a quadrangular shape. The horizontal magnet **42a** may be positioned so as to be parallel to the horizontal side **41a** of the reed switch **41**. The vertical magnet **42b** may be positioned so as to be parallel to the vertical side **41b** of the reed switch **41**. The horizontal side of the reed switch **41** may be larger than the vertical side of the reed switch **41**.

(325) The horizontal magnet **42a** and the vertical magnet **42b** are bar-type magnets. As shown in the vertical sectional view of FIG. **28**, the length of the horizontal magnet **42a** and the vertical magnet **42b** may be greater than the height of the horizontal magnet **42a** and the vertical magnet **42b**.

(326) In a state in which the door **20** is closed, the horizontal magnet **42a** may be positioned while extending leftward and rightward at the rear of the reed switch **41**. The vertical magnet **42b** may be positioned while extending forward and rearward at the left side or the right side of the reed switch **41**. That is, the two magnets **42a** and **42b** may be positioned so as to surround the reed switch **41** while being spaced apart from each other. Consequently, the horizontal magnet **42a** and the vertical magnet **42b** may be provided in a state in which the horizontal magnet **42a** and the vertical magnet **42b** have different horizontal angles with respect to the reed switch **41**.

(327) In a state in which the door **20** is closed, therefore, the two magnets **42a** and **42b** simultaneously provide effective magnetic forces to the reed switch **41**.

(328) In addition, the distance between the horizontal magnet **42a** and the hinge shaft **23** may be less than the distance between the vertical magnet **42b** and the hinge shaft **23**. That is, the turning radius of the horizontal magnet **42a** is less than the turning radius of the vertical magnet **42b**. In addition, the distance between the reed switch **41** and the hinge shaft **23** may be set to be approximate to the turning radius of the horizontal magnet **42a**.

(329) As the door **20** is opened, therefore, the horizontal magnet **42a** is turned toward the reed switch **41**, and the vertical magnet **42b** is turned away from the reed switch **41**. This means that the overlapping area between the horizontal magnet **42a** and the reed switch **41** is greater than the overlapping area between the vertical magnet **42b** and the reed switch **41**.

(330) Consequently, the magnetic force generated by the horizontal magnet **42a** is basically sensed by the reed switch **41**, and the magnetic force generated by the vertical magnet **42b** auxiliarily affects the reed switch **41**.

(331) Since the intensity of the magnetic force is inversely proportional to the square of the distance, the intensity of the magnetic force is very rapidly reduced as the magnet **42a** moves away from the reed switch **41**. This means that the intensity of the magnetic force may be greatly changed even if the magnet moves a very short distance. For this reason, it is very difficult to precisely set a critical point at which the magnetic force has no effect, i.e. a door opening angle, at a position at which the magnetic force has an effect. In other words, in a case in which only one magnet, for example only the horizontal magnet **42a**, is provided, it is difficult to set the critical point, since the intensity of the magnetic force is rapidly changed before and behind the critical point.

(332) In this embodiment, the intensity of the magnetic force may be gently changed by the vertical magnet **42b** until the critical point is reached, and the intensity of the magnetic force may be sharply changed by the vertical magnet **42b** after the critical point is exceeded. That is, the vertical magnet **42b** continuously provides an auxiliary magnetic force until the critical point is reached, whereby a rapid change in the magnetic force may be limited until the critical point is reached.

(333) Meanwhile, in this embodiment, it is possible to easily set the critical point, i.e. the door opening angle, to about 90 degrees. That is, it is possible to easily adjust the door opening angle

that is sensed by the reed switch **41**.

(334) Specifically, the horizontal magnet **42a**, shown in FIG. **28**, may be moved forward and rearward in a state in which the vertical magnet **42b** is fixed. That is, the vertical magnet **42b** may be a fixed magnet, and the horizontal magnet **42a** may be a moving magnet.

(335) When the horizontal magnet **42a** is moved rearward, the distance between the horizontal magnet **42a** and the hinge shaft **23** is decreased. In other words, the distance between the horizontal magnet **42a** and the reed switch **41** is decreased in a state shown in FIG. **4** (i.e. in a state in which the door is open by 90 degrees). In order to prevent the application of the magnetic force of the horizontal magnet **42a** to the reed switch **41**, therefore, it is necessary to open the door **20** further. That is, the horizontal magnet **42a** may be horizontally moved so as to be close to the hinge shaft **23** such that the door opening angle exceeds 90 degrees.

(336) On the other hand, the horizontal magnet **42a** may be moved forward. That is, the horizontal magnet **42a** may be horizontally moved so as to be distant from the hinge shaft **23**. In this case, the distance between the horizontal magnet **42a** and the reed switch **41** may be increased. In a state in which the door opening angle is less than 90 degrees, the effect of the horizontal magnet **42a** may be eliminated.

(337) Eventually, it is possible to flexibly set the door opening angle using the fixed magnet **42b**, which is an auxiliary magnet, and the moving magnet **42a**, which is a main magnet.

(338) Meanwhile, in the above embodiment of the sensor **40**, the reed switch is provided under the magnet. Alternatively, the reed switch may be provided above the magnet. For example, the magnet may be provided at a door decoration part, which defines the upper surface of the door **20**, and the reed switch may be provided so as to face the door decoration part. In any case, the vertical distance between the magnet and the reed switch may be fixed regardless of the door opening angle, and the horizontal distance between the magnet and the reed switch may be changed as the door opening angle is changed.

(339) In this embodiment, it is possible to simply and precisely sense whether the door is open or closed at the predetermined door opening angle using the reed switch and the magnet, which are very simple. In addition, the door opening angle may be set differently for individual refrigerator models. In this case, the position of one of the magnets may be changed.

(340) Meanwhile, as to the sensor **40**, a hall sensor may be used in place of the reed switch.

(341) The magnet may be provided above or under the hall sensor such that the magnet and the hall sensor are arranged vertically. For example, at the position at which the door is open by 90 degrees, the magnet may be positioned vertically above or below the hall sensor. When the door opening angle is increased to 90 degrees, therefore, the hall sensor recognizes the magnet and thus senses that the door is open.

(342) For example, the magnet may always be positioned vertically above or under the hall sensor until the door is open by 90 degrees. That is, the magnet may be formed in an arc shape, or a plurality of magnets may be provided such that the hall sensor can always recognize the magnet(s) until the door is open by 90 degrees. When the door is open by 90 degrees, the magnet may escape from a region that is sensed by the hall sensor, with the result that the open state of the door may be sensed.

(343) Hereinafter, control construction elements that can be applied to an embodiment of the present invention will be described in detail with reference to FIG. **29**.

(344) A refrigerator according to an embodiment of the present invention includes a main controller **300**. The main controller **300** may control the basic operation of the refrigerator.

(345) The refrigerator according to an embodiment of the present invention may further include a motor assembly **160**. The motor assembly **160** may include a motor **162** and a motor controller **165**. The motor **162** may be driven in forward and reverse directions. For example, the motor **162** may be driven in the forward direction (the clockwise direction) to move the transfer member **171** forward. On the other hand, the motor **162** may be driven in the reverse direction to move the

transfer member **171** rearward. The electric driving unit, i.e. the motor, may be driven in the forward direction to generate force necessary to move the drawer forward. The force by which the drawer is moved forward may be released when the motor is driven in the reverse direction.

(346) The direction in which the motor **162** is driven, the duty ratio applied to the motor **162**, and the driving and stoppage of the motor **162** may be controlled by the motor controller **165**.

(347) The motor assembly **160** may include a connection member **163**, which is configured to move forward and rearward as previously described. The maximum protruding length of the connection member **163** corresponds to the ready position of the transfer member, and the minimum protruding length of the connection member **163** corresponds to the initial position of the transfer member. Consequently, the connection member **163** moves between the maximum protruding length and the minimum protruding length.

(348) Therefore, it is possible to determine whether the connection member **163** of the motor assembly **160** is at a position corresponding to the initial position of the drawer or at a position corresponding to the ready position of the drawer. That is, the motor assembly **160** may be provided with two hall sensors **166** and **167**.

(349) As shown in FIGS. **6** and **7**, the motor assembly **160** may be provided with a magnet **168**. The magnet **168** may be configured so as to move in the housing **161** when the connection member **163** moves. When the first hall sensor **166** recognizes the magnet **168**, therefore, it is determined that the transfer member is at the initial position. On the other hand, when the second hall sensor **167** recognizes the magnet **168**, it is determined that the transfer member is at the ready position.

(350) It is possible to determine whether the motor assembly **160** is operating normally using the hall sensors **166** and **167** and the magnet **168**, which will be described hereinafter in detail when describing the control method of the refrigerator.

(351) When it is sensed that the door is open through the sensor **40**, the motor controller **165** operates the motor **162**. That is, the motor controller **165** drives the electric driving unit. Here, the sensor **40** may be a sensor for sensing that the door is open at a predetermined opening angle (for example, 90 degrees). That is, for example, when the sensor **40** senses that the door is open by 90 degrees, the motor controller **165** operates the motor **162** such that the drawer is withdrawn from the initial position to the ready position. Consequently, the motor controller **165** controls the driving of the electric driving unit to withdraw the drawer.

(352) When the sensor **40** senses that the door is open, which is a drawer withdrawal condition, a door opening signal or a signal for driving the motor assembly **160** in the forward direction may be generated and transmitted to the motor controller **165** via the main controller **300**. Of course, the door opening signal or the signal for driving the motor assembly **160** in the forward direction may be directly transmitted to the motor controller **165**. Meanwhile, the refrigerator according to the embodiment of the present invention may further include a door switch **50**. The door switch **50** may be an element that is generally used in refrigerators. When the door switch **50** senses that the door is open, a lighting device **60**, configured to illuminate the storage compartment, may be operated. The door switch **50** may be provided separately from the sensor **40**.

(353) Hereinafter, a control method that can be applied to an embodiment of the present invention will be described in detail with reference to FIGS. **30** to **36**.

(354) First, an initial step (S**10**) will be described in detail with reference to FIG. **30**. The initial step may be a step at which the motor **162** is initially driven when the refrigerator is powered on. That is, the initial step may be a step at which the motor **162** is initially driven to move the transfer member to the initial position such that an electric drawer system is initialized.

(355) When the initial step (S**10**) is started, therefore, the motor is operated (S**30**). That is, the motor is operated to return the transfer member **171** to the initial position. At this time, the motor may be driven, for example, in the counterclockwise direction or in the reverse direction. In the following description, the transfer member **171** moves rearward when the motor is driven in the counterclockwise direction, and the transfer member **171** moves forward when the motor is driven

in the clockwise direction.

(356) The operation of the motor may be controlled using a predetermined duty ratio. Upon determining through the first hall sensor **166** that the transfer member **171** has returned to the initial position, the operation of the motor is stopped. For example, when an ON signal is generated by the first hall sensor **166**, the operation of the motor is stopped (**S40**).

(357) In addition, the operation of the motor may be controlled to be stopped when the operation time of the motor exceeds a predetermined time. For example, the predetermined time may be 5 seconds. Here, the predetermined time may be set so as to be greater than the allowable maximum motor operation time in consideration of a margin. Consequently, the operation of the motor may be performed until an ON signal is generated by the first hall sensor **166** before the predetermined time is reached.

(358) Consequently, when the operation time of the motor is equal to or greater than the predetermined time or when it is sensed that an ON signal has been generated by the second hall sensor **167** (**S50**) after the motor is stopped (**S40**), it may be determined that an error has occurred in the electric drawer system. When the error has occurred, therefore, a notification step (**S100**) may be performed. That is, a step of displaying the error may be performed.

(359) The ON signal may be generated by the second hall sensor **167** at the ready position of the transfer member **171**, rather than at the initial position of the transfer member **171**. When the ON signal is generated by the second hall sensor **167** at the initial step (**S10**), therefore, it may be determined that an error has occurred in the entire electric drawer system including the hall sensors.

(360) In addition, even when the motor is operated for a predetermined time or more, it may be determined that an error has occurred in the entire electric drawer system, as previously described. This is because when the transfer member returns to the initial position, the load of the drawer is not applied to the motor **162**.

(361) Meanwhile, when an error has occurred, the notification step (**S100**) may be performed, as shown in FIG. **31**. At the notification step (**S100**), it is determined again whether the error has occurred (**S110**). At this time, the drawer in which the error has occurred may be determined. That is, in a case in which electric drawers are mounted in the left and right freezing compartments, it may be determined whether the error has occurred in the left drawer or the right drawer. In a case in which the error has occurred in the electric drawer, an error display step (**S120**) may be performed. At this time, an error code may be displayed on a display unit.

(362) After the error is displayed or if it is determined that no error has occurred, the state of the refrigerator is switched to a ready state (**S200**). The ready state (**S200**) may be a state in which the driving of the motor is stopped.

(363) As shown in FIG. **32**, determination steps (**S210** and **S220**) of determining a condition for automatically withdrawing the drawer in the ready state (**S200**) may be performed. The condition for automatically withdrawing the drawer may be sensing whether the door is open through the sensor **40**. Specifically, when the reed switch **41** senses that the door is open (**S210**), it may be determined that the above condition has been satisfied.

(364) Consequently, the motor **162** may remain in the ready state (**S200**) until it is sensed that the door is open through the sensor **40**.

(365) Meanwhile, as previously described, sensing whether the door is open through the sensor must be performed based on sensing whether the door is open through the door switch **50**. When it is sensed that the door is open through the door switch **50**, therefore, a withdrawal step (**S300**) may be performed.

(366) When the sensor **40** senses that the door is open but the door switch **50** does not sense that the door is open, it is determined that an error has occurred in the sensor **40** (**S230**), and the error may be displayed on the display unit. Subsequently, the state of the refrigerator may be switched to the ready state (**S200**).

(367) At the withdrawal step (**S300**), the motor **162** may be operated to push the drawer **30**

forward. At the withdrawal step, therefore, a relatively high load is applied to the motor **162**. In particular, in a case in which a large amount of goods is stored in the drawer **30**, a higher load is applied to the motor **162**. At the withdrawal step (S300), therefore, the motor may be controlled to generate high output based on the load. That is, control may be performed so as to increase the duty ratio as the load is increased.

(368) Specifically, the withdrawal step (S300) may include a step (S310) of calculating a signal (frequency generator; FG), which is generated when the motor is rotated. FG may be calculated at predetermined time intervals, e.g. every 100 ms.

(369) In addition, the withdrawal step (S300) may further include a motor withdrawal driving step (S320) of driving the motor in the clockwise direction while changing the duty ratio based on the calculated FG. That is, a plurality of load conditions may be set based on the calculated FG, and the motor may be driven using a predetermined duty ratio in each load condition.

(370) For example, in a case in which the calculated FG is 0 to 50, the motor may be driven at a duty ratio of 180. In addition, in a case in which the calculated FG is 51 to 100, the motor may be driven at a duty ratio of 200. As the calculated FG is increased, the duty ratio may also be increased. For example, in a case in which the calculated FG is 251 or more, the motor may be driven at a duty ratio of 250.

(371) In other words, in a case in which the calculated FG is greater, it is determined that the load is higher, with the result that the output of the motor is increased.

(372) The motor withdrawal driving step (S320) may be performed until an ON signal is generated by the second hall sensor **167**. In addition, the motor withdrawal driving step (S320) may be performed until a predetermined time is reached. For example, the predetermined time may be 3 seconds.

(373) Meanwhile, an obstacle that disturbs the withdrawal of the drawer may arise while the motor withdrawal driving step (S320) is being performed. That is, a very heavy object may be placed in front of the drawer, or foreign matter may be introduced into the rail **120** such that the moving rail **121** cannot move. In this case, a high load may be applied to the motor if the motor is continuously driven in the clockwise direction. That is, the motor may be broken or damaged. When an obstacle is present, therefore, the motor withdrawal driving step (S320) may be stopped, and a returning step (S500) may be performed.

(374) Of course, an obstacle determination step (S350) of determining whether an obstacle is present may be performed before the returning step (S500) is performed.

(375) The obstacle determination step (S350) may be performed during the motor withdrawal driving step. The obstacle determination step (S350) may be performed when the FC calculation cycle reaches a predetermined number of times. For example, the obstacle determination step (S350) may be performed when a fourth FG calculation is performed. That is, the first three FG calculations may not be considered in the determination of the obstacle. The reason for this is that a relatively high load may be applied to the motor at the motor withdrawal driving step due to an initial static frictional force. Consequently, the obstacle determination step (S350) may be performed after execution a predetermined number of times. The obstacle determination step (S350) may be performed until the motor withdrawal driving step (S320) is finished.

(376) In a case in which the calculated FG is greater than a predetermined obstacle FG, it is determined at the obstacle determination step (S350) that there is no obstacle, and the motor withdrawal driving step may be continuously performed. On the other hand, in a case in which the calculated FG is equal to or less than the predetermined obstacle FG, it is determined at the obstacle determination step (S350) that an obstacle is present. That is, it may be determined that an overload is generated in the motor due to an obstacle, whereby the motor is not driven normally. When it is determined that there is an obstacle, therefore, the driving of the motor is stopped, and the returning step (S500) is performed.

(377) Meanwhile, when it is determined that there is no obstacle and the motor withdrawal driving

step is finished, an error determination step (S340) may be performed. It may be determined whether a predetermined time, for example 3 seconds or more, elapses before the motor withdrawal driving step is finished. The predetermined time may be the allowable maximum time. In a case in which 3 seconds or more elapse, it may be determined that an error has occurred. In addition, in a case in which an ON signal is generated by the first hall sensor **166**, it may be determined that an error has occurred. When the motor withdrawal driving step (S320) is finished, i.e. the withdrawal step (S300) is finished, a stopping step (S400) may be performed.

(378) As shown in FIG. **34**, the stopping step (S400) is not a step of stopping the driving of the motor but a step of stopping the withdrawal of the drawer. In other words, the motor withdrawal driving step (S320) is a step of moving the transfer member forward. When it is determined that the transfer member has moved to the ready position through the second hall sensor **167**, the forward movement of the transfer member is stopped. That is, the stopping step (S400) is a step of stopping the forward movement of the transfer member.

(379) As previously described, the motor is driven in the clockwise direction (i.e. the forward direction) at the withdrawal step (S300). In the same manner, the motor may remain driven in the clockwise direction even at the stopping step (S400). Since the drawer is not pushed at the stopping step (S400), however, the driving of the motor may be maintained at the minimum output.

(380) At the stopping step (S400), the motor may be controlled to be driven (S420) at the minimum duty ratio that can be applied to the motor. That is, in the driving of the motor connected to the automatic withdrawal of the drawer, the motor may be controlled to be driven at the smallest duty ratio that can be applied to the motor. This is performed in order to prevent the drawer from being automatically inserted before the user manipulates the drawer due to the elastic force of the automatic closing unit (i.e. the elastic device) at the stopping step (S400).

(381) Meanwhile, the stopping step (S400) may be performed for a predetermined time. For example, the stopping step may be performed for 10 seconds (S430). After the predetermined time, the returning step (S500) may be performed.

(382) When the user further withdraws the drawer within the predetermined time, the connection between the elastic device and the drawer is released. The connection between the elastic device and the drawers that have not been further withdrawn is maintained. After the predetermined time, therefore, the drawers that have not been further withdrawn may be returned rearward by the elastic device. At this time, the return speed of the drawers is not faster than the return speed of the transfer member. The reason for this is that when the drawers are returned by the elastic device, the drawers follow the transfer member, which is returned.

(383) The characteristics in which the drawers can be individually further withdrawn and automatically returned lead to convenience in using the drawers and reduced loss of cool air. The reason for this is that the opening region of the drawer that is used may be further extended by these characteristics. In addition, the drawers that are not used may be inserted to the initial position such that the loss of cool air is further reduced than at the ready position.

(384) Meanwhile, at the stopping step (S400), the user may push the drawer **30** in a direction in which the drawer **30** is inserted. In this case, the returning step (S500) may be performed even during the stopping step. To this end, an FG calculation step (S410) may be performed even during the stopping step (S400). In addition, a step (S440) of determining whether the user has pushed the drawer in the direction in which the drawer is inserted based on the calculated FG may be performed. For example, in a case in which the calculated FG is less than a predetermined FG, it may be determined that the user has pushed the drawer.

(385) Even in this case, the determination step (S440) may be performed after a predetermined number of FG calculations.

(386) As previously described, the stopping step (S400) may be performed normally, for example for 10 seconds. Subsequently, the motor may be driven to return the transfer member **171** to the initial position, which is the returning step (S500). At the returning step (S500), the force applied to

the drawer is removed. At the returning step (S500), therefore, the drawer may return automatically due to the elastic restoring force of the automatic closing unit.

(387) As shown in FIG. 35, the returning step (S500) includes a step (S510) of stopping the driving of the motor. That is, a step of temporarily stopping the driving of the motor in order to change the direction in which the motor is driven may be performed. The motor may be a brushless direct current (BLDC) motor. At this time, a brake input may be performed in order to stop the driving of the motor. The step (S510) may be performed, for example, for 10 ms. The step (S510) may be performed in order to prevent impact from being applied to the motor due to the abrupt change in rotational direction of the motor.

(388) Meanwhile, the returning step (S500) may be performed in a manner similar to the initial step (S10).

(389) That is, when an ON signal is generated by the first hall sensor 166 or the motor is driven in the counterclockwise direction for a predetermined time (S530), the driving of the motor may be stopped (S540). Here, the predetermined time may be the allowable maximum time, for example 5 seconds. When the motor is driven for 5 seconds, it may be determined that an error has occurred in the motor (S550). Of course, even when an ON signal is generated by the second hall sensor 167, it may be determined that an error has occurred in the motor (S550). In this case, the notification step (S100) may be performed.

(390) When the driving of the motor is stopped (S540), the ready state (S200) is performed.

(391) The duty ratio at the returning step (S500) may be higher than the duty ratio at the initial step (S10). In a case in which the FG sensed at the withdrawal step, i.e. the total FG, is less than a predetermined FG, however, the motor may be driven at the same duty ratio as at the initial step (S10). This is because in this case, the motor is driven in the forward direction or in the reverse direction in a state in which the obstacle is substantially eliminated. Consequently, the transfer member may smoothly return even when a relatively low duty ratio is applied.

(392) Meanwhile, when the door is abruptly closed, the drawer at the ready position may collide with the basket of the door. For this reason, a step (S700) of urgently returning the drawer may be performed. For example, when the door is abruptly closed during the withdrawal of the drawer, the drawer may collide with the door. In addition, when the door is abruptly closed in the state in which the drawer is withdrawn, the drawer may collide with the door.

(393) In this case, therefore, control logic for urgently returning the drawer may be performed.

(394) That is, the step (S700) of urgently returning the drawer may be performed during the withdrawal step (S300) and the stopping step (S400). Of course, it is not necessary to perform the urgent returning step in a case in which the drawer is positioned at the initial position.

Consequently, a step (S600) of determining whether an urgent returning condition is satisfied may be performed first. When the urgent returning condition is satisfied, the urgent returning step (S700) may be performed.

(395) As previously described, the urgent returning condition may be satisfied during the withdrawal step or the stopping step. Specifically, the urgent returning condition may be satisfied when the sensor 40 senses that the door is being closed. That is, the urgent returning condition may be satisfied when the reed switch 41 senses that the door is being closed.

(396) The urgent returning step may be performed in the same manner as the returning step. That is, the urgent returning step may include a step (S720) of driving the motor in the counterclockwise direction (i.e. the reverse direction), a step (S740) of stopping the motor, and an error determination step (S750). Of course, when it is determined that an error has occurred, the notification step (S100) may be performed.

(397) Meanwhile, the urgent returning step may be started in a state in which the motor is driven in the clockwise direction. In the same manner, therefore, the driving of the motor may be stopped for a predetermined time, for example 10 ms, and then the motor may be driven in the counterclockwise direction.

(398) It is possible to minimize the load applied to the motor using the above-described control method. In addition, in a case in which a high load is applied to the motor due to an obstacle, the returning step or the urgent returning step may be performed in order to prevent the high load from being continuously applied to the motor.

(399) The following list of embodiment is provided to further illustrate some particular embodiments according to the present disclosure. This list is provided for illustrative purposes only and not meant to limit the disclosure to the embodiments contained in this list.

(400) The following is the first list of embodiments.

(401) 1. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a sensor configured to sense that the door is open when the door is turned by a predetermined angle; an electric driving unit configured to withdraw the drawer forward when it is sensed that the door is open; and a rail configured to allow the drawer to move forward and rearward relative to the storage compartment.

(402) 2-1. The refrigerator according to the first embodiment, wherein the door and the drawer are provided separately, whereby a force applied to open the door is independent of withdrawal of the drawer.

(403) 2-2. The refrigerator according to the first embodiment, wherein the electric driving unit is configured to automatically withdraw the drawer to a ready position spaced apart forward from an initial position of the drawer by a predetermined distance.

(404) 2-3. The refrigerator according to any one of the first to 2-2 embodiments, wherein the electric driving unit comprises a motor assembly and a moving frame, and the moving frame is configured to be movable forward and rearward between the initial position and the ready position by driving of the motor assembly.

(405) 3. The refrigerator according to the 2-3 embodiment, wherein the moving frame is connected with the drawer so as to apply a force to the drawer in a direction in which the drawer is withdrawn from the initial position to the ready position.

(406) 4. The refrigerator according to the third embodiment, wherein the moving frame is disconnected from the drawer so as not to apply a force to the drawer in a direction in which the drawer is inserted from the ready position to the initial position.

(407) 5. The refrigerator according to the 2-3 embodiment, wherein the electric driving unit further comprises a connection member for interconnecting the motor assembly and the moving frame, and a distance between the moving frame and the motor assembly is changed in proportion to a distance by which the connection member is withdrawn from the motor assembly.

(408) 6. The refrigerator according to the 2-3 embodiment, further comprising a sidewall defining the storage compartment and a support cover coupled to the sidewall for movably supporting the drawer along the rail.

(409) 7. The refrigerator according to the sixth embodiment, wherein a predetermined space is defined between the sidewall and the support cover, and the motor assembly is mounted to an inside surface of the support cover such that the motor assembly is positioned in the predetermined space.

(410) 8. The refrigerator according to the seventh embodiment, wherein the rail is mounted to an outside surface of the support cover.

(411) 9. The refrigerator according to the seventh embodiment, wherein the support cover is provided at an upper part and a lower part of the inside surface thereof with guide bars, and the moving frame is supported so as to be movable forward and rearward between the upper guide bar and the lower guide bar.

(412) 10. The refrigerator according to the seventh embodiment, wherein a slit is formed through the support cover, and the moving frame is provided with a transfer member, the transfer member extending through the slit so as to be connected with the drawer.

(413) 11. The refrigerator according to the tenth embodiment, wherein the drawer is provided with a first catching member, the first catching member being formed at a front of the transfer member so as to correspond to the transfer member, and, when the transfer member moves forward, the first catching member is pushed forward such that the drawer is moved in a direction in which the drawer is withdrawn.

(414) 12. The refrigerator according to the 2-3 embodiment, wherein the drawer comprises a basket and a drawer frame provided outside the basket.

(415) 13. The refrigerator according to the twelfth embodiment, wherein the moving frame is selectively connected with the drawer frame, and when the moving frame is connected with the drawer frame, movement of the moving frame is converted into movement of the drawer.

(416) 14. The refrigerator according to the thirteenth embodiment, wherein the drawer comprises a plurality of vertically arranged drawers, and the moving frame is selectively connected with the drawer frame of each of the drawers.

(417) 15. The refrigerator according to the fourteenth embodiment, wherein the rail is configured to allow the drawer to move forward and rearward between a maximum withdrawal position, which is spaced apart forward from the ready position by a predetermined distance, and the initial position.

(418) 16. The refrigerator according to the fifteenth embodiment, wherein a connection between the moving frame and the drawer frame is released from the ready position to the maximum withdrawal position such that the drawer is manually withdrawn.

(419) 17. The refrigerator according to the sixteenth embodiment, wherein the motor assembly is continuously driven in a state in which the door remains open such that the moving frame remains at the ready position.

(420) 18. The refrigerator according to the seventeenth embodiment, wherein the drawer is manually inserted from the maximum withdrawal position to the ready position, at which the drawer frame is coupled again to the moving frame.

(421) 19. The refrigerator according to the eighteenth embodiment, wherein, when it is sensed that the door is closed, the motor assembly is operated to return the moving frame from the ready position to the initial position.

(422) 20. The refrigerator according to the nineteenth embodiment, wherein, when it is sensed through the sensor that the door has been opened by a predetermined angle or more in a state in which the door is closed, it is determined that the door is open, and, when it is sensed through the sensor that the door has been closed by a predetermined angle or less in a state in which the door is open, it is determined that the door is closed.

(423) 21. The refrigerator according to any one of the 2-3 to twentieth embodiments, further comprising an elastic device configured to be elastically deformed when the drawer moves from the initial position to the ready position and configured to provide an elastic restoring force to the drawer when the drawer moves from the ready position to the initial position.

(424) 22. The refrigerator according to the twenty-first embodiment, wherein the elastic device comprises: a housing having a slot formed therein in a longitudinal direction and a spring mounted therein; and a hanging member configured to move along the slot to elastically deform and elastically restore the spring, the hanging member being selectively connected with the drawer.

(425) 23. The refrigerator according to the twenty-second embodiment, wherein the drawer is provided with a second catching member configured to be selectively connected with the hanging member, and the hanging member and the second catching member are connected with each other at the ready position such that the drawer returns to the initial position due to an elastic restoring force of the spring.

(426) 24. The refrigerator according to the twenty-third embodiment, wherein the slot is provided at a front end thereof with a first inclined slot for limiting movement of the hanging member and releasing a connection between the hanging member and the second catching member in a state in which the hanging member maximally elastically deforms the spring.

- (427) 25. The refrigerator according to the twenty-fourth embodiment, wherein the hanging member moves into the first inclined slot at a predetermined position between the ready position and the maximum withdrawal position of the drawer such that the connection between the hanging member and the second catching member is released.
- (428) 26. The refrigerator according to the twenty-fifth embodiment, wherein, as the drawer returns to the ready position after being withdrawn, the hanging member escapes from the first inclined slot and is connected with the second catching member.
- (429) 27. The refrigerator according to the twenty-sixth embodiment, wherein the slot is provided at a rear end thereof with a second inclined slot for limiting the movement of the hanging member and releasing the connection between the hanging member and the second catching member in a state in which the hanging member maximally elastically restores the spring.
- (430) 28. The refrigerator according to the twenty-seventh embodiment, wherein the hanging member moves into the second inclined slot at a predetermined position between the ready position and the initial position of the drawer such that the connection between the hanging member and the second catching member is released.
- (431) 29. The refrigerator according to the twenty-eighth embodiment, wherein, as the drawer is withdrawn from the initial position to the ready position, the hanging member escapes from the second inclined slot and is connected with the second catching member.
- (432) 30. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a sensor configured to sense that the door is open when the door is turned by a predetermined angle; an electric driving unit configured to move the drawer forward to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open; a rail configured to allow the drawer to move forward and rearward relative to the storage compartment; and an elastic device selectively connected to the drawer for providing an elastic restoring force to the drawer when the drawer returns from the ready position to the initial position.
- (433) 31. The refrigerator according to the thirtieth embodiment, wherein the sensor is configured to sense that the door is closed when the door is turned and closed by a predetermined angle after sensing that the door is open.
- (434) 32. The refrigerator according to the thirty-first embodiment, wherein the predetermined angle at which it is sensed that the door is open is equal to the predetermined angle at which it is sensed that the door is closed.
- (435) 33. The refrigerator according to the thirty-first embodiment, wherein the predetermined angle at which it is sensed that the door is open is greater than the predetermined angle at which it is sensed that the door is closed.
- (436) 33. The refrigerator according to any one of the thirtieth to thirty-third embodiment, further comprising: a door switch for sensing that the door is open when tight contact between the door and the cabinet is released and sensing that the door is closed when the tight contact between the door and the cabinet is performed, wherein the door switch is provided separately from the sensor.
- (437) The following is the second list of embodiments.
- (438) 1. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a sensor for sensing whether the door is open; an electric driving unit comprising a motor assembly and a moving frame configured to move forward and rearward by driving of the motor assembly, the moving frame moving forward to move the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open; and a rail configured to allow the drawer to move forward and rearward relative to the storage compartment, wherein the electric driving unit is driven such that the moving frame returns rearward after moving forward to move the drawer to the ready position.

(439) 2-1. The refrigerator according to the first embodiment, wherein it is sensed that the door is open when the door is turned by a predetermined angle.

(440) 2-2. The refrigerator according to the first embodiment, wherein the electric driving unit remains driven in a forward direction for a predetermined time in a state in which the moving frame moves forward to move the drawer to the ready position and is then driven in a reverse direction to return the moving frame.

(441) 2-3. The refrigerator according to any one of the first to 2-2 embodiments, wherein the moving frame is connected with the drawer so as to apply a force to the drawer in a direction in which the drawer is withdrawn from the initial position to the ready position.

(442) 3. The refrigerator according to the 2-3 embodiment, wherein the moving frame is disconnected from the drawer so as not to apply a force to the drawer in a direction in which the drawer is inserted from the ready position to the initial position.

(443) 4. The refrigerator according to any one of the first to third embodiments, wherein the electric driving unit further comprises a connection member for interconnecting the motor assembly and the moving frame, and a distance between the moving frame and the motor assembly is changed in proportion to a distance by which the connection member is withdrawn from the motor assembly.

(444) 5. The refrigerator according to any one of the first to third embodiments, further comprising a sidewall defining the storage compartment and a support cover coupled to the sidewall for movably supporting the drawer along the rail.

(445) 6. The refrigerator according to the fifth embodiment, wherein a predetermined space is defined between the sidewall and the support cover, and the motor assembly is mounted to an inside surface of the support cover such that the motor assembly is positioned in the predetermined space.

(446) 7. The refrigerator according to the sixth embodiment, wherein the support cover is provided at an upper part and a lower part of the inside surface thereof with guide bars, and the moving frame is supported so as to be movable forward and rearward between the upper guide bar and the lower guide bar.

(447) 8. The refrigerator according to the sixth embodiment, wherein a slit is formed through the support cover, and the moving frame is provided with a transfer member, the transfer member extending through the slit so as to be connected with the drawer.

(448) 9. The refrigerator according to the eighth embodiment, wherein the drawer is provided with a first catching member, the first catching member being formed at a front of the transfer member so as to correspond to the transfer member, and, when the transfer member moves forward, the first catching member is pushed forward such that the drawer is moved in a direction in which the drawer is withdrawn.

(449) 10. The refrigerator according to any one of the first to ninth embodiments, wherein the drawer comprises a basket and a drawer frame provided outside the basket.

(450) 11. The refrigerator according to the tenth embodiment, wherein the moving frame is selectively connected with the drawer frame, and, when the moving frame is connected with the drawer frame, movement of the moving frame is converted into movement of the drawer.

(451) 12. The refrigerator according to the eleventh embodiment, wherein the drawer comprises a plurality of vertically arranged drawers, and the moving frame is selectively connected with the drawer frame of each of the drawers.

(452) 13. The refrigerator according to the twelfth embodiment, wherein the rail is configured to allow the drawer to move forward and rearward between a maximum withdrawal position, which is spaced apart forward from the ready position by a predetermined distance, and the initial position.

(453) 14. The refrigerator according to the thirteenth embodiment, wherein a connection between the moving frame and the drawer frame is released from the ready position to the maximum withdrawal position such that the drawer is manually withdrawn.

(454) 15. The refrigerator according to any one of the first to fourteenth embodiments, wherein the electric driving unit is driven such that the moving frame moves to the initial position regardless of whether it is sensed that the door is open or closed after moving to the ready position.

(455) 16. The refrigerator according to the fifteenth embodiment, wherein, when the moving frame returns to the initial position, a connection between the moving frame and the drawer frame is released such that the drawer remains at the ready position.

(456) 17. The refrigerator according to the sixteenth embodiment, wherein, when the moving frame returns to the initial position, the drawer is manually inserted from a maximum withdrawal position, at which the drawer is maximally withdrawn forward, to the ready position of the drawer.

(457) 18. The refrigerator according to the sixteenth embodiment, wherein, when the moving frame moves from the initial position to the ready position, the moving frame is connected with the drawer, whereby the moving frame pushes the drawer.

(458) 19. The refrigerator according to any one of the first to eighteenth embodiments, further comprising an elastic device configured to be elastically deformed when the drawer moves from the initial position to the ready position and configured to provide an elastic restoring force to the drawer when the drawer moves from the ready position to the initial position.

(459) 20. The refrigerator according to the nineteenth embodiment, wherein the elastic device comprises: a housing having a slot formed therein in a longitudinal direction and a spring mounted therein; and a hanging member configured to move along the slot to elastically deform and elastically restore the spring, the hanging member being selectively connected with the drawer.

(460) 21. The refrigerator according to the twentieth embodiment, wherein the drawer is provided with a second catching member configured to be selectively connected with the hanging member, and the hanging member and the second catching member are connected with each other as the drawer is inserted such that the drawer returns to the initial position due to an elastic restoring force of the spring.

(461) 22. The refrigerator according to the twenty-first embodiment, wherein the slot is provided at a front end thereof with a first inclined slot for limiting movement of the hanging member and releasing a connection between the hanging member and the second catching member in a state in which the hanging member maximally elastically deforms the spring.

(462) 23. The refrigerator according to the twenty-second embodiment, wherein the hanging member moves into the first inclined slot at the ready position of the drawer such that the connection between the hanging member and the second catching member is released.

(463) 24. The refrigerator according to the twenty-second embodiment, wherein the slot is provided at a rear end thereof with a second inclined slot for limiting the movement of the hanging member and releasing the connection between the hanging member and the second catching member in a state in which the hanging member maximally elastically restores the spring.

(464) 25. The refrigerator according to the twenty-fifth embodiment, wherein, after the hanging member is connected with the second catching member, the hanging member moves into the second inclined slot at a predetermined position before the drawer is inserted to the initial position such that the connection between the hanging member and the second catching member is released.

(465) 26. The refrigerator according to the twenty-fifth embodiment, wherein, as the drawer is withdrawn from the initial position to the ready position, the hanging member escapes from the second inclined slot and is connected with the second catching member.

(466) 27. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a sensor configured to sense whether the door is open or closed based on a turning angle of the door; an electric driving unit comprising a motor assembly and a moving frame configured to move forward and rearward by driving of the motor assembly, the moving frame moving forward to move the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open; a rail configured to

allow the drawer to move forward and rearward relative to the storage compartment; and an elastic device selectively connected to the drawer for providing an elastic restoring force to the drawer when the drawer returns to the initial position after being withdrawn, wherein the electric driving unit is driven such that the moving frame returns to the initial position after moving to the ready position.

(467) 28. The refrigerator according to the twenty-seventh embodiment, wherein a spring of the elastic device is elastically deformed as the drawer is withdrawn from the initial position, and the elastic device is disconnected from the drawer while remaining elastically deformed when the drawer is withdrawn to the ready position.

(468) The following is the third list of embodiments.

(469) 1. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a sensor for sensing whether the door is open; a drawer provided in the storage compartment, the drawer being configured to move to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open; an elastic device configured to be elastically deformed when the drawer moves from the initial position to the ready position and configured to provide an elastic restoring force to the drawer such that the drawer moves from the ready position to the initial position; an electric driving unit for moving the drawer from the initial position to the ready position and elastically deforming the elastic device; and a rail configured to allow the drawer to move forward and rearward relative to the storage compartment.

(470) 2. The refrigerator according to the first embodiment, wherein the elastic device comprises: a hanging member configured to be selectively connected with the drawer; and a spring configured to be elastically deformed and elastically restored depending on a direction in which the hanging member moves and a distance by which the hanging member moves.

(471) 3. The refrigerator according to the second embodiment, wherein the drawer is provided with a catching member configured to be selectively connected with the elastic device.

(472) 4. The refrigerator according to the third embodiment, wherein the elastic device further comprises a housing for receiving the spring, the housing being provided with a slot for guiding movement of the hanging member.

(473) 5. The refrigerator according to the fourth embodiment, wherein the hanging member protrudes from the housing in a horizontal direction such that the hanging member is selectively connected with the drawer.

(474) 6. The refrigerator according to any one of the first to fifth embodiments, wherein the electric driving unit comprises a motor assembly and a moving frame, and the moving frame is configured to be movable forward and rearward between the initial position and the ready position by driving of the motor assembly.

(475) 7. The refrigerator according to the sixth embodiment, wherein the moving frame comprises a transfer member for transferring a force to the drawer, and the transfer member pushes the drawer such that the drawer moves from the initial position to the ready position.

(476) 8. The refrigerator according to the third embodiment, wherein the electric driving unit comprises a motor assembly and a transfer member configured to be moved forward and rearward by driving of the motor assembly, and the transfer member moves the catching member of the drawer forward via the hanging member.

(477) 9. The refrigerator according to the eighth embodiment, wherein the hanging member protrudes from the housing of the elastic device so as to be connected to the catching member of the drawer, and the transfer member pushes the hanging member in a direction perpendicular to a direction in which the hanging member protrudes.

(478) 10. The refrigerator according to the ninth embodiment, wherein the hanging member is provided at one side thereof with a location part on which the transfer member is located in contact.

(479) 11. The refrigerator according to the tenth embodiment, wherein movement of the transfer

member is transferred to the hanging member as the transfer member moves forward while being located on the location part, and a connection between the transfer member and the hanging member is released when the transfer member moves rearward while being separated from the location part.

(480) 12. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a sensor for sensing whether the door is open; a drawer provided in the storage compartment such that the drawer moves to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open, the drawer comprising a catching member; an elastic device comprising a hanging member configured to be selectively connected with the catching member and a spring configured to be elastically deformed and elastically restored by a movement of the catching member; an electric driving unit comprising a transfer member for pushing the hanging member to move the drawer from the initial position to the ready position and a motor assembly for electrically moving the transfer member; and a rail configured to allow the drawer to move forward and rearward relative to the storage compartment.

(481) 13. The refrigerator according to the twelfth embodiment, wherein a connection between the catching member and the hanging member is maintained within a section between the initial position and the ready position.

(482) 14. The refrigerator according to the thirteenth embodiment, wherein the connection between the catching member and the hanging member is released as the drawer is withdrawn further forward from the ready position such that the drawer is manually withdrawn.

(483) 15. The refrigerator according to the thirteenth embodiment, wherein the electric driving unit is driven to move the transfer member rearward such that the drawer moves from the ready position to the initial position.

(484) 16. The refrigerator according to the fifteenth embodiment, wherein the hanging member moves the catching member rearward as the spring is elastically restored such that the drawer returns from the ready position to the initial position.

(485) 17. The refrigerator according to any one of the first to sixteenth embodiments, wherein the elastic device further comprises a housing for receiving the spring, the housing being provided with a slot for guiding movement of the hanging member.

(486) 18. The refrigerator according to the sixteenth embodiment, wherein the hanging member protrudes from the housing toward the drawer such that the hanging member is selectively connected with the drawer.

(487) 19. The refrigerator according to the eighteenth embodiment, wherein the transfer member is positioned between the housing and the catching member, the transfer member being configured to push the hanging member forward at a rear of the hanging member.

(488) 20. The refrigerator according to the twelfth embodiment, wherein the drawer comprises a plurality of vertically arranged drawers, the transfer member being provided at each of the drawers, and the electric driving unit comprises a moving frame configured to simultaneously move the transfer members by driving of the motor assembly.

(489) 21. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a moving frame configured to be selectively connected with the drawer, the moving frame being movable forward and rearward; an electric driving unit for moving the moving frame, connected with the drawer, forward to withdraw the drawer from an initial position to a ready position and moving the moving frame rearward to release the connection between the drawer and the moving frame when the door is opened; and a controller for controlling driving of the electric driving unit.

(490) 22. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the

storage compartment; and an electric driving unit for generating a driving force such that the drawer is automatically withdrawn from an initial position to a ready position regardless of application of a user's force to open the door when the door is opened, wherein the drawer is manually inserted from the ready position to the initial position regardless of the driving force of the electric driving unit.

(491) 23. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a moving frame configured to be selectively connected with the drawer, the moving frame being movable forward and rearward; and an electric driving unit for moving the moving frame, connected with the drawer, forward to withdraw the drawer from an initial position to a ready position when the door is opened, wherein the connection between the drawer and the moving frame is released such that the drawer is manually withdrawn from the ready position to a maximum withdrawal position.

(492) 24. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; an electric driving unit for driving the drawer such that the drawer is withdrawn from an initial position to a ready position when the door is opened; and a controller for controlling driving of the electric driving unit.

(493) 25. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; an electric driving unit for driving the drawer such that the drawer is withdrawn from an initial position to a ready position when the door is opened; an elastic device configured to be elastically deformed when the drawer is withdrawn and to provide an elastic restoring force to the drawer when the drawer is inserted; and a controller for controlling driving of the electric driving unit.

(494) 26. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; an electric driving unit for driving the drawer such that the drawer is withdrawn from an initial position to a ready position when the door is opened; and an elastic device configured to be elastically deformed when the drawer is withdrawn from the initial position to the ready position and to insert the drawer from the ready position to the initial position using an elastic restoring force such that an insertion speed of the drawer is different from a withdrawal speed of the drawer.

(495) 27. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; an electric driving unit for driving the drawer such that the drawer is withdrawn from an initial position to a ready position when the door is opened; and an elastic device configured to be connected with the drawer such that the elastic device is elastically deformed when the drawer is withdrawn from the initial position to the ready position and configured to be disconnected from the drawer when the drawer is withdrawn from the ready position to a maximum withdrawal position.

(496) 28. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; and a support assembly configured to be coupled to the drawer for supporting the drawer so as to be movable forward and rearward relative to the storage compartment, the support assembly being separably coupled to a left sidewall or a right sidewall of the storage compartment, wherein the support assembly comprises: a support cover configured to be coupled to the sidewall of the storage compartment; an electric driving unit configured to be driven to automatically withdraw the drawer, the electric driving unit being mounted to an inside surface of the support cover facing the sidewall of the storage compartment; a rail mounted to an

outside surface of the support cover for supporting the drawer so as to be movable forward and rearward; and a moving frame configured to transfer a driving force of the electric driving unit to the drawer, the moving frame being movably provided at the support cover separately from the rail.

(497) 29. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a plurality of drawers provided in the storage compartment, the drawers being arranged vertically; a support cover configured to be separably coupled to a one-side sidewall of the storage compartment; a plurality of rails mounted to an outside surface of the support cover for supporting the drawers so as to be movable forward and rearward; an electric driving unit configured to be driven to automatically withdraw the drawers; and a single moving frame configured to simultaneously transfer a driving force of the electric driving unit to the drawers, the single moving frame being movably provided at the support cover separately from the rail.

(498) 30. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a support cover configured to be fixedly coupled to a one-side sidewall of the storage compartment; a rail mounted to the support cover for supporting the drawer so as to be movable forward and rearward; a sensor for sensing that the door is open when the door is opened by a predetermined opening angle; an electric driving unit; a moving frame configured to be movable forward and rearward relative to the support cover by driving of the electric driving unit; and a controller for controlling the electric driving unit to move the moving frame forward such that the drawer is withdrawn from an initial position to a ready position when it is sensed that the door is open.

(499) 31. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment; a support cover configured to be fixedly coupled to a one-side sidewall of the storage compartment, the storage compartment having a recess or a through part; an electric driving unit comprising a motor and a housing for receiving the motor, the housing being fixed to an inside surface of the support cover, the housing being inserted into the recess or inserted through the through part when the support cover is coupled to the sidewall of the storage compartment; a moving frame configured to be movable forward and rearward by driving of the electric driving unit; and a controller for controlling the electric driving unit to move the moving frame forward such that the drawer is withdrawn from an initial position to a ready position when it is sensed that the door is open.

(500) The following is the fourth list of embodiments.

(501) 1. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a magnet provided at the door, the magnet being configured to turn about a rotary shaft of the door while maintaining a predetermined turning radius as the door is opened; and a reed switch provided above or under the magnet such that the reed switch is spaced apart from the magnet, the reed switch being fixed to the cabinet regardless of a hinged rotation of the door, the reed switch having a critical point of effective magnetic intensity for contact point switching when an opening angle of the door reaches a predetermined opening angle.

(502) 2. The refrigerator according to the first embodiment, wherein the magnet is configured such that a vertical distance between the magnet and the reed switch is not changed while a horizontal distance between the magnet and the reed switch is changed as the opening angle of the door is changed.

(503) 3. The refrigerator according to the second embodiment, wherein the magnet is mounted in a door decoration part defining a lower surface of the door, and the reed switch is provided under the magnet.

(504) 4. The refrigerator according to the third embodiment, wherein the rotary shaft of the door is

formed as a vertical shaft, about which the door is hingedly rotated leftward and rightward such that the door is opened and closed, the rotary shaft of the door being provided on a left side or a right side of the cabinet such that the rotary shaft of the door is spaced apart forward from the cabinet.

(505) 5. The refrigerator according to the fourth embodiment, wherein the reed switch is spaced apart forward from the cabinet, the reed switch being biased from the rotary shaft of the door to left and right middle parts of the cabinet.

(506) 6. The refrigerator according to the fifth embodiment, wherein the reed switch is provided at a hinge cover forming the rotary shaft of the door, the hinge cover being horizontal with respect to a ground.

(507) 7. The refrigerator according to the fifth embodiment, wherein the magnet is configured to pass a vertical upper part of the reed switch such that the horizontal distance between the magnet and the reed switch is increased as the opening angle of the door is increased.

(508) 8. The refrigerator according to the first embodiment, wherein it is sensed whether the door is open or closed as a result of the contact point switching of the reed switch.

(509) 9. The refrigerator according to any one of the first to eighth embodiments, wherein the magnet comprises a plurality of magnets having different horizontal angles relative to the reed switch.

(510) 10. The refrigerator according to the ninth embodiment, wherein the magnet comprises a horizontal magnet provided parallel to a front surface of the cabinet in a longitudinal direction and a vertical magnet provided substantially perpendicular to the horizontal magnet, in a state in which the door is closed.

(511) 11. The refrigerator according to the tenth embodiment, wherein the magnet is a bar-type magnet, and a length of the magnet is greater than a height of the magnet in a state in which the magnet is mounted.

(512) 12. The refrigerator according to the tenth embodiment, wherein the reed switch is configured to have a quadrangular shape having horizontal sides and vertical sides.

(513) 13. The refrigerator according to the twelfth embodiment, wherein a length of the horizontal sides is greater than a length of the vertical sides.

(514) 14. The refrigerator according to the twelfth embodiment, wherein the length of the horizontal magnet and the vertical magnet is greater than the length of the horizontal sides of the reed switch.

(515) 15. The refrigerator according to the fourteenth embodiment, wherein, when the opening angle of the door is 90 degrees, an angle between the horizontal magnet and each horizontal side of the reed switch is substantially 90 degrees.

(516) 16. The refrigerator according to the tenth embodiment, wherein the turning radius of the horizontal magnet about the rotary shaft of the door is less than the turning radius of the vertical magnet.

(517) 17. The refrigerator according to the tenth embodiment, wherein the horizontal magnet is mounted such that a distance between the horizontal magnet and the rotary shaft of the door is decreased in order to increase the opening angle of the door corresponding to the critical point, and the horizontal magnet is mounted such that the distance between the horizontal magnet and the rotary shaft of the door is increased in order to decrease the opening angle of the door corresponding to the critical point.

(518) 18. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a sensor comprising a magnet provided at the door, the magnet being configured to turn about a rotary shaft of the door while having a predetermined turning radius as the door is opened and a reed switch fixed to the cabinet, a contact point of the reed switch being switched at a critical point of effective magnetic intensity due to the magnet, the sensor being configured to sense that the door is open when an

opening angle of the door reaches a predetermined opening angle; an electric driving unit for moving the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open; and a rail configured to allow the drawer to move forward and rearward relative to the storage compartment.

(519) 19. The refrigerator according to the eighteenth embodiment, wherein the magnet is configured such that a vertical distance between the magnet and the reed switch is not changed while a horizontal distance between the magnet and the reed switch is changed as the opening angle of the door is changed.

(520) 20. The refrigerator according to the nineteenth embodiment, wherein the magnet is mounted in a door decoration part defining a lower surface of the door, and the reed switch is provided under the magnet.

(521) 21. The refrigerator according to the twentieth embodiment, wherein the magnet comprises a plurality of magnets having different horizontal angles relative to the reed switch.

(522) The following is the fifth list of embodiments.

(523) 1. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment so as to be movable forward and rearward; and a rail for supporting the drawer such that the drawer moves forward and rearward relative to the storage compartment, wherein the drawer comprises: a basket for receiving goods; and a drawer frame provided with a basket location part, on which the basket is located, and a rail coupling part located on the rail so as to be coupled to the rail.

(524) 2. The refrigerator according to the first embodiment, wherein the basket is configured to be located on the basket location part vertically downward so as to be coupled to the drawer frame and is configured to move vertically upward such that the coupling between the basket and the drawer frame is released.

(525) 3. The refrigerator according to the second embodiment, wherein the drawer frame is provided in a middle part thereof with an opening, and the basket location part is formed around the opening.

(526) 4. The refrigerator according to the third embodiment, wherein the basket comprises a lower basket configured to be inserted through the opening and an upper basket configured to be located on the basket location part, and the lower basket and the upper basket are integrally formed.

(527) 5. The refrigerator according to the fourth embodiment, wherein the rail coupling part is provided at each of left and right sides of the drawer frame such that the rail coupling part extends forward and rearward, and a horizontal width of the upper basket is greater than a horizontal width of the lower basket such that the upper basket covers the rail coupling part from above.

(528) 6. The refrigerator according to any one of the first to fifth embodiments, wherein the rail comprises a moving rail configured to move forward and rearward together with the drawer.

(529) 7. The refrigerator according to the sixth embodiment, wherein the rail coupling part is formed in a sectional shape of a channel such that the rail coupling part is located while surrounding the moving rail.

(530) 8. The refrigerator according to the seventh embodiment, wherein the moving rail is provided at a rear thereof with a catching part, the catching part being configured to catch a rear end of the rail coupling part so as to limit a rearward movement and an upward movement of the rail coupling part.

(531) 9. The refrigerator according to the eighth embodiment, wherein the rear end of the rail coupling part is inserted into the catching part.

(532) 10. The refrigerator according to the eighth embodiment or the ninth embodiment, wherein an elastic protrusion is provided at a front end of the moving rail, and the rail coupling part is provided at a front end thereof with a mounting hole, into which the elastic protrusion is inserted while being elastically restored after being elastically deformed.

(533) 11. The refrigerator according to the tenth embodiment, wherein, as the front end of the rail coupling part moves downward in a state in which the rear end of the rail coupling part is inserted in the catching part, the elastic protrusion is elastically deformed and is then inserted and coupled into the mounting hole.

(534) 12. The refrigerator according to the eleventh embodiment, wherein the moving rail is provided at each side of the drawer, and the drawer frame is fixed to the moving rail at four support points, respectively positioned at a front, rear, left, and right, by the elastic protrusion and the catching part.

(535) 13. The refrigerator according to the sixth embodiment, wherein the rail comprises a fixed rail fixed in the storage compartment, the fixed rail being provided under the moving rail, and the moving rail is configured to be slidable relative to the fixed rail.

(536) 14. The refrigerator according to the fifth embodiment, wherein the drawer further comprises a drawer decoration part provided at a lower part of a front of the basket such that the drawer decoration part extends leftward and rightward so as to cover the rail coupling part at a front of the drawer.

(537) 15. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer provided in the storage compartment so as to be movable forward and rearward, the drawer comprising a basket for receiving goods and a drawer frame provided with a rail coupling part; and a rail coupled to the rail coupling part for supporting the drawer such that the drawer moves forward and rearward relative to the storage compartment, wherein the rail coupling part is formed in a channel shape such that the rail coupling part is located on the rail downward from above so as to surround the rail, and the rail is provided at a rear thereof with a catching part, into which a rear end of the rail coupling part is inserted, and the rail is provided at a front thereof with an elastic protrusion, which is inserted into a mounting hole provided at a front end of the rail coupling part.

(538) 16. The refrigerator according to the fifteenth embodiment, wherein the rail comprises a moving rail and a fixed rail provided under the moving rail for slidably supporting the moving rail, and the rail coupling part is coupled to the moving rail.

(539) 17. The refrigerator according to the sixteenth embodiment, wherein the mounting hole is formed in a side flange configured to cover an outside surface of the moving rail, and the elastic protrusion is elastically deformed toward left and right middles of the drawer and is then restored and inserted into the mounting hole.

(540) 18. The refrigerator according to the sixteenth embodiment, wherein an elastic protrusion bracket for forming the elastic protrusion at the fixed rail is mounted to a front end of the fixed rail, and a cutout part for allowing the elastic protrusion to be elastically deformed is formed between the elastic protrusion bracket and the elastic protrusion.

(541) 19. The refrigerator according to the fifteenth embodiment, wherein the rail comprises a rail bracket for fixing the fixed rail to a sidewall of the storage compartment.

(542) 20. The refrigerator according to the nineteenth embodiment, wherein the rail bracket comprises a front rail bracket and a rear rail bracket respectively provided at a front and a rear of the fixed rail such that the front rail bracket and the rear rail bracket are spaced apart from each other by a predetermined distance.

(543) 21. The refrigerator according to the twentieth embodiment, wherein the drawer frame is provided with a catching member protruding toward the sidewall of the storage compartment, and a transfer member protrudes from the sidewall of the storage compartment toward the drawer frame to push the catching member at a rear of the catching member.

(544) 22. The refrigerator according to the twentieth embodiment, further comprising: an electric driving unit for moving the catching member based on a movement of the transfer member to move the drawer to a ready position spaced apart forward from an initial position by a predetermined distance, wherein the transfer member and the catching member are configured to move between

the front rail bracket and the rear rail bracket within a section between the initial position and the ready position of the drawer.

(545) 23. A refrigerator comprising: a cabinet having a storage compartment; a door hingedly connected to the cabinet for opening and closing the storage compartment; a sensor for sensing whether the door is open; a drawer provided in the storage compartment so as to be movable forward and rearward, the drawer comprising a basket for receiving goods and a drawer frame provided with a rail coupling part and a catching member protruding toward a sidewall of the storage compartment; a rail coupled to the rail coupling part for supporting the drawer such that the drawer moves forward and rearward relative to the storage compartment, the rail being supported by the sidewall of the storage compartment via a front rail bracket and a rear rail bracket; and an electric driving unit comprising a transfer member protruding from the sidewall of the storage compartment toward the drawer frame, the electric driving unit moving the catching member based on a movement of the transfer member to move the drawer to a ready position spaced apart forward from an initial position by a predetermined distance when it is sensed that the door is open, wherein the transfer member is configured to move between the front rail bracket and the rear rail bracket within a section between the initial position and the ready position of the drawer in order to avoid an interference between the transfer member and the front and rear rail brackets.

(546) The following is the sixth list of embodiments.

(547) 1. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: determining a condition for automatically withdrawing the drawer in a ready state (a determination step); upon determining at the determination step that the condition is satisfied, driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position (a withdrawal step); continuously driving the motor in the one direction to stop withdrawal of the drawer after the withdrawal step (a stopping step); and driving the motor in a reverse direction to return the transfer member rearward after the stopping step (a returning step).

(548) 2. The control method according to the first embodiment, further comprising driving the motor in the reverse direction to move the transfer member to the initial position in order to start the ready state when the refrigerator is powered on (an initial step).

(549) 3. The control method according to the second embodiment, wherein, at the initial step, the motor is driven in the reverse direction until a first hall sensor for sensing the initial position generates a signal or until a predetermined time has lapsed.

(550) 4. The control method according to the third embodiment, wherein, when driving of the motor is stopped at the initial step, the ready state is started in order to perform the determination step.

(551) 5. The control method according to the third embodiment, further comprising announcing that the refrigerator is operating abnormally when a second hall sensor for sensing the ready position generates a signal, or when the motor is driven until a predetermined time has lapsed, at the initial step (a notification step).

(552) 6. The control method according to the first embodiment, wherein, at the withdrawal step, the motor is driven until a second hall sensor for sensing the ready position generates a signal or until a predetermined time has lapsed.

(553) 7. The control method according to the sixth embodiment, wherein the motor is a brushless direct current (BLDC) motor, and, at the withdrawal step, a signal (frequency generator; FG) generated during rotation of the motor is calculated and a duty ratio to drive the motor is controlled to be different based on the calculated FG.

(554) 8. The control method according to the seventh embodiment, wherein the duty ratio is controlled to be increased as the calculated FG is increased.

(555) 9. The control method according to the sixth embodiment, wherein, when the calculated FG is less than an obstacle FG corresponding to an obstacle blocking withdrawal of the drawer during the withdrawal step, the returning step is performed without the stopping step being performed.

(556) 10. The control method according to the sixth embodiment, further comprising notifying that the refrigerator is operating abnormally when a first hall sensor for sensing the initial position generates a signal, or when the motor is driven until a predetermined time has lapsed, at the withdrawal step (a notification step).

(557) 11. The control method according to the first embodiment, wherein the stopping step is performed at a minimum duty ratio to drive the motor for a predetermined time.

(558) 12. The control method according to the eleventh embodiment, wherein the FG is calculated at the stopping step, and, when the FG calculated during the stopping step is a predetermined FG, the returning step is performed.

(559) 13. The control method according to the first embodiment, wherein the returning step comprises stopping the driving of the motor for a predetermined time.

(560) 14. The control method according to the first embodiment, wherein at the returning step, the motor is driven in the reverse direction such that the driving of the motor is stopped until a first hall sensor for sensing the initial position generates a signal or until a predetermined time has lapsed.

(561) 15. The control method according to the fourteenth embodiment, further comprising notifying that the refrigerator is operating abnormally when a second hall sensor for sensing the ready position generates a signal, or when the motor is driven until the predetermined time has lapsed, at the returning step (a notification step).

(562) 16. The control method according to the fourteenth embodiment, wherein the driving of the motor is stopped, and switching to the ready state is performed.

(563) 17. The control method according to any one of the first to sixteenth embodiments, wherein the refrigerator further comprises a door for opening and closing the storage compartment and a sensor for sensing that the door is open when an opening angle of the door is a predetermined opening angle, and the condition for automatically withdrawing the drawer comprises generating a door opening signal through the sensor.

(564) 18. The control method according to the seventeenth embodiment, wherein the refrigerator further comprises a door switch for sensing whether the door is in tight contact with the cabinet to sense whether the door is open or closed, and the condition for automatically withdrawing the drawer further comprises sensing whether the door is open through the door switch.

(565) 19. The control method according to the seventeenth embodiment, further comprising stopping the withdrawal step or the stopping step and driving the motor in the reverse direction to return the transfer member rearward when it is sensed through the sensor that the door is closed during the withdrawal step or the stopping step (an urgent returning step).

(566) 20. The control method according to the nineteenth embodiment, wherein, when the urgent returning step starts to be performed in a state in which the motor is driven in the one direction, the driving of the motor is stopped for a predetermined time and is then driven in the reverse direction.

(567) 21. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: determining a condition for automatically withdrawing the drawer in a ready state (a determination step); upon determining at the determination step that the condition is satisfied, driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position (a withdrawal step); and driving the motor in a reverse direction to return the transfer member rearward (a returning step).

(568) 22. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member

for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position when a door for opening and closing the storage compartment is opened (a withdrawal step); and stopping the withdrawal step and driving the motor in a reverse direction to return the transfer member rearward when the door is closed during the withdrawal step (a returning step).

(569) 23. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position when a door for opening and closing the storage compartment is opened (a withdrawal step); and driving the motor in a reverse direction to return the transfer member rearward in order to release a connection between the drawer and the transfer member after the withdrawal step (a returning step).

(570) 24. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position when a door for opening and closing the storage compartment is opened (a withdrawal step); continuously driving the motor in the one direction to stop the withdrawal of the drawer after the withdrawal step (a stopping step); and driving the motor in a reverse direction to return the transfer member rearward in order to release a connection between the drawer and the transfer member after the stopping step (an urgent returning step).

(571) 25. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position when a door for opening and closing the storage compartment is opened (a withdrawal step); driving the motor in a reverse direction to return the transfer member rearward in order to release a connection between the drawer and the transfer member after the withdrawal step (a returning step); and stopping the withdrawal step and driving the motor in the reverse direction to return the transfer member rearward when the door is closed during the withdrawal step (an urgent returning step).

(572) 26. A control method of a refrigerator comprising a motor, a drawer provided in a storage compartment defined in a cabinet so as to be movable forward and rearward, and a transfer member for pushing the drawer to automatically withdraw the drawer from an initial position to a ready position by driving of the motor, the control method comprising: driving the motor in one direction to move the transfer member forward such that the drawer is withdrawn to the ready position when a door for opening and closing the storage compartment is opened (a withdrawal step); continuously driving the motor in the one direction to stop the withdrawal of the drawer after the withdrawal step (a stopping step); driving the motor in a reverse direction to return the transfer member rearward in order to release a connection between the drawer and the transfer member after the stopping step (a returning step); and stopping the withdrawal step or the stopping step and driving the motor in the reverse direction to return the transfer member rearward when the door is closed during the withdrawal step or the stopping step (an urgent returning step).

(573) The following is the seventh list of embodiments.

(574) 1. A refrigerator comprising: a cabinet having a storage compartment with a food introduction port formed in a front thereof; a door hingedly connected to the cabinet for opening

and closing the storage compartment; a plurality of drawers disposed in the storage compartment, the drawers being arranged vertically; a moving frame extending vertically so as to correspond to a height at which the drawers are disposed, the moving frame being configured to selectively push the drawers such that the drawers are moved toward the food introduction port; an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port; and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

(575) 2. The refrigerator according to the first embodiment, wherein the moving frame is provided with a plurality of transfer members configured to be selectively connected to the respective drawers.

(576) 3. The refrigerator according to the second embodiment, wherein the drawers are provided with catching members configured to be pushed by the respective transfer members.

(577) 4. The refrigerator according to the third embodiment, wherein a connection between the moving frame and the drawers is released when the moving frame returns rearward.

(578) 5. The refrigerator according to the fourth embodiment, wherein the controller controls the electric driving unit to return the moving frame rearward after the moving frame withdraws the drawers.

(579) 6. The refrigerator according to any one of the first to fifth embodiments, further comprising rails for supporting the drawers such that the drawers move relative to the storage compartment.

(580) 7. The refrigerator according to the sixth embodiment, further comprising: a support cover configured to be coupled to a left or right sidewall of the storage compartment, wherein the rails are mounted to an outside surface of the support cover.

(581) 8. The refrigerator according to the seventh embodiment, wherein the moving frame is movably provided at the support cover separately from the rails.

(582) 9. The refrigerator according to any one of the first to fifth embodiments, wherein the moving frame is a single moving frame configured to simultaneously transfer a driving force of the electric driving unit to the drawers.

(583) 10. The refrigerator according to the ninth embodiment, wherein the electric driving unit comprises: a motor assembly; a housing for receiving the motor assembly; and a connection member for interconnecting the motor assembly and the moving frame.

(584) 11. The refrigerator according to the tenth embodiment, wherein the moving frame is moved in proportion to a distance by which the connection member is withdrawn from the housing.

(585) 12. The refrigerator according to the ninth embodiment, further comprising a support cover having one side surface, to which rails for supporting the drawers are coupled, and the other side surface, coupled to a sidewall defining the storage compartment, the support cover being configured to movably support the drawers along the rails.

(586) 13. The refrigerator according to the twelfth embodiment, wherein the motor assembly is mounted between the sidewall and the support cover.

(587) 14. The refrigerator according to the thirteenth embodiment, wherein the moving frame is configured to move forward and rearward between the sidewall and the support cover.

(588) 15. The refrigerator according to the fourteenth embodiment, wherein the support cover is provided at an inside surface thereof with a guide bar connected with the moving frame for movably supporting the moving frame.

(589) 16. The refrigerator according to the fifteenth embodiment, wherein the moving frame is provided with a sliding support part for surrounding the guide bar.

(590) 17. The refrigerator according to the fifteenth embodiment, wherein the sliding support part comprises at least two sliding support parts provided at an upper end, a middle part, and a lower end of the moving frame.

(591) 18. The refrigerator according to the fifteenth embodiment, wherein a liner is provided between the guide bar and the sliding support part.

(592) 19. The refrigerator according to the eighteenth embodiment, wherein the liner is provided at an upper inside surface, a lower inside surface, a left inside surface, and a right inside surface thereof with friction avoidance recesses for reducing a frictional area between the guide bar and the liner.

(593) 20. The refrigerator according to the thirteenth embodiment, further comprising a moving frame cover coupled to the support cover for covering the moving frame.

(594) 21. The refrigerator according to the twentieth embodiment, wherein the moving frame is provided with a roller such that the moving frame is slidably supported by the moving frame cover via the roller.

(595) 22. The refrigerator according to the ninth embodiment, wherein the moving frame is formed in a plate shape having a vertical height larger than a width in a direction in which the moving frame moves and a thickness smaller than the width.

(596) 23. The refrigerator according to the twenty-second embodiment, further comprising a transfer member for transferring movement of the moving frame to the drawers through a through part formed in the support cover.

(597) 24. The refrigerator according to the twenty-third embodiment, wherein the transfer member extends from the moving frame in a thickness direction of the moving frame such that the transfer member extends through the through part.

(598) 25. A refrigerator comprising: a cabinet having a storage compartment with a food introduction port formed in a front thereof; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer disposed in the storage compartment; a moving frame configured to selectively push the drawer such that the drawer is moved toward the food introduction port; an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port; and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open, wherein the drawer remains separated from the moving frame at a position at which a front part of the drawer is withdrawn after escaping from the food introduction port.

(599) 26. The refrigerator according to the twenty-fifth embodiment, wherein the electric driving unit is driven such that the drawer is withdrawn from an initial position to a ready position at which the front part of the drawer does not escape from the food introduction port.

(600) 27. The refrigerator according to the twenty-sixth embodiment, wherein the withdrawal of the drawer from the initial position to the ready position is performed by a driving force of the electric driving unit regardless of application of a user's force to open the door.

(601) 28. The refrigerator according to the twenty-sixth embodiment, wherein a connection between the moving frame and the drawer is maintained such that moving frame pushes the drawer to the ready position.

(602) 29. The refrigerator according to the twenty-sixth embodiment, wherein a connection between the moving frame and the drawer is released when the moving frame returns rearward.

(603) 30. The refrigerator according to the twenty-fifth embodiment, wherein the drawer is configured to be manually withdrawn from a ready position of the drawer to a maximum withdrawal position of the drawer.

(604) 31. The refrigerator according to any one of the twenty-fifth to thirtieth embodiments, wherein the drawer is configured to be manually inserted.

(605) 32. The refrigerator according to the thirty-first embodiment, wherein the electric driving unit is driven to move the moving frame toward the food introduction port and is then driven to return the moving frame such that the drawer is manually inserted.

(606) 33. A refrigerator comprising: a cabinet having a storage compartment with a food introduction port formed in a front thereof; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer disposed in the storage compartment; a moving frame configured to selectively push the drawer such that the drawer is moved toward the food

introduction port; an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port; a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open; and an elastic device coupled to one side of the drawer and an inside wall of the storage compartment for selectively generating an elastic restoring force, wherein the drawer is returned by the elastic restoring force.

(607) 34. The refrigerator according to the thirty-third embodiment, wherein the elastic device is elastically deformed as the drawer is withdrawn and provides the elastic restoring force to the drawer when the drawer is inserted.

(608) 35. The refrigerator according to the thirty-third embodiment, wherein the elastic device is selectively connected with the drawer.

(609) 36. The refrigerator according to the thirty-fifth embodiment, wherein the drawer is provided with a catching member, and the elastic device comprises a hanging member configured to be selectively connected with the catching member.

(610) 37. The refrigerator according to the thirty-sixth embodiment, wherein the elastic device further comprises a spring configured to be elastically deformed and elastically restored depending on displacement of the hanging member.

(611) 38. The refrigerator according to the thirty-seventh embodiment, wherein the electric driving unit drives the drawer such that the drawer is withdrawn from an initial position to a ready position.

(612) 39. The refrigerator according to the thirty-eighth embodiment, wherein a connection between the elastic device and the drawer is maintained at the ready position of the drawer.

(613) 40. The refrigerator according to the thirty-ninth embodiment, wherein the drawer is configured to be manually withdrawn from the ready position to a maximum withdrawal position.

(614) 41. The refrigerator according to the fortieth embodiment, wherein the connection between the elastic device and the drawer is released when the drawer is withdrawn forward from the ready position.

(615) 42. The refrigerator according to the forty-first embodiment, wherein the elastic device is elastically deformed by a driving force of the electric driving unit when the drawer is withdrawn to the ready position, and the elastic device is elastically deformed by a user's force to manually withdraw the drawer when the drawer is withdrawn from the ready position until the connection between the elastic device and the drawer is released.

(616) 43. The refrigerator according to the forty-second embodiment, wherein the elastic device remains elastically deformed when the connection between the elastic device and the drawer is released.

(617) 44. The refrigerator according to the thirty-eighth embodiment, wherein the connection between the elastic device and the drawer is maintained at the initial position of the drawer.

(618) 45. The refrigerator according to the forty-fourth embodiment, wherein the connection between the elastic device and the drawer is maintained from the initial position to the ready position of the drawer.

(619) 46. The refrigerator according to the forty-fourth embodiment, wherein the connection between the elastic device and the drawer is released at the ready position of the drawer.

(620) 47. The refrigerator according to the thirty-eighth embodiment, wherein the connection between the elastic device and the drawer is released at the initial position of the drawer, and the elastic device and the drawer are connected with each other before the drawer is withdrawn to the ready position.

(621) 48. The refrigerator according to the forty-seventh embodiment, wherein the connection between the elastic device and the drawer is released again at the ready position of the drawer.

(622) 49. The refrigerator according to the thirty-seventh embodiment, wherein the elastic device further comprises a housing for receiving the spring, the housing being provided with a slot for guiding movement of the hanging member.

(623) 50. The refrigerator according to the forty-ninth embodiment, wherein the hanging member

protrudes in a direction parallel to the housing or in a direction perpendicular to the housing such that the hanging member is selectively connected with the drawer.

(624) 51. A refrigerator comprising: a cabinet having a storage compartment with a food introduction port formed in a front thereof; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer disposed in the storage compartment; a moving frame coupled to the drawer for moving the drawer forward and rearward; an electric driving unit coupled to the moving frame; and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open or closed, wherein the controller controls the electric driving unit to be driven at a higher speed when the door is closed than when the door is opened.

(625) 52. The refrigerator according to the fifty-first embodiment, wherein the electric driving unit is driven to withdraw the drawer from an initial position to a ready position when it is sensed that the door is open.

(626) 53. The refrigerator according to the fifty-second embodiment, wherein the withdrawal of the drawer from the initial position to the ready position is automatically performed by a driving force of the electric driving unit regardless of application of a user's force to open the door.

(627) 54. The refrigerator according to the fifty-first embodiment, wherein the drawer is configured to be automatically inserted to the initial position by an elastic restoring force provided by the elastic device.

(628) 55. The refrigerator according to the fifty-fourth embodiment, wherein the electric driving unit is driven to return the moving frame rearward when it is sensed that the door is closed.

(629) 56. The refrigerator according to the fifty-fifth embodiment, wherein the moving frame is controlled such that a speed at which the moving frame moves rearward is higher than a speed at which the moving frame moves forward.

(630) 57. The refrigerator according to the fifty-sixth embodiment, wherein the electric driving unit is driven to return the moving frame rearward after moving the moving frame forward.

(631) 58. The refrigerator according to the fifty-fourth embodiment, wherein the elastic device is elastically deformed as the drawer is withdrawn from the initial position, and the connection between the drawer and the elastic device is released in a state in which the elastic device remains elastically deformed at the ready position.

(632) 59. The refrigerator according to the fifty-eighth embodiment, wherein the electric driving unit is driven to return the moving frame rearward after moving the moving frame forward.

(633) 60. The refrigerator according to the fifty-ninth embodiment, wherein the electric driving unit is driven to move the returned moving frame rearward when it is sensed that the door is closed, and the drawer moves rearward such that the drawer is reconnected with the elastic device as a result of rearward movement of the moving frame.

(634) 61. A refrigerator comprising: a cabinet having a storage compartment with a food introduction port formed in a front thereof; a door hingedly connected to the cabinet for opening and closing the storage compartment; a drawer disposed in the storage compartment; a fixed rail coupled to a sidewall of the storage compartment for supporting a load of the drawer; a moving rail movably coupled to the fixed rail, the moving rail being coupled to a side surface of the drawer; a moving frame for selectively pushing the drawer to move the drawer toward the food introduction port; an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port; and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

(635) 62. The refrigerator according to the sixty-first embodiment, wherein the moving frame is configured to be movable relative to the sidewall of the storage compartment separately from the rail.

(636) 63. The refrigerator according to the sixty-second embodiment, further comprising: a support cover configured to be coupled to the sidewall of the storage compartment, wherein the fixed rail is

mounted to an outside surface of the support cover.

(637) 64. The refrigerator according to the sixty-third embodiment, wherein the moving frame is movably mounted to an inside surface of the support cover.

(638) 65. The refrigerator according to the sixty-fourth embodiment, wherein the drawer is provided with a catching member, and the refrigerator further comprises a transfer member selectively connected with the catching member for selectively transferring movement of the moving frame to the catching member.

(639) 66. The refrigerator according to the sixty-fifth embodiment, wherein the support cover is provided with a slit, and the transfer member is configured to extend through the slit so as to interconnect the moving frame and the catching member.

(640) 67. The refrigerator according to the sixty-fifth embodiment, wherein the transfer member and the catching member are connected with each other such that only a horizontal force is transferred to the catching member.

(641) 68. The refrigerator according to the sixty-seventh embodiment, wherein the transfer member is configured to push the catching member at a rear of the catching member.

(642) 69. The refrigerator according to the sixty-sixth embodiment, wherein the fixed rail is mounted to the support cover via a front rail bracket and a rear rail bracket, and the slit is formed between the front rail bracket and the rear rail bracket such that a forward and rearward movement of the transfer member is not impeded by the front rail bracket and the rear rail bracket.

(643) 70. The refrigerator according to the sixty-fifth embodiment, wherein the drawer is provided at a side surface thereof with a rail coupling part located on the moving rail so as to be connected with the moving rail.

(644) 71. The refrigerator according to the seventieth embodiment, wherein the catching member protrudes from the rail coupling part in a lateral direction such that the catching member and the moving rail are arranged side by side in a leftward and rightward direction.

(645) 72. The refrigerator according to the sixty-third embodiment, wherein the support cover and the sidewall are provided with a plurality of fastening parts such that a load applied to the support cover is transferred to the sidewall via the fastening parts.

(646) 73. The refrigerator according to the seventy-second embodiment, wherein a load of the drawer is transferred to the support cover via the moving rail and the fixed rail, and the moving frame is selectively connected with the drawer only in a horizontal direction such that the load of the drawer is not transferred to the moving frame.

(647) 74. A refrigerator comprising: a cabinet having a storage compartment with a food introduction port formed in a front thereof; a door hingedly connected to the cabinet for opening and closing the storage compartment; a support cover mounted to an inside wall of the storage compartment so as to define a sidewall of the storage compartment, the support cover being provided with a through part; a plurality of rails mounted to an outside surface of the support cover, the rails being arranged vertically; a plurality of drawers disposed in the storage compartment such that the drawers are inserted or withdrawn through the food introduction port along the rails, the drawers being arranged vertically; a moving frame disposed inside the support cover, the moving frame extending vertically so as to correspond to a height at which the drawers are disposed, the moving frame being configured to selectively push the drawers through a transfer member extending through the through part such that the drawers are moved toward the food introduction port; an electric driving unit coupled to the moving frame for moving the moving frame toward the food introduction port inside the support cover; and a controller for controlling the electric driving unit to move the moving frame when it is sensed that the door is open.

(648) 75. The refrigerator according to the seventy-fourth embodiment, wherein the support cover is coupled to the sidewall of the storage compartment so as to cover the electric driving unit and the moving frame such that the electric driving unit and the moving frame are prevented from being exposed in the storage compartment.

(649) 76. The refrigerator according to the seventy-fourth embodiment, wherein the sidewall is provided with a through part, through which the electric driving unit is inserted, or a recess, into which the electric driving unit is inserted.

(650) 77. The refrigerator according to the seventy-sixth embodiment, wherein the sidewall is a partition wall for partitioning the storage compartment into left and right storage compartments, and the support cover is mounted to the partition wall.

(651) 78. The refrigerator according to the seventy-fourth embodiment, wherein the through part or the recess is formed between an upper rail and a lower rail.

(652) 79. The refrigerator according to the seventy-fourth embodiment, wherein the electric driving unit comprises: a motor assembly comprising a motor; a housing for receiving the motor assembly; and a connection member coupled to the moving frame, the connection member being configured such that a length by which the connection member protrudes from the housing is variable.

(653) 80. The refrigerator according to the seventy-ninth embodiment, wherein the support cover is provided with a motor avoidance recess for receiving at least a portion of the motor.

(654) 81. The refrigerator according to any one of the seventy-fourth to eightieth embodiments, wherein the rails, the moving frame, and the electric driving unit are mounted to the support cover so as to constitute a single moving assembly, and the support cover is separably coupled to the sidewall, whereby the moving assembly is separably coupled to the sidewall.

(655) 82. The refrigerator according to the eighty-first embodiment, wherein the moving assembly comprises an elastic device mounted to the support cover for selectively providing an elastic restoring force to each of the drawers.

(656) 83. The refrigerator according to the eighty-first embodiment, wherein each of the drawers comprises: a basket for receiving goods; and a drawer frame for receiving the basket downward from above, the drawer frame being provided with a rail coupling part coupled to a corresponding one of the rails.

(657) 84. The refrigerator according to the eighty-third embodiment, wherein the rail coupling part is formed in a channel shape such that the rail coupling part is located on the rail downward from above so as to surround the rail.

(658) 85. The refrigerator according to the eighty-fourth embodiment, wherein the rail is provided at a rear thereof with a catching part, into which a rear end of the rail coupling part is inserted, and the rail is provided at a front thereof with an elastic protrusion, which is inserted into a mounting hole provided in a front end of the rail coupling part.

(659) 86. The refrigerator according to the eighty-fifth embodiment, wherein the rail comprises a moving rail and a fixed rail provided under the moving rail for slidably supporting the moving rail, and the rail coupling part is coupled to the moving rail.

(660) 87. The refrigerator according to the eighty-sixth embodiment, wherein the mounting hole is formed in a side flange configured to cover an outside surface of the moving rail, and the elastic protrusion is elastically deformed toward left and right middles of the drawer and is then restored and inserted into the mounting hole.

(661) 88. The refrigerator according to the eighty-fourth embodiment, wherein the rail coupling part is provided with a first catching member configured to be selectively connected with the transfer member.

(662) 89. The refrigerator according to the eighty-eighth embodiment, further comprising: an elastic device mounted to the support cover for selectively providing an elastic restoring force to each of the drawers, wherein the rail coupling part is provided with a second catching member configured to be selectively connected with the elastic device.

(663) 90. The refrigerator according to the eighty-ninth embodiment, wherein the second catching member is provided above the first catching member such that the second catching member is separably coupled to the rail coupling part.

Claims

1. A refrigerator comprising: a cabinet with a storage space; a drawer that is disposed in the storage space, and that is configured to move from a first position to a second position; a brushless direct current (BLDC) motor that is configured to supply a driving force to move the drawer; a sensor that is configured to sense a position of the drawer; and a controller that is configured to control the BLDC motor to maintain the driving force for a predetermined time based on the sensor sensing that the drawer is in the second position.
2. The refrigerator according to claim 1, wherein the controller is configured to perform a stopping operation to cause the drawer to stop at the second position, and is configured to control the BLDC motor to maintain the driving force during the predetermined time.
3. The refrigerator according to claim 1, wherein the controller is configured to calculate a frequency generator (FG) signal generated during a rotation of the BLDC motor, and control a duty ratio for driving the BLDC motor based on the FG signal.
4. The refrigerator according to claim 3, wherein the controller is configured to minimize the duty ratio of the BLDC motor while maintaining the driving force.
5. The refrigerator according to claim 3, wherein the controller is configured to perform a returning operation to cause the drawer to return to the first position based on the calculated FG signal being less than a predetermined FG signal.
6. The refrigerator according to claim 5, wherein the controller is configured to perform the returning operation based on the calculated FG signal being less than the predetermined FG signal a predetermined number of times.
7. The refrigerator according to claim 5, wherein the controller is configured to stop the drawer before performing the returning operation.
8. The refrigerator according to claim 1, wherein the sensor comprises a first sensor that is configured to sense the first position and a second sensor that is configured to sense the second position.
9. The refrigerator according to claim 8, wherein the controller is configured to stop the BLDC motor based on the first sensor sensing that the drawer is in the first position while the controller is performing a returning operation.
10. The refrigerator according to claim 9, wherein the controller is configured to display a drawer error based on the BLDC motor being stopped and the second sensor sensing that the drawer is in the second position.
11. A refrigerator comprising: a cabinet with a storage space; a drawer that is disposed in the storage space and that is configured to move from a first position to a second position; a brushless direct current (BLDC) motor that is configured to supply a driving force to move the drawer; a sensor that is configured to sense a position of the drawer; a display unit that is configured to display an error signal; a controller that is configured to control the BLDC motor in sequence of an initial operation, a withdrawing operation, a stopping operation, and a returning operation, and wherein the controller is configured to control the BLDC motor to maintain the driving force for a first predetermined time based on the sensor sensing that the drawer is in the second position, and wherein the controller is configured to stop the driving force of the BLDC motor and control the display unit to display an error signal based on an operation time of the BLDC motor for the initial operation being greater than a second predetermined time.
12. The refrigerator according to claim 11, wherein the controller is configured to control the display unit to display the error signal based on the sensor sensing that the drawer is in the second position during the initial operation.
13. The refrigerator according to claim 11, wherein the controller is configured to increase a duty ratio of the BLDC motor during the withdrawing operation based on a load of the BLDC motor

increasing.

14. The refrigerator according to claim 11, wherein the controller is configured to calculate a frequency generator (FG) signal generated during a rotation of the BLDC motor and increase the duty ratio for driving the BLDC motor based on an increase of the calculated FG signal.

15. The refrigerator according to claim 14, wherein the controller is configured to change from the withdrawing operation to the returning operation based on the calculated FG signal being less than a predetermined FG signal.

16. The refrigerator according to claim 14, wherein the controller is configured to determine whether a user is pushing the drawer during the stopping operation based on the calculated FG signal.

17. The refrigerator according to claim 16, wherein the controller is configured to change from the stopping operation to the returning operation based on determining that the user is pushing the drawer.

18. The refrigerator according to claim 11, further comprising: a door that is configured to rotate relative to the cabinet by a hinge and that is configured to open and close the storage space; and a rotation sensor that is configured to sense a rotation angle of the door.

19. The refrigerator according to claim 18, wherein the drawer is configured to start moving from the first position to the second position based on the rotation sensor sensing a first rotation angle of the door.

20. The refrigerator according to claim 19, wherein the drawer is configured to move to the first position based on the rotation sensor sensing a second rotation angle of the door.
